

DOCTORAL THESIS

Paris-Sorbonne University

Master 2 Economic Intelligence - Intelligence Warfare

AI FOR AMERICANS FIRST

*AI Protectionism, Energy and Semiconductors: US/Europe Divergence Trajectories
2024-2030*

Integrated Geostrategic and Economic Analysis

**78.9% of global operational AI
compute = USA**

**1.59x EU/US energy cost (PPP-
adjusted)**

**3.64:1 US/EU CACI ratio (Power
Mode)**

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Paris - February 2026

11 Chapters | 4 Prospective Scenarios | 4 Geographic Zones | 108 Figures

Keywords: artificial intelligence, technological protectionism, semiconductors, export controls, sovereign compute, AI geopolitics, France, United States, China.

Executive Summary

This thesis analyzes the emergence of a US AI protectionist regime (the implicit 'AI for Americans First' doctrine) and its geostrategic consequences for France, Europe, and the rest of the world over the 2024-2030 period. The study employs an original empirical framework (public dashboard snapshot April 2026) and proposes a unified theoretical model: the Compute-Adjusted Competitive Index (CACI).

As of April 2026, the United States concentrates 78.9% of global operational AI compute. The raw US/EU compute ratio stands at 17.6:1, which the CACI Power Mode formula translates into a 3.64:1 advantage. While the EU maintains high sovereignty over its physical install base (78.6%), its vulnerability lies in the operational cloud layer (hosted mainly on AWS/Azure/GCP).

Four prospective scenarios for 2026-2030 are developed (Status Quo, Fracture, Asymmetric Partnership, Forced Sovereignty), identifying a tipping point in 2028 where EU compute/energy saturation meets the potential activation of US Cloud Sovereignty Mandates.

Strategic recommendations focus on five axes: compute, energy, alliances, regulation, and talent. France's unique advantages (nuclear fleet, Mistral AI, ASML partnership) position it as a credible pillar for targeted strategic autonomy. The goal is not autarky but the capacity for sovereign choice.

Table of Contents

1. Chapter I - Introduction and Theoretical Framework
2. Chapter II - Methodology
3. Chapter III - Empirical Diagnosis 2020-2026
4. Chapter IV - Mechanisms of US Competitive Advantage
5. Chapter V - Prospective Scenarios 2026-2030
6. Chapter VI - Consequences for France and Europe
7. Chapter VI bis - Consequences for South America and Brazil
8. Chapter VI ter - Consequences for Asia and China
9. Chapter VI quater - Consequences for Africa
10. Chapter VII - Strategic Recommendations
11. General Conclusion

Note: a dynamic Table of Contents is included below. Press Ctrl+A then F9 in Word to refresh.

CHAPTER I - INTRODUCTION AND THEORETICAL FRAMEWORK

CHAPTER I

Introduction and Theoretical Framework

1.1 Research Question

Artificial intelligence is reshaping the foundations of global economic competitiveness. Since the launch of ChatGPT in November 2022 and the spectacular acceleration of investments in foundation models, generative AI has established itself as a cross-cutting transformative factor of the economy, simultaneously affecting finance, industry, healthcare, transportation and services. Yet this transformation rests on a precise material substrate: considerable computing capacity, powered by cutting-edge semiconductors and abundant electrical energy. Mastery of this triptych - compute, chips, energy - has become a geostrategic issue of the first order.

In this context, the United States has progressively erected a control regime over access to frontier AI technologies. As early as October 2022, the Bureau of Industry and Security (BIS) of the Department of Commerce imposed restrictions on the export of advanced GPUs to China. In January 2025, the Biden administration extended these controls to more than 120 countries via the AI Diffusion Rule, creating a tier-based segmentation system that conditions access to the most powerful AI chips on the degree of geopolitical alignment with Washington. The Trump administration, taking office in January 2025, replaced this framework with a more explicitly competitive approach, culminating in July 2025 with the publication of America's AI Action Plan, then in January 2026 with the imposition of 25 percent tariffs on certain advanced AI semiconductors (Nvidia H200, AMD MI325X) under Section 232.[1]

These measures, officially motivated by national security imperatives, de facto produce a structural competitive advantage for American companies: they benefit from unlimited access to frontier compute, while actors from other regions - including European allies - see their capacities capped, made more expensive, or conditioned. We are thus witnessing the emergence of a new type of technology protectionism, where the tax is not merely tariff-based but also regulatory, logistical and strategic.

This study poses the following question: to what extent does US technology protectionism on AI - export controls, tariffs, domestic compute prioritization - create a structural competitiveness divergence between the United States and Europe, and what are the measurable consequences for France by 2030?

This research question breaks down into three sub-questions articulating the empirical, prospective and normative dimensions of the analysis:

(a) What is the current AI computing capacity gap (compute gap) between the United States and the European Union, and how does this gap evolve under different scenarios of American trade and technology policy?

(b) How does the asymmetry of compute access translate into sectoral productivity, model training costs, and market share in AI services?

(c) Do European energy constraints and the rise of AI robotics amplify the divergence, and can France leverage its nuclear advantage to mitigate this structural disadvantage?

The originality of this study lies in its integrated approach. The existing literature treats separately export controls (Carnegie Endowment, CSIS, Hudson Institute), data center energy projections (IEA), semiconductor market dynamics (McKinsey, SIA, Deloitte), compute sovereignty (Hawkins, Lehdonvirta and Wu, 2025), and AI-related competitive barriers (Bruegel, OECD). No work proposes the complete causal trajectory we seek to establish here: US protectionism then restriction of European compute then productivity divergence then strategic dependence, with energy and robotics as amplifying factors.

1.2 Operational Definitions

The analysis deployed in this study rests on a set of concepts that require rigorous definition, as their usage varies across disciplinary contexts.

Frontier compute. We define frontier compute as the installed computing capacity in the form of AI accelerators - GPUs (Nvidia A100, H100, H200, B200), ASICs (Google TPU), or specialized circuits - deployed in industrial-scale data centers. This capacity is measured in FLOPs (floating-point operations per second) aggregated at the national or regional level, in H100-equivalents (the metric used by the Epoch AI dataset), or in GW of IT load. Frontier compute is the fundamental input for the development and deployment of frontier AI: without it, it is impossible to train competitive foundation models or operate large-scale inference services.

AI technology protectionism. AI technology protectionism designates the set of state measures - export controls, tariffs, quotas, export licenses, logistical prioritization, restrictions on model weights and APIs - that create an asymmetry of access to compute, models, and AI services between geographic regions. This concept extends the classical notion of trade protectionism by integrating the intangible dimension (software, models, cloud services) and the infrastructural dimension (energy, chip production capacity). Unlike traditional tariff barriers, AI technology protectionism can operate through non-tariff channels - for example, prioritizing GPU deliveries to domestic companies in a context of global shortage.

Compute gap. The compute gap measures the ratio of effectively available AI computing capacity between two regions. We define it as the ratio between installed and accessible AI FLOPs (or H100-equivalents) for economic actors in region A and those in region B, normalized by active population or GDP. A compute gap of x15 between the United States and the EU means that, per unit of GDP, American actors have fifteen times more AI computing power than their European counterparts. This ratio constitutes a synthetic indicator of structural advantage in AI.

Compute sovereignty. We adopt the definition proposed by Hawkins, Lehdonvirta and Wu (2025), who decompose compute sovereignty into three levels: (1) the quantity of AI compute physically present on national territory, (2) the nationality of the companies owning the data centers, and (3) the nationality of the

accelerator suppliers whose chips power these data centers.[2] A country can have significant computing capacity on its soil while being dependent on foreign operators and chip suppliers, which limits its real sovereignty. This concept is essential for understanding why the presence of AWS or Azure data centers in Europe does not, in itself, constitute a form of European sovereignty over compute.

Geopolitical vendor lock-in. We propose the concept of geopolitical vendor lock-in to designate the structural dependence of an economic ecosystem on foreign technology providers whose access can be restricted, made more expensive, or conditioned by political decision of a third-party government. This concept extends the classical notion of vendor lock-in in IT (cloud provider migration costs, for example) by adding a geopolitical dimension: the risk that access to critical infrastructure may be used as a negotiating lever between states. The Starlink-Ukraine episode of March 2025, where American control of a communication system was perceived as an instrument of pressure, strikingly illustrates this type of risk.[3]

Compute-Adjusted Competitiveness Index (CACI). We construct, as the central original indicator of this study, a Compute-Adjusted Competitiveness Index combining the four dimensions identified above into a weighted geometric composite. In its absolute formulation, called Power Mode, the indicator is defined as $CACI = F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$, where F is the installed AI compute (H100-equivalents, total or sovereign), L is the relevant labor force (millions of AI-skilled workers), R is a geopolitical access index (0 to 1), and E is the average industrial electricity price (USD per MWh). The four exponents (40, 20, 15, 25) sum to one and reflect the relative importance attributed to compute, labor, geopolitical access and energy cost. A second formulation, called Intensity Mode, divides the result by GDP to obtain a competitiveness density per unit of national wealth. The full methodological derivation is presented in Chapter II and the indicator is updated in real time on the public dashboard.[a]

Two variants of the indicator are used in parallel. Physical CACI integrates all compute physically present on the territory (F_{total}), regardless of ownership. Sovereign CACI restricts F to compute owned and operated by domestic actors (F_{dom}), thereby capturing the actual sovereign autonomy. The gap between the two variants measures the structural dependence of a region on foreign hyperscalers. For the European Union, this gap is the most direct quantitative measure of the strategic vulnerability that this study seeks to characterize.

1.3 Theoretical Framework

Our analysis is grounded in four complementary theoretical currents, which provide the conceptual tools necessary to articulate the technological, economic and geopolitical dimensions of the phenomenon under study.

1.3.1 AI as a General Purpose Technology

The first theoretical anchor is the General Purpose Technologies (GPT) theory formalized by Bresnahan and Trajtenberg (1995). A GPT is a technology characterized by three properties: its pervasiveness (it is used as an input in numerous downstream sectors), its inherent potential for technical improvement, and its innovation complementarities (R&D productivity in user sectors increases as a consequence of GPT improvement).[4] The model predicts that GPTs generate increasing returns to scale and that their diffusion

throughout the economy is a source of generalized productivity gains. However, Bresnahan and Trajtenberg also emphasize that a decentralized economy may have difficulty fully exploiting a GPT potential, as market transactions between the GPT producer and its users may lead to too little, too late innovation.

Brynjolfsson, Rock and Syverson (2019) applied this framework to contemporary AI, demonstrating that AI, and in particular machine learning, satisfies Bresnahan and Trajtenberg three criteria to be qualified as a GPT.[5] Their productivity J-curve model explains why productivity gains linked to a GPT may initially be invisible in statistics: firms must first invest massively in intangible assets (reorganization, training, process reengineering) before reaping the benefits.

This framework is fundamental to our analysis because it implies that early and massive access to AI compute - that is, the capacity to invest in the GPT at large scale from the earliest deployment phases - generates cumulative advantages that are difficult to reverse. Innovation complementarities create a path dependence dynamic: actors who access abundant compute early develop superior models, capture usage data, generate revenues they reinvest in compute, and thus widen a gap that self-reinforces over time. Any policy restricting compute access for a given region therefore has non-linear consequences: it does not merely delay adoption, it structurally compromises it.

1.3.2 Weaponized Interdependence and Control of Global Networks

The second theoretical anchor is the weaponized interdependence theory developed by Farrell and Newman (2019). These authors demonstrate that global economic networks, far from creating symmetrical relations of mutual interdependence as classical liberal theory posited, tend to produce asymmetric structures in which certain nodes (hubs) become far more connected than others. States exercising political jurisdiction over these central nodes can instrumentalize them for coercive purposes, via two mechanisms: the panopticon effect (strategic information collection) and the chokepoint effect (capacity to cut or restrict flows).[6]

The application to the AI value chain is remarkably pertinent. The United States controls several critical chokepoints: AI accelerator design (Nvidia holds over 80 percent of the GPU market for data centers), cloud infrastructure (AWS, Azure and Google Cloud represent approximately 70 percent of the global market), and the most advanced foundation models (OpenAI, Anthropic, Google DeepMind). Control of these chokepoints allows the US government to modulate global access to AI compute as a geopolitical lever, in exactly the same way that the SWIFT system was used as a lever in the financial domain. Farrell and Newman themselves recognized, in an update published in Foreign Affairs in December 2025, that semiconductors and AI had become a major application terrain for their theory, the Trump administration having explicitly used export controls on AI chips as bargaining currency in negotiations with China.[7]

1.3.3 Concentration Economics and Innovation Rents

The third theoretical anchor mobilizes the literature on digital market concentration and barriers to entry in the AI ecosystem. Martens (2024), in a policy brief for Bruegel, demonstrates that the training costs of foundation models grow exponentially, constituting an insurmountable barrier to entry for most actors.[8] He estimates that a compute farm on the order of a trillion dollars is conceivable in the medium term, an

investment threshold completely beyond the reach of public funding and the vast majority of companies. Only the GAMMAN (Google, Apple, Meta, Microsoft, Amazon, Nvidia) have the resources to access it.

The OECD (2025) confirms this analysis in its report on competition in AI infrastructure, identifying barriers to entry at multiple levels of the supply chain: extremely high capital requirements, massive economies of scale, switching costs between providers, and absence of interoperability standards.[9] The Federal Reserve Board (October 2025), in a comparative analysis of AI competitiveness in advanced economies, shows that the United States concentrates more than 75 percent of global venture capital investment in generative AI, and that Europe lags significantly not only in investment but also in enterprise adoption, with high energy costs constituting an additional brake.[10]

This concentration dynamic has a direct consequence for our analysis: US technology protectionism does not merely limit European access to compute, it reinforces the position of dominant actors who are precisely those benefiting from domestic exemption. It is a dual advantage mechanism: reduction of constraints for US companies, increase of constraints for their competitors.

1.3.4 Digital Sovereignty and European Strategic Autonomy

The fourth theoretical anchor mobilizes the European current of digital sovereignty, analyzed notably by Mugge (2024) in the Journal of European Public Policy. Mugge identifies three fundamental tensions in Europe AI sovereignty ambition: does sovereignty oppose the EU to other AI powers, or citizens to large platforms? Does it aim at economic competitiveness or rights protection? And who really benefits from it - European champions or the entire ecosystem?[11] These tensions are precisely those that American protectionism exacerbates: by compressing European technological sovereignty space, it forces the EU to arbitrate between these contradictory objectives in a context of urgency.

Hawkins, Lehdonvirta and Wu (2025) provide a decisive empirical contribution by measuring compute sovereignty through the infrastructure of the nine main global cloud providers, which represent approximately 70 percent of the global market. Their results reveal that the degree of sovereignty varies considerably depending on the level of analysis (territorial, corporate, or hardware), and that most European countries exhibit a sovereignty deficit at at least two of these three levels.[12] McKinsey (December 2025) estimates the European sovereign AI opportunity at 480 billion euros annually by 2030, contingent on a scenario of strong technological sovereignty and high AI adoption.[13]

1.4 Targeted Literature Review and Gap Identification

The state of the literature in February 2026 reveals a rapidly forming field of research, but one that remains highly fragmented. We identify five main corpora, each covering one dimension of our research question.

Corpus 1: Export controls and AI geopolitics. The Carnegie Endowment for International Peace provides the most detailed analysis of US AI export control policy. Winter-Levy and Phillips-Robins (May 2025) described the Biden AI Diffusion Rule as a compromise between three objectives - control, promotion and geopolitical leverage - and analyzed replacement options under Trump.[14] CSIS, the Hudson Institute and Pillsbury Law have documented the legal mechanisms and recent developments, notably the Section 232 tariffs of January

2026. Contrary Research (November 2025) offers a particularly rich analysis of the underlying temporal dilemma: if AGI is five years away, export controls strengthen American dominance; if it is ten years or more, they accelerate Chinese technological autonomization.[15]

Corpus 2: Energy and data center infrastructure. The IEA special report Energy and AI (April 2025) constitutes the global reference for data center energy projections. It establishes that global data center electricity consumption will reach 945 TWh in 2030 in the base scenario, compared to 415 TWh in 2024, with AI as the primary driver of this growth.[16] In the United States, data centers will consume more electricity than all energy-intensive industry combined by 2030. In Europe, growth will be plus 45 TWh (plus 70 percent), with a risk of delay on approximately 20 percent of projects linked to electricity grid constraints. IEA-4E published in parallel a critical review of estimation models, highlighting the magnitude of uncertainties (2030 projections vary by a factor of 40 across studies).[17]

Corpus 3: Semiconductor market. McKinsey (January 2026) significantly revised upward its estimates of the semiconductor market size by integrating captive designers (Apple, Amazon, Tesla) and fabless operators whose value does not appear in traditional statistics. Their base estimate goes from 775 billion USD in 2024 to 1,600 billion USD in 2030, a CAGR of 13 percent.[18] SIA reports record sales of 627.6 billion USD in 2024, while Deloitte (February 2026) anticipates that generative AI chips alone will represent nearly half of the sector revenue in 2026.[19]

Corpus 4: European AI sovereignty and competitiveness. Several recent publications document the European deficit. The European Parliament (2025) finds that only 11 percent of small European enterprises use AI, compared to 58 percent of small American businesses.[20] Accenture (November 2025) reports that 62 percent of European organizations are seeking sovereign solutions in the face of geopolitical uncertainty, but only 19 percent see it as a competitive advantage. The majority (48 percent) is motivated by regulatory compliance obligations, suggesting a defensive rather than strategic approach.[21] The Draghi report (September 2024) on European competitiveness had already identified the digital investment deficit as a structural factor in Europe decline.

Corpus 5: Competitive barriers and concentration. Bruegel, the OECD and the Federal Reserve Board converge on the observation of growing concentration of the AI ecosystem around a small number of actors. The CERRE report (June 2025) on competition policy for cloud and AI identifies migration barriers, cloud credit practices likely to create lock-in, and growing dependence of small AI developers on hyperscalers for access to accelerated compute.[22] The structural collaboration between AI startups and Big Tech, illustrated by the Mistral-Microsoft agreement, testifies to this dependence.

The gap this study aims to fill. Examination of these five corpora reveals an uncovered analytical space: no study proposes an integrated trajectory linking US technology protectionism to EU competitiveness divergence via the compute-energy-semiconductor triptych. Existing works treat each dimension in isolation - export controls without energy, energy without semiconductors, semiconductors without productivity. Moreover, the specific angle of AI robotics as an amplifier of energy demand is virtually absent from the literature. It is precisely this gap that our study intends to fill, by proposing a unified analytical framework and an original indicator - the Compute-Adjusted Competitiveness Index (CACI) - enabling measurement and

projection of this divergence. The indicator and the underlying dataset are publicly accessible and updated in real time on the project dashboard.[a]

1.5 Report Structure

The report is organized in eight chapters. After this introduction, Chapter II presents the methodology, detailing the multi-scenario approach, data sources, CACI construction and methodological limitations. Chapter III establishes the 2020-2026 empirical diagnosis, covering energy trajectories, the semiconductor market, installed compute and the export controls chronology. Chapter IV analyzes the mechanisms of US competitive advantage. Chapter V presents four 2026-2030 scenarios structured around two axes of uncertainty. Chapter VI details the consequences for France and Europe, differentiated by actor type and sector. Chapter VII formulates strategic recommendations at three time horizons. Chapter VIII concludes by synthesizing the study contributions and identifying avenues for future research.

Notes

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23. [a] Public dashboard: <https://moOgly.github.io/America-First-IA/dashboard/>. Working Paper: https://moOgly.github.io/America-First-IA/pdf/Working_Paper_CACI_AI_Competitiveness.pdf. The CACI Power Mode formula $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$ reproduces the live calculation; the headline ratio of 3.46:1 between USA and EU (aggregated) is computed on the live dataset as of April 2026.

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CHAPTER II - METHODOLOGY

CHAPTER II

Methodology

2.1 General approach: mixed-method multi-scenario prospective analysis

This study combines a retrospective empirical analysis (2020-2026 diagnosis) with a prospective scenario-based projection (2026-2030). This two-pronged architecture responds to the nature of the phenomenon under investigation: AI technology protectionism is simultaneously an observable fact (export controls, Section 232 tariffs) and an ongoing process whose future trajectory depends on discretionary political variables, European strategic responses and partially unpredictable technological developments.

The retrospective component employs a descriptive quantitative method, based on the aggregation and cross-referencing of data from institutional sources (IEA, SIA/WSTS, Eurostat, EIA), industry reports (McKinsey, Deloitte, Epoch AI) and regulatory documents (Federal Register, BIS, White House). The objective is to establish a rigorous, sourced factual foundation covering three dimensions: data center energy consumption, the semiconductor market, and installed AI compute capacity by region.

The prospective component draws on the scenario planning tradition as formalized by Schwartz (1991) and practiced at Royal Dutch/Shell since the 1970s.[1] This method, belonging to the Intuitive Logics school (Bradfield et al., 2005), consists of constructing plausible and internally consistent scenarios - not to predict the future, but to explore the space of possibilities and evaluate the robustness of different strategies against divergent environmental developments.[2] It is particularly suited to situations characterized by high political and technological uncertainty, where classical econometric models reach their limits, which is precisely the case with AI technology protectionism.

Justification of the methodological choice. Three reasons underpin the choice of scenario methods over pure econometric modelling. First, the key analytical variables are largely political and discretionary: the decision of a US president to impose or not impose GPU quotas on Europe cannot be modelled through a regression function. Second, the interactions between the energy, technological and geopolitical dimensions are non-linear and systemic: a restriction on GPUs can, through cascade effects, alter energy investment flows, data center location decisions and the competitive structure of entire sectors. Third, the available data on installed compute by region are partial and heterogeneous: no unified public database of AI FLOPs by country exists, making rigorous econometric calibration premature.

The chosen method therefore combines the quantitative rigour of the empirical diagnostic (sourced data, time series, measurable ratios) with the qualitative flexibility of scenario construction, in the spirit of what Schoemaker (1995) calls a disciplined heuristic.[3] The scenarios are not probabilistic forecasts but coherent strategic narratives, each based on explicit assumptions and developing their consequences through measurable metrics.

2.2 Data sources: classification and critical evaluation

The study draws on three categories of sources, whose reliability and potential biases must be explicitly acknowledged. This methodological transparency conforms to the recommendations of the OECD/JRC Handbook on Constructing Composite Indicators (Nardo et al., 2008), which prescribes systematic documentation of sources, their limitations and their biases in any composite indicator construction.[4]

2.2.1 Primary sources (official and regulatory documents)

This category includes normative or institutional texts: presidential proclamations (Section 232), BIS rules (AI Diffusion Rule, Entity List), IEA reports, European Parliament publications (EPRS), official statistical data (SIA/WSTS for semiconductors, Eurostat and EIA for energy, RTE for France). These sources offer the highest factual reliability but may contain institutional framing biases: the IEA tends to favour moderate scenarios, the European Parliament to emphasize EU sovereignty risks.

2.2.2 Academic sources and think tanks

This category includes peer-reviewed articles (Farrell and Newman, 2019; Bresnahan and Trajtenberg, 1995; Brynjolfsson et al., 2019; Mugge, 2024) and publications from recognized think tanks (Bruegel, Carnegie Endowment, CSIS, OECD, Federal Reserve Board). The former provide robust theoretical grounding; the latter offer empirically founded policy analyses but are potentially influenced by each institution ideological orientation. We prioritize cross-referencing sources of different orientations (Bruegel / Carnegie / Fed) to limit this bias.

2.2.3 Industry and consulting sources

McKinsey, Deloitte, Accenture, Epoch AI and CFG Europe provide market data, sectoral projections and capacity estimates unavailable in public sources. These sources present a potential systematic bias: consulting firms have an interest in amplifying trends (to justify transformation engagements) and market estimates are often optimistic. We mitigate this bias by triangulating figures with institutional data and explicitly flagging discrepancies between sources. For example, 2024 semiconductor sales are 627.6 billion USD according to the SIA (traditional scope) but 775 billion USD according to McKinsey (expanded scope), a 24 percent gap reflecting methodological differences, not inconsistencies.[5]

2.2.4 AI compute data: the Epoch AI GPU Clusters dataset

Measuring installed AI compute by country constitutes the central methodological challenge of this study. We rely primarily on the Epoch AI GPU Clusters dataset (Pilz, Rahman, Sanders and Heim, 2025), which catalogues over 500 supercomputers and GPU clusters worldwide for the period 2019-2025.[6] This dataset, available under an open Creative Commons Attribution licence, constitutes the most comprehensive and systematically documented source on global AI compute infrastructure to date. It is used as a reference by the Stanford AI Index Report (2025), by several government reports, and by institutions such as OpenAI and DeepMind.

The dataset covers for each cluster: country of location, chip type (H100, A100, GB200, TPU, etc.), computational performance in 16-bit FLOP/s, number of H100 equivalents, operational date, sector (private/public), electrical power (MW) and estimated hardware cost. This granularity enables aggregation by country and year, directly meeting the needs of our variable $F(r)$ in the CACI.

Limitations of the Epoch AI dataset. Three limitations must be highlighted. First, coverage is estimated at 10-20 percent of global aggregate AI compute performance (March 2025), with significant heterogeneity across companies and chip types: approximately 20-37 percent of NVIDIA H100s, 12 percent of A100s, but less than 4 percent of Google TPUs and a negligible fraction of custom chips from AWS, Microsoft or Meta.[7] Second, Chinese systems are anonymized (names removed, values rounded to one significant figure), limiting analytical precision for China. Third, the physical location of a cluster does not determine access: many clusters are accessible via cloud services from other countries.

We complement these data with the OECD Working Paper by Lehdonvirta, Wu, Hawkins et al. (October 2025), which develops a methodology for estimating public cloud AI compute availability by country, counting cloud regions from major providers equipped with AI accelerators (A100, H100, GB200) across 39 economies.[8] This complementary approach distinguishes installed compute (Epoch AI) from accessible compute (OECD), a distinction crucial for the CACI.

2.3 Scenario construction

Scenario construction follows a four-step protocol, inspired by the 2x2 matrix methodology (Schwartz, 1991; van der Heijden, 2004) and adapted to the geostrategic context of AI.[9]

Step 1 - Identification of driving forces. Predetermined elements (whose evolution is reasonably predictable regardless of scenario) include: continued growth in global AI compute demand; structural increase in data center energy consumption; European dependence on Asian and American foundries for leading-edge chips; concentration of the global cloud around three US hyperscalers; and exponential increase in frontier model training costs.

Critical uncertainties (whose evolution depends on political choices, strategic reactions or technological disruptions) are grouped along two dimensions. Dimension 1 - Intensity of US technology protectionism: this dimension covers a spectrum from maintaining current restrictions to aggressive hardening (GPU quotas for the EU, restrictions on APIs and models, explicit prioritization of deliveries to US companies). Dimension 2 - European response capacity: this dimension covers a spectrum from passive posture (marginal adaptation, acceptance of dependence) to active response (Compute Zones with derogated energy, accelerated AI Factories, SMR nuclear for data centers, alternative partnerships Japan-Korea-Taiwan, AI Act revision).

Step 2 - 2x2 matrix and scenario generation. Crossing the two uncertainty dimensions generates a four-scenario matrix. The choice of four rather than three scenarios is deliberate. Schwartz (1991) and Shell method practitioners recommend never building three scenarios, as the human mind tends to treat the middle scenario as the most likely, thereby reducing the exercise utility.[10] The 2x2 matrix forces the analyst to explore extreme quadrants, precisely where strategic ruptures play out.

Step 3 - Narrative development and quantification. Each scenario is developed following a standardized protocol comprising three components: a strategic narrative describing the plausible sequence of events between 2026 and 2030; a quantification of key metrics calibrated on 2020-2026 empirical data and projected according to the scenario assumptions; and early warning indicators (leading indicators) enabling identification, from 2026-2027 onward, of which scenario reality is converging toward.

Step 4 - Sensitivity analysis and robustness. For each recommendation formulated in Chapter VII, we evaluate its robustness across all four scenarios. A recommendation is considered robust if it produces positive or neutral results in at least three of the four scenarios.

2.4 Key metrics and original indicator: the CACI

We define six metrics to be calculated or estimated in the empirical diagnostic (Chapter III), then projected in each scenario (Chapter V). Together, they form a dashboard of US/EU divergence in AI: M1 (compute gap, US/EU installed AI FLOPs ratio normalized by GDP), M2 (relative FLOP cost for training), M3 (cloud dependence, share of EU AI workloads on US infrastructure), M4 (sectoral AI productivity), M5 (energy constraint, data center demand/capacity ratio) and M6 (AI relocations).

2.4.1 Theoretical foundations of the CACI

Grounding in the literature. The construction of a compute-centred AI competitiveness composite indicator responds to a need identified by several converging research streams. Since 2023-2024, academic and institutional literature increasingly emphasizes that computational capacity has become the most discriminating factor of production for frontier AI (Sevilla et al., 2022; Epoch AI, 2025; Pilz et al., 2025). US export controls (BIS, October 2022; updated 2023 and 2025) explicitly place advanced compute at the heart of geopolitical competition, while Hawkins, Lehdonvirta and Wu (2025) introduce the concept of compute sovereignty as a structuring dimension of strategic autonomy.[11]

Yet existing AI competitiveness indices do not place compute at the centre of their construction. The IMF AI Preparedness Index (Cazzaniga et al., 2024), covering 174 countries, aggregates four dimensions (digital infrastructure, human capital, innovation/economic integration, regulation/ethics) without directly measuring installed compute capacity.[12] The Tortoise Media Global AI Index (2024), ranking 83 countries on 122 indicators across three pillars (Implementation, Innovation, Investment), includes an infrastructure/supercomputing component but subsumes it within a weighted additive index where compute is merely one factor among many.[13] The Stanford AI Index (2025) provides rich compute trend data but proposes no composite index. None of these instruments formalizes the mechanism by which compute, adjusted for its energy cost and related to an economy absorptive capacity, determines AI competitiveness.

The CACI aims to fill this gap by proposing a parsimonious yet theoretically grounded indicator that captures the multiplicative interaction between four factors: accessible compute, the relevant labor force, geopolitical access conditions and the energy cost.

2.4.2 Formal definition

The CACI for a region r at period t is defined in two complementary modes. In Power Mode (absolute): $CACI(r,t) = F(r,t)^{0.40} \times L(r,t)^{0.20} \times R(r,t)^{0.15} / E(r,t)^{0.25}$. In Intensity Mode (per unit of national wealth): $CACI(r,t) = F(r,t)^{0.40} \times L(r,t)^{0.20} \times R(r,t)^{0.15} / (E(r,t)^{0.25} \times GDP(r,t))$.

$F(r,t)$ - installed and accessible AI compute capacity in region r at period t . We aggregate the Epoch AI GPU Clusters dataset by country, in H100-equivalents (the metric used by Epoch AI), and complement it with OECD estimates of accessible cloud compute. F admits two specifications used in parallel: F_total (all clusters located in the territory regardless of ownership, defining the Physical CACI) and F_dom (clusters owned and operated by domestic actors, defining the Sovereign CACI).

$L(r,t)$ - working population with AI competencies (proxy: STEM graduates plus certified AI training plus LinkedIn Economic Graph AI skill density). This factor captures absorptive capacity in the sense of Cohen and Levinthal (1990): abundant compute without human capital to exploit it does not produce competitiveness.[14]

$R(r,t)$ - geopolitical access index (between 0 and 1) capturing the country position in the export-control regime. $R = 1$ for the United States (unlimited access to frontier compute), $R = 0.95$ for Tier-1 close allies (UK), $R = 0.9$ for the EU and France (Tier-1 with selective restrictions), $R = 0.85$ for India (Tier-2), $R = 0.7$ for China (heavily restricted).

$E(r,t)$ - average effective energy cost of compute in region r , in USD per MWh, adjusted for hyperscaler Power Purchase Agreements. The dashboard reference values for the April 2026 snapshot are USA 85, China 92, France 115, Germany 140, UK 190 USD per MWh. These figures correspond to PPA-adjusted prices effectively faced by data centers, not to raw Eurostat industrial tariffs (which are typically higher in Europe). EU effective costs in this PPA-adjusted scope are 1.4 to 1.7 times US levels.[15]

$GDP(r,t)$ - gross domestic product of region r (World Bank, Eurostat). Used only in Intensity Mode to normalise across economies of very different sizes.

Justification of the multiplicative form. The choice of multiplicative (geometric) rather than additive (arithmetic) aggregation is deliberate and rests on three arguments. First, General Purpose Technology theory (Bresnahan and Trajtenberg, 1995) posits strong complementarity between innovation inputs: compute without affordable energy or qualified human capital produces no competitiveness gains, justifying a form where weakness in one factor penalizes the whole. Second, the OECD/JRC Handbook (2008, p. 33) recommends geometric aggregation when components are not perfectly substitutable: unlike the arithmetic mean, the geometric mean does not allow a very high score on one dimension to fully compensate a very low score on another. Third, recent work on AI index construction (Koronakos, Kritikos and Sotiros, 2024; analysis of the Tortoise GAI via Choquet integral) confirms that AI competitiveness dimensions exhibit interactions (complementarities and redundancies) that make simple linear aggregation problematic.[16]

Economic interpretation. In logarithmic form, the CACI in Intensity Mode can be written $\ln(CACI) = 0.40 \ln(F) + 0.20 \ln(L) + 0.15 \ln(R) - 0.25 \ln(E) - \ln(GDP)$. This transformation linearizes the relationship and allows each coefficient to be interpreted as an elasticity. The indicator is designed for bilateral comparison: the ratio $CACI(US)/CACI(EU)$ measures relative competitive advantage. The April 2026 snapshot yields a ratio of 3.46:1 in Power Mode (F_total) - this is the headline figure of this study.

2.4.3 Calibration protocol and data sources

CACI calibration follows a four-step protocol consistent with OECD/JRC Handbook (2008) recommendations: identification and collection of raw data; treatment of missing values and normalization; aggregation; and sensitivity analysis.

$F(r,t)$ - Installed AI compute capacity. Estimation follows a bottom-up approach. The main layer comes from the Epoch AI GPU Clusters dataset (April 2026 version), providing 16-bit FLOP/s and H100-equivalent performance aggregated by country for 2019-2026. National shares of operational compute on the April 2026 snapshot are: USA 76.9 percent, China 12.8 percent, EU (13 main economies) 4.4 percent, Norway 1.1 percent, Japan 1.0 percent.[17] When all reported clusters are included (operational plus planned), the USA share moves to about 50 percent, with the Gulf (UAE, Saudi Arabia) and Korea capturing a large share of the projected 2027-2030 buildout. The complementary layer comes from the OECD (Lehdonvirta et al., 2025), cataloguing cloud regions with AI accelerators across 39 economies, capturing the accessibility dimension that physically installed compute does not fully reflect.

F aggregation procedure. For each country-year, we extract from the Epoch AI dataset the sum of H100-equivalents from all clusters located in the country, filtering for confirmed systems (certainty greater than or equal to Likely); apply an extrapolation factor to correct for the dataset under-coverage (10 to 20 percent of global compute), using Epoch AI sectoral coverage estimates by chip type; and add estimated accessible cloud capacities via the OECD methodology for countries where foreign cloud compute represents a significant share of effective capacity. Raw data and calculation code (Python/pandas) are documented in the methodological appendix and on the public dashboard.[18]

$E(r,t)$ - Energy cost. The dashboard CSV reference values used throughout this study, in USD per MWh and PPA-adjusted, are: USA 85, China 92, France 115, Germany 140, UK 190, India 88. These figures incorporate negotiated large-consumer tariffs (hyperscaler PPAs), using IEA (2025, Energy and AI) estimates of data center energy mix by region. They are systematically lower than raw Eurostat industrial tariffs in Europe, because hyperscaler PPAs and direct nuclear/renewable sourcing materially reduce the effective cost faced by AI data centers. The Federal Reserve Board (October 2025) documents a significant negative correlation between energy costs and AI adoption at the European firm level.[19]

$GDP(r,t)$ and $L(r,t)$. GDP is available from the World Bank (World Development Indicators) and Eurostat. The dashboard April 2026 snapshot uses GDP values in trillion USD: USA 29.3, China 18.7, EU 18.9, France 3.16, Germany 4.68, UK 3.6. The AI human capital proxy $L(r,t)$ combines three sub-indicators: number of STEM graduates (OECD Education at a Glance), AI skills density as measured by LinkedIn Economic Graph (profiles with AI skills relative to working population) and AI certifications (estimates based on certified AWS/Google/Microsoft cloud programs by country). The dashboard snapshot uses, in millions of AI-skilled workers: USA 3.5, China 4.8, EU 3.1, France 1.5, Germany 1.9, UK 1.8. We acknowledge a bias favouring English-speaking countries and economies where LinkedIn is dominant, and document this bias effect in the sensitivity analysis (section 2.4.5).[20]

CACI normalization and benchmark selection. The raw CACI produces heterogeneous absolute values whose direct interpretation is challenging. In accordance with standard practice (Nardo et al., 2008; Saisana and

Tarantola, 2002), we apply a min-max normalization to the leader: each country normalised CACI is expressed as a percentage of the United States raw CACI, so that $CACI_norm(USA) = 100$ by construction.

This normalization choice is methodologically grounded on three arguments. First, the United States constitutes a natural benchmark given its dominant position across all four CACI factors: 76.9 percent of global operational AI compute (Epoch AI, April 2026), the lowest effective industrial energy costs among major OECD economies (USD 85 per MWh, PPA-adjusted, versus USD 115 in France and USD 140 in Germany - dashboard CSV, April 2026), the highest nominal GDP, and an unmatched AI human capital ecosystem. Second, normalization to the leader is the established convention for relative competitiveness indices: Balassa RCA (1965), the WEF GCI (Schwab, 2019) and UNIDO Competitive Industrial Performance Index all proceed by ratio to the sectoral or national leader. Third, this normalization eliminates interpretive ambiguities of absolute values and enables temporal comparisons: a France CACI moving from 25 to 35 between 2024 and 2028 signifies partial catch-up, even if both countries raw values have increased.

The most important methodological consequence of this normalization is the revelation of a structural gap that traditional indices conceal. The IMF AI Preparedness Index (Cazzaniga et al., 2024) assigns scores of 0.85 (USA) and 0.74 (France), a ratio of 1.15:1, suggesting near-parity. The Tortoise Global AI Index produces similar gaps. Yet the normalized CACI in Power Mode places France at about 25 and Germany at about 5 on a scale where the United States = 100, i.e. ratios of 4:1 and 19:1 respectively. The headline US/EU ratio (EU aggregated as 13 main economies) is 3.46:1. This divergence is not contradictory: it results from the fact that multi-dimensional indices linearly aggregate normative dimensions (regulation, ethics, publications) that arithmetically compensate for deficits on material factors (compute, energy). The CACI, by isolating these material factors, exposes the physical chasm that composite metrics average away.

This property of the CACI - revealing the true magnitude of the compute gap - constitutes the central analytical contribution of the index and the fundamental empirical thesis of this study. The CACI gap is not a normalization artifact: it is the scientific result.

2.4.4 Positioning relative to existing AI competitiveness indices

The IMF AI Preparedness Index (AIPI) covers 174 countries (2023) and aggregates four pillars: digital infrastructure, human capital and labour market policies, innovation and economic integration, regulation and ethics. The AIPI does not measure installed AI compute and does not weight by energy cost. Its expected correlation with the CACI is positive but imperfect: countries scoring highly on the AIPI (Singapore, Denmark, Netherlands) are not necessarily those with the most effective compute per unit of GDP.[21]

The Tortoise Media Global AI Index (GAI) ranks 83 countries on 122 indicators across three pillars (Implementation, Innovation, Investment). It includes an infrastructure/supercomputing component but aggregates it linearly with subjective weights (acknowledged by Tortoise as a limitation). The GAI is broader and more multidimensional than the CACI, but precisely because it is broad, it dilutes the compute signal in a many-indicator composite. As Koronakos et al. (2024) show, the GAI weighting subjectivity can invert country rankings depending on chosen weighting scenarios.

The Stanford AI Index (2025) constitutes the most comprehensive reference in terms of raw data (notable models by country, investment, publications, patents, compute trends). It does not propose a composite index but provides the time series used by many other indices. The Stanford AI Index public data are available via an open Google Drive folder.[22]

CACI specific value added. The CACI distinguishes itself through four properties: it places compute at the centre rather than the periphery of the indicator, reflecting compute now-dominant role in frontier AI; it explicitly integrates energy cost as a bottleneck, consistent with IEA (2025) findings; it uses a theoretically grounded multiplicative aggregation rather than additive; and it is parsimonious (four variables), making it transparent and reproducible, at the cost of lesser comprehensiveness.

2.4.5 CACI limitations and sensitivity analysis

First limitation - opacity of $F(r,t)$. Installed compute measurement depends on incomplete private data. The Epoch AI dataset covers only 10-20 percent of global compute, with uneven coverage across sectors and companies. National attribution is itself debatable: a growing share of compute is held by a few private hyperscalers operating globally. Mitigation: we systematically document uncertainty margins, present ranges rather than point values, and verify result stability when F varies by plus or minus 30 percent. The Physical/Sovereign decomposition partially addresses the attribution problem.

Second limitation - qualitative heterogeneity of compute. The CACI aggregates FLOPs without distinguishing GPU generations (an H200 does not equal an A100 in energy efficiency and real performance). Mitigation: the Epoch AI dataset provides performance in H100 equivalents, offering partial normalization. We propose a GPU generation weighting factor in the appendix and show that adjustment marginally modifies rankings.

Third limitation - the human capital proxy $L(r,t)$. The STEM plus LinkedIn plus certifications combination presents a bias favouring English-speaking countries. This bias likely underestimates China and some Asian economies. Mitigation: we replicate the analysis using the IMF AIPI Human Capital sub-index as an alternative proxy and show ranking sensitivity to this choice.

Fourth limitation - indicator staticity. The CACI measures a state at a given moment, not a dynamic. This is why we calculate it across multiple years (2022, 2024, 2026) and project it in each scenario, enabling trajectory tracking and gap evolution assessment.

Fifth limitation - endogeneity. Countries gaining AI productivity invest massively in compute, creating a reverse causality risk: the CACI might capture the consequence rather than the cause of competitiveness. This study, within its dissertation framework, does not formally instrument this relationship (no instrumental variables or GMM strategy). However, we note that the exogenous shock of the October 2022 BIS rules offers a natural quasi-experiment that could ground a causal identification strategy in Difference-in-Differences. We identify the formal treatment of this endogeneity (DiD, IV or Arellano-Bond GMM) as a priority research avenue for any publishable extension of this work.[23]

Sixth limitation - normalization bias for small economies. Comparative analysis of the CACI against the IMF AIPI and Tortoise GAI indices reveals an instructive anomaly: certain countries with low GDP and a reduced AI workforce obtain disproportionately high CACI scores. South Africa is emblematic: ranked last or second -

to-last on the IMF AIPI and Tortoise indices, it can appear as a global leader on the Intensity Mode CACI when the GDP normalisation amplifies a small-denominator effect. This phenomenon is a classic case of normalization bias identified in the composite indicator literature. The OECD/JRC Handbook (Nardo et al., 2008, pp. 27-29) warns that normalization by GDP or population can produce misleading results for small economies. Our mitigation is threefold: a critical mass threshold on F (excluding economies below 10 000 H100-equivalents); a scaling factor $\alpha(r) = \min(1, F(r) / F_{\text{median}})$ for sensitivity tests; and a dual ranking (Power Mode for absolute power, Intensity Mode for density), with the explicit caveat that bilateral comparisons in this study focus on USA, EU, France and China.[24]

2.4.6 Reproducible numbers - April 2026 snapshot

To allow the reader to replay the calculation from the public dashboard CSVs, we expose here the inputs and outputs used throughout this study. The four input variables are sourced from the dashboard data folder (energy_prices.csv, gdp_data.csv, workforce_data.csv, gpu_clusters.csv) at <https://mo0ogly.github.io/America-First-IA/dashboard/data/>.

Inputs (April 2026 snapshot). USA: $F_{\text{total}} = 39.65$ million H100-eq, $L = 3.5$ million, $R = 1.00$, $E = 85$ USD/MWh, GDP = 29.3 trillion USD. EU (13 main economies aggregated): $F_{\text{total}} = 2.62$ million H100-eq, $L = 3.1$ million, $R = 0.90$, $E = 135$ USD/MWh, GDP = 18.9 trillion USD. France: $F_{\text{total}} = 2.44$ million H100-eq, $L = 1.5$ million, $R = 0.90$, $E = 115$ USD/MWh, GDP = 3.16 trillion USD. Germany: $F_{\text{total}} = 0.05$ million H100-eq, $L = 1.9$ million, $R = 0.90$, $E = 140$ USD/MWh, GDP = 4.68 trillion USD. China: $F_{\text{total}} = 0.40$ million H100-eq, $L = 4.8$ million, $R = 0.70$, $E = 92$ USD/MWh, GDP = 18.7 trillion USD.

Outputs (Power Mode F_{total} , normalised at USA = 100). USA = 100.0; EU(13) = 28.9; France = 25.3; India = 22.2; China = 15.7; UK = 7.0; Germany = 5.4. UAE = 55.7 in Physical CACI but only 6.0 in Sovereign CACI (F_{dom} restricted to clusters owned by domestic entities G42, MGX, Mubadala, Khazna, TII), illustrating the Physical/Sovereign decomposition: 99.6 percent of UAE F_{total} is owned by US-side actors (Stargate UAE, Microsoft, OpenAI). Headline ratios: US/EU(13) = 3.46:1; US/France = 3.96:1; US/Germany = 18.59:1; US/China = 7.46:1.

These outputs are reproduced live on the public dashboard. Any subsequent dashboard update (new clusters, energy revisions, GDP refresh) will mechanically refresh the figures; the methodology does not change. The Power Mode F_{total} ratio of 3.46:1 reported on the cover and throughout this study corresponds exactly to this snapshot.

2.5 Scope and delimitations

Geographic scope. The analysis focuses on the bilateral United States / European Union relationship, with a specific focus on France. China is treated as a contextual variable (primary target of US export controls, factor of pressure on chip production capacities) but is not subject to in-depth analysis. Japan, South Korea and Taiwan appear as semiconductor supply chain actors.

Temporal scope. The diagnostic covers 2020-2026, scenarios cover 2026-2030. The 2030 horizon is chosen as it corresponds to the convergence of several deadlines: IEA projections for data center energy, expected

maturity of the EU Chips Act, France 2030 SNIA objectives and potential arrival of the first operational SMR nuclear reactors.

Technological scope. The study covers frontier AI (foundation models, compute-intensive) and its material prerequisites (GPU/ASIC, data centers, energy). It integrates AI robotics as an amplifying factor for energy demand. It does not address edge embedded AI (smartphones, IoT), except insofar as it constitutes a specific objective of the French SNIA.

2.6 General methodological limitations

Radical political uncertainty. Technology protectionism depends on discretionary political decisions with structurally low predictability. A change of US administration in 2028, an unexpected US-EU trade agreement, or an escalation of the US-China conflict could invalidate certain assumptions. This is precisely why we propose four scenarios rather than a single trajectory.

Technological disruptions. The DeepSeek episode (January 2025), where a Chinese model achieved near-frontier performance with substantially reduced training budget, illustrates the possibility of efficiency breakthroughs that would alter the problem terms. The IEA (2025, Energy and AI) devotes a case study to DeepSeek and concludes that even with significant efficiency improvements, demand growth absorbs gains (Jevons rebound effect).[25]

Compute data opacity. The exact number of GPUs deployed per hyperscaler, the precise geographic distribution of data centers and GPU volumes exported by region are partially or fully confidential data. Our installed compute estimates carry significant margins of error, which we systematically document.

Consulting source bias. As noted in section 2.2, industry sources carry a systematic optimism bias. We mitigate this through triangulation but cannot eliminate it entirely.

These limitations do not compromise the analysis validity. The scenario method is precisely designed to function in high-uncertainty environments, where the objective is not prediction but structured exploration of possibilities. As Schwartz notes, scenarios are not forecasts; they are plausible stories that help you think.[26] Our contribution lies in the rigour of the framing, the explicitness of assumptions, the transparency of data sources and the originality of the CACI indicator, rather than in the precision of numerical projections.

Notes

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4. Nardo, M., Saisana, M., Saltelli, A. and Tarantola, S. (2008), *Handbook on Constructing Composite Indicators: Methodology and User Guide*, OECD Publishing, Paris.

5. The SIA/McKinsey gap is explained by scope: McKinsey (January 2026, 'Hiding in Plain Sight') includes the value of captive designers (Apple, Amazon, Tesla) and fabless operators whose sales do not appear in WSTS statistics.
6. Pilz, K.F., Rahman, R., Sanders, J. and Heim, L. (2025), 'Trends in AI Supercomputers,' arXiv:2504.16026, April 2025. Dataset accessible at <https://epoch.ai/data/gpu-clusters> under Creative Commons Attribution licence.
7. Epoch AI (2025), GPU Clusters Documentation. Coverage estimated at 20-37 percent of NVIDIA H100s, 12 percent of A100s and 18 percent of AMD MI300X, but less than 4 percent of Google TPUs.
8. Lehdonvirta, V., Wu, B., Hawkins, Z.J., Caira, C. and Russo, L. (2025), 'Measuring Domestic Public Cloud Compute Availability for Artificial Intelligence,' OECD AI Papers No. 49, <https://doi.org/10.1787/8602a322-en>.
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13. Tortoise Media (2024), *The Global Artificial Intelligence Index 2024*. See also Koronakos, G., Kritikos, M. and Sotiros, D. (2024), 'Mitigating Subjectivity and Bias in AI Development Indices,' *Expert Systems with Applications*.
14. Cohen, W.M. and Levinthal, D.A. (1990), 'Absorptive Capacity: A New Perspective on Learning and Innovation,' *Administrative Science Quarterly*, 35(1), pp. 128-152.
15. Effective costs are derived from the dashboard `energy_prices.csv` and reflect PPA-adjusted prices. Raw Eurostat industrial tariffs in Europe (consumption band IE) can be 1.5 to 2 times higher; the gap between raw and effective prices is documented in Chapter III. The Federal Reserve Board (October 2025) documents a significant negative correlation between energy costs and AI adoption at the European firm level.
16. Geometric aggregation is also used by the UNDP Human Development Index since 2010, for analogous reasons of non-substitutability between dimensions. See OECD/JRC Handbook (2008), pp. 31-33.
17. Operational shares computed by aggregating Epoch AI clusters with status Existing/Operational/Online/In progress on the April 2026 snapshot.
18. The Python code and data are reproducible. The Epoch AI dataset is downloadable in CSV at https://epoch.ai/data/gpu_clusters.csv (daily refresh). The dashboard CSVs are mirrored at <https://mo0gly.github.io/America-First-IA/dashboard/data/>.
19. The LinkedIn bias is documented: in countries where LinkedIn is rarely used (China, Russia, some Southeast Asian economies), AI skills density is mechanically underestimated.
20. IMF (2024), AI Preparedness Index Dashboard. Singapore, Denmark, Netherlands and the US occupy top positions; China ranks 31st (score 0.63).
21. Maslej, N. et al. (2025), 'Artificial Intelligence Index Report 2025,' Stanford Institute for Human-Centered AI, April 2025.
22. Tortoise Media (2024), op. cit. Stanford AI Index public data: open Google Drive folder.
23. The most promising causal identification strategy would exploit the exogenous shock of the October 2022 BIS rules in a Difference-in-Differences framework. See Arellano, M. and Bond, S. (1991), 'Some Tests of Specification for Panel Data,' *Review of Economic Studies*, 58(2), pp. 277-297.
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25. IEA (2025), *Energy and AI*, devotes a case study to DeepSeek and concludes that even with significant efficiency improvements, demand growth absorbs gains (Jevons rebound effect).
26. Schwartz (1991), op. cit., p. 38. Author translation.

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CHAPTER III - EMPIRICAL DIAGNOSIS 2020-2026

CHAPTER III

Empirical Diagnosis 2020-2026

This chapter establishes the factual foundation of the analysis. It covers four interdependent dimensions: the energy trajectory of data centers, the evolution of the semiconductor market, the geographical distribution of installed AI compute, and the timeline of American regulatory measures. The data presented here constitute the predetermined elements (in the Schwartz sense) that structure the prospective scenarios in Chapter V. All time series are sourced, and where data diverge between sources, the discrepancy is made explicit. All ratios involving compute have been recalibrated on the live April 2026 snapshot of the public dashboard.

3.1 Data Center Energy Trajectory: Demand Doubling in Six Years

3.1.1 Global Consumption 2020-2024

The International Energy Agency (IEA, April 2025) estimates global data center electricity consumption at approximately 415 TWh in 2024, representing 1.5 percent of global electricity consumption.[1] This consumption has grown at an average of 12 percent per year since 2017, a rate four times higher than total electricity consumption growth. In 2020, it stood at approximately 270 TWh; the increase thus represents +54 percent over four years.

The geographical distribution is highly concentrated. The United States absorbs 45 percent of global data center consumption (approximately 180 TWh in 2024), followed by China (25 percent, ~102 TWh) and Europe (15 percent, ~70 TWh for the EU according to the European Commission).[2] The four main European countries (Germany, France, United Kingdom, Netherlands) total approximately 41 TWh, or 10 percent of the global total. Total installed global data center capacity reached nearly 100 GW in 2024. Nearly half of US capacity is concentrated in just five regional clusters.

3.1.2 Projections 2024-2030: The IEA Central Scenario

The IEA central scenario (Base Case) projects global consumption of 945 TWh by 2030, representing a doubling compared to 2024 and equivalent to Japan current electricity consumption. AI is identified as the most important driver of this growth. AI share of data center consumption, estimated at 5 to 15 percent in recent years, could reach 35 to 50 percent by 2030.[3]

Growth is geographically uneven. The United States adds approximately +240 TWh (+130 percent compared to 2024), China +175 TWh (+170 percent), and Europe only +45 TWh (+70 percent). The IEA notes a 20 percent risk of project delays in Europe, linked to grid connection constraints and permitting timelines.[4] In the United States, data centers will account for nearly half of electricity demand growth by 2030; the country will consume more electricity for its data centers than for aluminum, steel, cement, and chemical production combined.

3.1.3 The Energy Factor as Competitive Advantage

The energy asymmetry constitutes a structural advantage for the United States. The IEA notes that natural gas currently provides 40 percent of US data center electricity, renewables 24 percent, nuclear 20 percent, and coal 15 percent. In Europe, the energy mix is more constrained by climate objectives and higher energy costs. Industrial electricity prices are typically 2 to 3 times higher in Europe than in the United States on the unadjusted Eurostat tariff. Once Power Purchase Agreements negotiated by hyperscalers are factored in, the gap narrows to roughly 1.4 to 1.7 times (1.59x for the EU aggregate at the April 2026 dashboard reference of 135 USD/MWh in the EU vs. 85 USD/MWh in the United States). As documented by the Federal Reserve Board (October 2025), there is a significant negative correlation between energy costs and AI adoption at the European enterprise level, even after PPA correction.[5]

France, however, has a specific asset in this context: its nuclear electricity mix (approximately 65 to 70 percent of production), which offers relatively inexpensive, decarbonized, and baseload-available electricity. RTE estimates an additional need of +10 GW for data centers in France by 2030, which remains compatible with installed capacity subject to investments in the transmission network.

3.1.4 Amplifying Effect: AI Robotics and Industrial Energy Demand

A factor still poorly quantified in the literature is the impact of AI robotics on energy demand. Autonomous robots require both embedded compute (edge computing) and centralized compute (cloud AI for training and model updates). The widespread deployment of AI automation in industry could add 20 to 30 percent additional energy demand in affected sectors, on top of data center growth. This factor is integrated as a sensitivity variable in our scenarios (Chapter V), as no precise and reliable estimate yet exists. The IEA nonetheless devotes a section of its report to the Jevons rebound effect: even with significant efficiency gains (illustrated by the DeepSeek case), demand growth absorbs the gains.[6]

3.2 Semiconductor Market: AI-Driven Growth

3.2.1 Global Sales Trajectory 2020-2025

The global semiconductor market has followed a remarkable trajectory since 2020. After a record of 555.9 billion USD in 2022, the sector contracted by 8.2 percent in 2023 (526.8 billion USD), before a spectacular rebound in 2024: global sales reached 630.5 billion USD (revised SIA/WSTS data, March 2025), a 19.1 percent year-on-year increase and a new all-time record.[7]

The year 2025 confirmed the acceleration. The SIA announced in February 2026 global sales of 791.7 billion USD in 2025, a 25.6 percent increase over 2024.[8] The fourth quarter of 2025 (236.6 billion USD) was 37.1 percent higher than the same quarter in 2024. The SIA now projects sales close to 1 trillion USD in 2026, a symbolic threshold that seemed out of reach two years ago.

Two segments are driving this growth. Logic chips (processors, GPUs, ASICs) reached 301.9 billion USD in 2025 (+39.9 percent), becoming the top category by sales. Memory (DRAM, NAND) reached 223.1 billion USD (+34.8 percent). Together, these two segments represent over 66 percent of global sales and directly reflect AI infrastructure demand: servers, data centers, and HPC (High Performance Computing).

3.2.2 The Scope Question: SIA vs McKinsey

As noted in Chapter II, a notable methodological gap exists between SIA/WSTS data and McKinsey estimates. The SIA records semiconductor sales in the strict sense (components sold by manufacturers). McKinsey (January 2026) adopts an expanded scope including the value of captive designers (Apple, Amazon, Tesla, Google), i.e., the economic value of chips designed internally but manufactured by contract foundries. Under this expanded scope, McKinsey values the market at approximately 775 billion USD in 2024 and projects 1.5 to 1.8 trillion by 2030.[9] The expanded scope is more relevant to our analysis because it captures the full value of the AI chain, including hyperscaler design investments.

3.2.3 Regional Concentration of Manufacturing Capacity

The geographical distribution of foundry capacity constitutes a key vulnerability factor. Taiwan (TSMC) concentrates most of the leading-edge production (sub-7nm nodes). The United States represents approximately 10 percent of global installed capacity but is growing rapidly (the Chips Act has generated nearly 500 billion USD in announced private investment, with a goal of tripling US capacity by 2032).[10] The EU represents approximately 8 percent of global capacity, a figure the European Chips Act (43 billion EUR) aims to raise to 20 percent by 2030, a target deemed unrealistic by many observers. China, despite restrictions, is progressing from 21 percent toward a 30 percent target of installed capacity by 2030, primarily on mature nodes (28 nm and above).

3.3 Geographical Distribution of Installed AI Compute

3.3.1 American Dominance: Three Quarters of Operational Compute

The most significant data point for our analysis is the geographical distribution of AI-dedicated computing capacity. The work of Epoch AI (Pilz et al., April 2025) provides the foundational dataset, refreshed monthly on the public dashboard.[11] As of April 2026, the dashboard catalogues 786 clusters, of which 609 are operational and 171 are planned or announced.

The result is unequivocal. On operational compute alone, the United States concentrates 76.9 percent of global H100-equivalent capacity, China holds 12.8 percent, and the EU(13) accounts for 4.4 percent. The remaining countries combine to roughly 6 percent. The operational US/EU H100-equivalent ratio is therefore 17.6:1, a gap of considerable magnitude that has widened slightly since the May 2025 GeoCoded/Sanchez snapshot (74.5 / 14.1 / 4.8 percent).[12]

Two dynamics explain this concentration. First, the private sector share of global AI compute has grown from 40 percent in 2019 to 80 percent in 2025, and the technology companies investing massively in AI are almost exclusively American (Microsoft, Meta, Google, Amazon, xAI). Second, cluster sizes have exploded: systems with more than 10,000 chips were rare in 2019; in 2024, xAI deployed a 200,000-GPU Colossus cluster. The performance of leading AI supercomputers doubles every nine months.[13]

The picture shifts substantially when planned capacity is included. The dashboard F_total view, which counts both operational and announced clusters, shows the US share falling to 49.9 percent. The shift is not driven by foreign actors building independent capacity, but by US hyperscalers offshoring to allied jurisdictions: the

announced UAE 5 GW campus and Stargate UAE Phase 2 represent 22.9 million H100-equivalents, Saudi Arabia 7.2 million, and South Korea 5.3 million, all largely owned or operated by US firms. This distinction between Physical CACI (where the cluster sits) and Sovereign CACI (who controls it) is developed in Chapter V. The Chinese share collapses to 0.5 percent in F_{total} , an artifact of the Epoch AI methodology rather than a real position change: the dataset anonymises Chinese systems and rounds them aggressively, so announced Chinese capacity is structurally under-recorded. The EU(13) F_{total} share also dips to 3.3 percent because the Fluidstack 1 GW campus in France pulls the total up while the rest of the EU shows few large announcements.

3.3.2 Total Electrical Power of the AI Chip Fleet

Epoch AI (January 2026) estimates the total electrical capacity of the global AI chip fleet at approximately 30 GW at end-2025, comparable to the peak power consumption of New York State.[14] This estimate is based on quarterly AI chip sales from major manufacturers (Nvidia, AMD, Google TPU), multiplied by their nominal power (TDP) and a 2.5x factor for data center infrastructure. Global AI chip production doubles every seven months, a pace that exceeds all prior forecasts. The five largest US AI server investors (Microsoft, Google, Meta, Amazon, xAI) announced 320 billion USD in investments in 2025, compared to 230 billion USD in 2024.

3.3.3 Implications for the CACI Metric

These data allow a sharp calibration of the CACI defined in Chapter II. Using the geometric weighted formula $CACI = F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$ (Power Mode), with the April 2026 dashboard reference values: operational US compute of 2,759,968 PetaFLOP/s (FP16) versus 156,632 PetaFLOP/s for the EU(13), an EU/US PPA-adjusted energy ratio of 1.59x (135 vs. 85 USD/MWh), comparable workforce orders of magnitude (3.5 vs. 9.6 million across the EU(13)), and geopolitical access factors of $R(US) = 1.0$ versus $R(EU) = 0.9$, the $CACI(US)/CACI(EU)$ Power Mode ratio settles at 3.46:1.

The gap is substantially attenuated relative to the 17.6:1 raw H100-equivalent advantage because the geometric exponents (F at 0.40, L at 0.20, R at 0.15, E at 0.25) apply diminishing returns to the massive US compute lead and reward the EU partially on labour, geopolitical access and lower energy multipliers. In Intensity Mode (dividing additionally by GDP), the gap narrows further to roughly 1.4:1, which is why the headline ratio used throughout this study is the Power Mode ratio of 3.46:1. Chapter V projects the evolution of this ratio under each scenario.

3.4 Timeline of American Regulatory Measures (2022-2026)

The sequence of American semiconductor and AI control measures constitutes the guiding thread of our hypothesis. Four phases are identifiable, marking a progressive escalation in the scope and intensity of restrictions.

Phase 1 - The Initial Shock (October 7, 2022)

The BIS (Bureau of Industry and Security) of the Department of Commerce publishes a final interim rule that radically transforms American export controls on semiconductors. The measures cover three components: (i)

controls on advanced computing chips (GPUs above performance thresholds defined by TTP - Total Processing Performance), (ii) controls on semiconductor manufacturing equipment (SME, including EUV and DUV lithography), and (iii) restrictions on US person activities supporting advanced chip manufacturing in China.[15] The target is explicitly China: the stated objective is to prevent Chinese military modernization through AI compute access. The restrictions include three new Foreign Direct Product (FDP) rules that extend American jurisdiction to products manufactured outside the United States if US technologies are used in their production.

Phase 2 - Closing the Gaps (October 2023)

The BIS publishes two new interim rules that strengthen and broaden the October 2022 controls. Technical thresholds are adjusted to capture Nvidia chips specifically designed to circumvent restrictions (A800, H800). The geographic scope is extended to approximately 40 additional countries (Country Groups D:1, D:4, and D:5), with a differentiated licensing regime by country category.[16] Controls on manufacturing equipment are deepened. It is during this phase that European actors begin to perceive the indirect effects of restrictions, even though the EU is not the primary target.

Phase 3 - The AI Diffusion Rule (January 2025, Biden)

The Biden administration publishes in January 2025 the AI Diffusion Rule, which represents a qualitative shift. For the first time, restrictions apply not only to physical chips but also to AI model weights and cloud compute access. The rule classifies 120 countries into three categories: (i) trusted allies (broad access), (ii) intermediate countries (quotas), (iii) embargoed countries.[17] European reactions are strong: the European Parliament raises alarm about restrictions threatening the EU ability to train models on its AI Factories. France and Germany are classified as trusted partners, but other member states face caps on importable GPU volumes.

Phase 4 - The Trump Break: Section 232 and Explicit Protectionism (January 2026)

On January 14, 2026, President Trump signs Proclamation 11002, invoking Section 232 of the Trade Expansion Act of 1962.[18] This action marks a qualitative break with previous phases, for three reasons.

First, the legal instrument changes. Phases 1 to 3 fell under export controls (export regulation, defensive national security logic). Section 232 is a tariff instrument (import duties), whose logic is protectionist: it aims to protect domestic production, not merely restrict an adversary access.

Second, the tariff creates an explicit competitive advantage for American companies. The 25 percent tariff hits advanced GPUs (H200, MI325X) imported unless destined for American domestic uses: US data centers, R&D, startups, public sector, industrial and non-data-center consumer applications. Concretely, an American company using these chips on US soil does not pay the tariff; a foreign company importing the same chips for re-export to China pays 25 percent.[19]

Third, the proclamation announces an imminent expansion. The text provides that the Secretary of Commerce and USTR negotiate within 90 days with semiconductor-producing countries, and that broader tariffs, accompanied by a tariff offset program for companies investing in US production, could be imposed.

The Secretary of Commerce must provide by July 2026 a report on the semiconductor market used in American data centers.[20]

3.4.5 Interpretation: From Defensive Control to Offensive Protectionism

The 2022-2026 sequence reveals a qualitative transformation of American policy. The Biden phases (2022-2025) follow a denial strategy logic: preventing a designated adversary (China) from accessing key technologies, within a multilateral framework (coordination with Japan, Netherlands). The Trump phase (2025-2026) adds a capture strategy logic: generating revenue (25 percent tariff), prioritizing American companies (domestic exemptions), and using compute access as a negotiation lever with third countries.

This is precisely the transformation that validates the central hypothesis of our study: American technology protectionism is no longer limited to denying access to an adversary; it actively constructs a structural competitive advantage for American companies. The Section 232 domestic exemptions mean that, for the first time, the cost of accessing cutting-edge compute is legally differentiated by the user nationality. Even though direct effects on the EU remain limited in January 2026 (the tariff primarily targets re-exportation to China), the proclamation opens the way for an expansion that could directly affect Europe, precisely the territory of scenarios B, C, and D of our analysis.

3.5 Diagnostic Synthesis: The Predetermined Elements of 2026

The four dimensions of the diagnosis converge toward a coherent picture that structures the scenarios in Chapter V.

(1) AI compute demand is growing exponentially (chip sales doubling in two years, energy consumption doubling projected over six years), and this growth shows no signs of slowing.

(2) This demand is structurally concentrated in the United States. The April 2026 dashboard snapshot puts the US share of operational AI compute at 76.9 percent in H100-equivalents and at 49.9 percent if announced clusters are included; 45 percent of data center electricity consumption; and more than 80 percent of frontier-AI private investment. The concentration on operational compute is increasing over time.

(3) Europe starts from a structurally deficit position: roughly 4.4 percent of operational compute in H100-equivalents, around 15 percent of data center consumption, energy costs that run 1.4 to 1.7 times higher after PPA correction (and 2 to 3 times higher on unadjusted Eurostat industrial tariffs), and 72 to 80 percent dependency on US hyperscalers for AI cloud workloads. The EU investment plans (Chips Act, AI Factories, SNIA) carry budgets at least one order of magnitude smaller than the 2025 Big Tech announcements (320 billion USD for the five US frontier-AI investors alone).

(4) The American regulatory framework crossed a qualitative threshold in January 2026 with the shift from export controls to Section 232 tariffs, creating a legal mechanism for differentiating compute access costs by nationality. The proclamation explicitly signals the possibility of expansion, whose scope and timeline will depend on political variables, precisely the critical uncertainties that our scenarios explore.

The following chapter (Chapter IV) analyzes the mechanisms by which this compute asymmetry translates into measurable competitive advantage for American companies.

Table 4. Data center electricity consumption by region, 2020-2030.

Source: IEA (2025), Energy and AI.

Region	2020	2024	2030 (IEA)	Delta 24-30	Global Share 2024
United States	~120 TWh	~180 TWh	~420 TWh	+240 TWh	45 pct
China	~60 TWh	~102 TWh	~280 TWh	+175 TWh	25 pct
Europe (EU)	~45 TWh	~70 TWh	~115 TWh	+45 TWh	15 pct
World	~270 TWh	~415 TWh	~945 TWh	+530 TWh	100 pct

Table 5. Global semiconductor sales, 2020-2026.

Sources: SIA/WSTS (Feb. 2025, Feb. 2026). The 2026 projection is based on the WSTS autumn 2025 forecast (975.4 billion USD) and the SIA February 2026 statement.

Year	2020	2022	2023	2024	2025	2026 (proj.)
Sales (billion USD, SIA)	440	556	527	631	792	~975-1,000
Growth (pct)	+6.8	+3.3	-8.2	+19.1	+25.6	+23-26

Table 6. Indicators of US dominance in AI compute (April 2026 dashboard snapshot).

Sources: Epoch AI GPU Clusters dataset April 2026; IEA (2025); Sanchez/GeoCoded (2025).

Indicator	United States	China	EU(13)	US/EU Ratio
Operational H100-eq share	76.9 pct	12.8 pct	4.4 pct	17.6:1
F_total share (op + planned)	49.9 pct	0.5 pct	3.3 pct	15.1:1
Private sector share (2025)	~65 pct of global	~12 pct	~3 pct	~22:1
Data center consumption 2024	180 TWh	102 TWh	70 TWh	2.6:1
AI investment 2025	320 bn USD (5 Big Tech)	n.a.	~20 bn EUR (EU total)	>15:1
CACI Power Mode (April 2026)	100	15.7	28.9	3.46:1

Table 7. Timeline of American regulatory measures on semiconductors and AI (2022-2026).

Sources: BIS, White House, Pillsbury Law (2026), Gibson Dunn (2026).

Date	Admin.	Measure	Scope / Target
Oct. 2022	Biden	BIS export controls: advanced GPUs, SME, US persons	China (military)
Oct. 2023	Biden	Threshold reinforcement + 40-country extension + HBM	China + 40 countries (D:1/D:4/D:5)
Dec. 2024	Biden	Wave 3: 24 SME types, HBM, 140 entities, ECAD	China (full chain)
Jan. 2025	Biden	AI Diffusion Rule: model weights, cloud, 3-tier countries	120+ countries (incl. EU effects)

Jul. 2025	Trump	America AI Action Plan: deregulation, US compute	US (domestic strategy)
Sep. 2025	Trump	Announcement: China sales authorized against 25 pct revenue	China (monetization)
Jan. 2026	Trump	Section 232: 25 pct tariff advanced GPUs + BIS China license	Global (US domestic exemption)

Notes

1. IEA (2025), Energy and AI, Paris, IEA. <https://www.iea.org/reports/energy-and-ai>. The 415 TWh figure includes all data centers (cloud, enterprise, colocation), not just AI-dedicated data centers. The IEA notes that consumption has grown by 12 percent per year since 2017.
2. IEA (2025), op. cit., chapter 'Energy demand from AI.' European Commission (November 2025), 'In Focus: Data Centres - An Energy-Hungry Challenge,' energy.ec.europa.eu. The 70 TWh figure for the EU is an IEA estimate in the absence of precise consumption data.
3. IEA (2025), op. cit.; Carbon Brief (September 2025), 'AI: Five Charts that Put Data-Centre Energy Use - and Emissions - into Context.' The 35-50 percent range comes from a report prepared for the IEA (Kamiya and Coroama, 2025, IEA-4E).
4. IEA (2025), op. cit. The IEA further notes that in its Headwinds scenario, global consumption would only reach 790 TWh, 40 percent less than the central scenario, illustrating the scale of uncertainty.
5. Federal Reserve Board (October 2025), State of AI Competition in Advanced Economies. The report documents a significant correlation between industrial energy costs and AI adoption rates, particularly for large European companies. The PPA-adjusted reference values (USA 85, China 92, France 115, Germany 140, EU aggregate 135 USD/MWh) are documented on the public dashboard.
6. IEA (2025), op. cit., chapter 'Case study: DeepSeek and efficiency gains.' The IEA concludes that algorithmic efficiency gains, while significant, tend to be absorbed by demand growth (Jevons paradox).
7. SIA (February 2025, revised March 2025), 'Global Semiconductor Sales Increase 19.1 percent in 2024.' The initial figure of 627.6 billion USD was revised to 630.5 billion USD by the WSTS.
8. SIA (February 2026), 'Global Annual Semiconductor Sales Increase 25.6 percent to 791.7 Billion USD in 2025.' Q4 2025 sales (236.6 billion USD) were 37.1 percent higher than Q4 2024, marking a notable acceleration.
9. McKinsey (January 2026), 'Hiding in Plain Sight: The Semiconductor Industry Expanding Perimeter.' The gap with the SIA is explained by the inclusion of captive designers and vertically integrated fabless operators.
10. SIA (July 2025), State of the U.S. Semiconductor Industry Report. The SIA notes that the Chips and Science Act has generated nearly 500 billion USD in announced private investment and is expected to create or support more than 500,000 jobs.
11. Pilz, K.F., Rahman, R., Sanders, J. and Heim, L. (2025), 'Trends in AI Supercomputers,' arXiv:2504.16026. The dashboard refresh of April 2026 reaches 786 catalogued clusters, of which 609 are operational and 171 are planned, representing 10 to 20 percent of estimated global capacity.
12. Sanchez, C. (2025), 'GeoCoded Special Report: State of Global AI Compute (2025 Edition),' Sanchez.vc. The May 2025 estimate cross-references Epoch AI and Georgetown University data and produced shares USA 74.5 / China 14.1 / EU 4.8. The April 2026 dashboard snapshot is based on the same data feed but with eleven additional months of cluster commissioning, which explains the small drift in shares.
13. Pilz et al. (2025), op. cit. Leading AI supercomputer performance doubles every nine months, driven by a 1.6x per year increase in chip count and a 1.6x per year increase in per-chip performance.
14. Epoch AI (January 2026), 'Global AI Power Capacity Is Now Comparable to Peak Power Usage of New York State.' The 30 GW estimate is based on quarterly AI chip sales x TDP x 2.5x infrastructure factor.

15. BIS (October 7, 2022), 'Commerce Implements New Export Controls on Advanced Computing and Semiconductor Manufacturing Items to the People Republic of China,' Federal Register. See also GAO (2025), 'Export Controls: Commerce Implemented Advanced Semiconductor Rules,' GAO-25-107386.
16. Skadden (October 2023), 'BIS Updates October 2022 Semiconductor Export Control Rules.' The new rules notably capture the A800 and H800, Nvidia chips specifically designed to fall just below the October 2022 thresholds.
17. Carnegie Endowment (May 2025), Winter-Levy, H. and Phillips-Robins, A., 'What Is the AI Diffusion Rule?' The AI Diffusion Rule remained in effect only briefly before being partially replaced by the Trump approach, but it established the precedent of control over models and cloud, not just physical chips.
18. White House (January 14, 2026), Presidential Proclamation 11002, 'Adjusting Imports of Semiconductors, Semiconductor Manufacturing Equipment, and Their Derivative Products into the United States.' The proclamation invokes Section 232 of the Trade Expansion Act of 1962 (19 U.S.C. 1862).
19. Pillsbury Law (January 2026), 'Trump Admin Targets Advanced AI Semiconductors, Defers Broader Tariffs.' The legal analysis notes that the combination of Section 232 and the BIS rule of January 15, 2026 operationalizes the September 2025 announcement that the US government would collect 25 percent of AI chip sales to China.
20. White House (January 14, 2026), Fact Sheet: 'President Donald J. Trump Takes Action on Certain Advanced Computing Chips to Protect America Economic and National Security.' Gibson Dunn (January 2026), 'The Trump Administration New Tariffs on and Export Licensing Requirements for Advanced Semiconductors.'

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CHAPTER IV - MECHANISMS OF US COMPETITIVE ADVANTAGE

CHAPTER IV

Mechanisms of American Competitive Advantage Through AI

Chapter III established the facts: the United States concentrates 76.9 percent of operational global AI compute (49.9 percent if planned clusters are included), 45 percent of data center energy consumption, and controls 70 percent of the European cloud market through three hyperscalers. This chapter analyses the mechanisms through which this compute asymmetry translates into measurable competitive advantage. We identify four transmission channels: training costs as a barrier to entry, cloud concentration as a vector of dependence, regionally differentiated productivity, and the capture of innovation rents by first movers.

4.1 Training Costs as a Barrier to Entry

4.1.1 The Exponential Explosion of Training Costs

The most documented channel is that of foundation model training costs. Martens (Bruegel, October 2024) establishes that the cost of training a frontier model rose from 1,000 USD in 2017 to nearly 200 million USD in 2024, and could reach several billion by 2030.[1] This exponential growth follows the scaling laws identified by Kaplan et al. (2020): the improvement of cognitive performance in models is subject to constant returns in complementary inputs (parameters, training data, compute capacity). In other words, each marginal gain in performance requires a proportional increase in compute, data, and model size.

Cottier et al. (2024), cited by Martens, decompose training costs into five components: personnel, AI chips, server costs, network interconnection, and energy. They estimate infrastructure costs (hardware plus data center) at ten times the cost of training itself, with a depreciation rate of 140 percent per year (complete depreciation in 8.5 months, matching the renewal pace of chip generations). GPT-4 infrastructure by end-2023 was estimated at approximately 800 million USD. By extrapolation, infrastructure costs could reach 500 billion USD by 2030, replicated across half a dozen hyperscalers.[2]

4.1.2 The US Advantage: Abundant and Subsidised Compute

In this context, access to cutting-edge compute becomes the discriminating factor. American Big Tech (Microsoft, Meta, Google, Amazon, xAI) collectively invested 320 billion USD in 2025 in AI infrastructure (servers, data centers, chips), up from 230 billion USD in 2024.[3] Meta is developing an AI computing system with 350,000 H100 processors, estimated at over 10 billion USD. As Martens notes, this barrier to entry is completely beyond the reach of public funding; only the largest companies can aspire to it.[4]

For European actors, this asymmetry translates concretely. The cost of training a model comparable to GPT-4 is estimated at approximately 100 million USD in the United States (with direct GPU access, cheap energy, amortisation over a massive installed base). In Europe, the same training would cost 2 to 5 times more, due

to: (i) indirect compute access via US cloud (hyperscaler margins), (ii) energy costs that run 1.4 to 1.7 times higher after PPA correction (and 2 to 3 times higher on unadjusted Eurostat industrial tariffs - Chapter III), and (iii) the absence of comparable economies of scale. The FLOP cost gap (CACI metric M2) is estimated between 2.4x and 3.6x once cloud markups, energy and amortisation are stacked, even though the underlying PPA-adjusted energy ratio is only 1.59x.[5]

4.1.3 Forced Co-opetition: European Startups and US Big Tech

Exponential costs create a mechanism of structural dependence. As analysed by Martens (Bruegel), AI startups - including European ones - are forced to cooperate with American Big Tech to access computing infrastructure and distribution channels. This co-opetition (forced cooperation with a competitor) places startups in a subordinate position: they depend on US platforms to train their models, then find themselves in direct competition with the proprietary models of those same platforms. The UK Competition and Markets Authority (CMA, 2024) and the European Commission have identified these agreements as potentially anti-competitive.[6] Azoulay et al. (2024), cited by Martens, argue that resource sharing is the only path to avoid concentration, unless a firm uses openness as a competitive strategy, which Meta has partially done with LLaMA.

4.2 Cloud Concentration as a Vector of Geopolitical Dependence

4.2.1 The European Cloud Market: 70 Percent Under US Control

The second mechanism is the concentration of the European cloud market. Synergy Research Group (July 2025) establishes that the European cloud infrastructure services market reached 61 billion EUR in 2024 (multiplied by six since 2017) and should exceed 75 billion EUR in 2025 (+24 percent). Amazon (AWS), Microsoft (Azure), and Google (GCP) account for 70 percent of this market. The share of European providers dropped from 29 percent in 2017 to 15 percent in 2022, where it has stagnated since.[7]

Among European providers, SAP and Deutsche Telekom lead with approximately 2 percent market share each, followed by OVHcloud, Telecom Italia, and Orange. John Dinsdale, Synergy chief analyst, observes that US providers invest 10 billion EUR per quarter in European capex, and that the three hyperscalers now operate more than 140 hyperscale data centers in Europe.[8] The fastest growth (140 to 160 percent) concerns services specifically related to generative AI (GPU-as-a-Service, GenAI PaaS), a segment where European players are virtually absent.

4.2.2 From Commercial Lock-in to Geopolitical Lock-in

This concentration goes beyond a simple commercial issue. The concept of geopolitical vendor lock-in (defined in Chapter I) describes a situation where technical dependence on a foreign provider is compounded by a regulatory risk linked to the sovereign decisions of that provider home country. The CLOUD Act (2018) requires American cloud providers to furnish US authorities with data hosted on their servers, including those located in Europe. In May 2025, Microsoft France chief legal officer acknowledged under oath the inability to guarantee that French citizens data would never be transmitted to American authorities.[9]

In the AI context, geopolitical lock-in operates at three levels. At the infrastructure level (IaaS), AI workloads are deployed on servers physically controlled by entities subject to US jurisdiction. At the platform level (PaaS), training and inference frameworks are integrated into proprietary ecosystems (Azure ML, SageMaker, Vertex AI). At the model level (MaaS - Model-as-a-Service), access to state-of-the-art models (GPT-4, Claude, Gemini) goes through APIs controlled by US companies, with terms of service that can be modified unilaterally. Section 232 protectionism, by differentiating the cost of hardware access, reinforces all three levels simultaneously.

The Chapter I distinction between Physical CACI (F_{total}) and Sovereign CACI (F_{dom}) is critical here: while EU operational compute is largely owned by EU operators (OVH, Scaleway, Sesterce, EuroHPC, AI Factories), EU AI workloads run predominantly on US-owned infrastructure. The dependence is not on installed compute, but on used compute, and on the platforms and models layered above it.

4.2.3 The Draghi Report and the European Diagnosis

The Draghi Report (September 2024), commissioned by the European Commission, constitutes the most comprehensive institutional recognition of this dependence. Draghi observes that European restrictions on data storage and processing create high compliance costs and hinder the creation of large integrated datasets for AI model training. He recommends the creation of a European hyperscale cloud infrastructure, the consolidation of European cloud providers, and massive investment in AI, quantum computing, and EuroHPC supercomputers.[10] Martens (Bruegel, 2025) moderates this optimism, however, observing that EuroHPC supercomputers are not designed for generative AI and that upgrading to competitive standards exceeds the financial capacity of European budgets.[11]

4.3 Differentiated Productivity: The Abundant Compute Dividend

4.3.1 The Theoretical Macroeconomic Potential

The third transmission channel is AI impact on labor productivity and its regional asymmetry. The McKinsey Global Institute (May 2024, December 2025) estimates that generative AI could enable Europe to reach an annual productivity growth rate of 3 percent per year by 2030 in an accelerated adoption scenario with proactive workforce redeployment.[12] This figure is sufficient to close most of the productivity gap with the United States. However, in a slow adoption scenario, productivity growth would be only 0.3 percent, close to current levels and insufficient to fund the European social model.

The IMF (Working Paper, March 2025) provides a more granular analysis.[13] By cross-referencing task exposure indices to AI (Felten et al., Eloundou et al.) with sectoral data, the IMF estimates total factor productivity (TFP) gains significantly higher in advanced economies than in middle-income countries, due to their greater share of value added in high AI-exposure sectors (financial services, consulting, technology) and their higher wage levels, which further justify automation. Underlying microeconomic studies (randomized controlled trials) show gains ranging from 14 percent for low-skilled tasks to over 50 percent for software engineers.

4.3.2 The Gap Conditioned by Compute Access

Yet, the realisation of these productivity gains is conditioned by compute access. This is where the asymmetry diagnosed in Chapter III produces its effects. The French AI Commission (2024) estimates AI potential impact on French growth at 1.3 percentage points per year, by analogy with the effects of electricity, but this estimate assumes unconstrained access to cutting-edge compute. Similarly, McKinsey (December 2025) conditions the optimistic European scenario on accelerated adoption which implies massive infrastructure investments.[14]

The investment gap is considerable. McKinsey (January 2026) estimates that large European companies face an annual deficit of 700 billion USD in R&D and capital expenditure compared to their American counterparts. In digital technologies alone, US companies invested 2 trillion EUR more than European companies over the 2021-2025 period.[15] The annualised gap between January 2024 and September 2025 amounts to 580 billion EUR for corporate investment and 300 billion EUR for startups and scale-ups.

This investment deficit, combined with the compute asymmetry and energy costs, produces a conditional AI productivity gap. In our modelling, we distinguish between theoretical potential (identical for US and EU with comparable sectoral structures) and achievable potential, which depends on effective compute access. With a CACI(US)/CACI(EU) of 3.46:1 in Power Mode (Chapter III) on the April 2026 dashboard snapshot, the achievable European potential is structurally inferior. Translated into normalised CACI scores, the EU sits at 28.9 against a US benchmark of 100, France at 25.3, Germany at 5.4 - the gap is therefore not 7 to 12 times inferior as earlier analyses suggested, but is precisely 3.46 to 1 in compute-adjusted competitive intensity, with sharp variation across member states.

4.4 Innovation Rent Capture: A Self-Reinforcing Advantage

4.4.1 The First-Mover Advantage Mechanism in AI

The fourth channel is the most structurally significant in the long run. The theory of general purpose technologies (GPT, Bresnahan and Trajtenberg, 1995) predicts that early adopters of a general technology capture disproportionate innovation rents, which become self-reinforcing through three mechanisms.

First, data scale effects. Companies that deploy AI first accumulate usage data (feedback loops, fine-tuning) that improves the performance of their models. This data is non-rival but exclusive: once captured by an actor, it does not benefit competitors. McKinsey estimates that 75 percent of generative AI economic potential is concentrated in four key functions (customer service, software development, marketing, R&D), where usage data is particularly determinative.[16]

Second, platform network effects. US hyperscalers benefit from a virtuous circle: more clients leads to more revenue, which leads to more infrastructure investment, which leads to a better offering, which leads to more clients. This mechanism explains why, despite discussions on digital sovereignty, European providers share stagnates at 15 percent: the service quality gap (latency, range of services, technical support, integration ecosystem) is too large to be bridged by sovereignty arguments alone.

Third, talent capture. US companies offer compensation packages including salaries and equity that attract the best AI researchers and engineers worldwide, including Europeans. Martens notes that qualified AI

personnel are scarce and that companies compete fiercely to recruit them, with compensation packages difficult for European institutions to match. This AI brain drain weakens the European CACI L(r) factor and reinforces the American advantage in human capital.[17]

4.4.2 The Risk of Irreversibility

These three mechanisms produce a potentially irreversible cumulative effect. As the OECD analyses (Martens, 2021) by analogy with previous technological waves (broadband internet, cloud), latecomers in the adoption of a general technology become structurally dependent on foreign platforms, suffer compressed margins, and progressively lose their strategic autonomy. The 2022 semiconductor shortage, which cost the European automotive sector alone 100 billion EUR, illustrates the concrete consequences of unanticipated technological dependence.[18]

The trajectory is all the more concerning as the EU is absent from four of the eight segments of the generative AI value chain as mapped by McKinsey (2024): AI-design semiconductors (dominated by Nvidia), AI cloud platforms (AWS, Azure, GCP), foundation models (OpenAI, Google, Anthropic, Meta), and AI development tools. It is competitive only in downstream application segments: sectoral applications (SAP, Siemens), industrial integration, and specialised semiconductors (power semiconductors: Infineon, STMicroelectronics, NXP, with about 15 percent global market share).[19]

4.5 Synthesis: The Mechanics of US Competitive Advantage

The four channels identified form a reinforcing system. The compute asymmetry (Chapter III) feeds differentiated training costs (4.1), which strengthen dependence on US cloud (4.2), which constrains achievable productivity (4.3), which slows European investment and reinforces rent capture by American players (4.4), which in turn widens the compute gap. With the April 2026 snapshot of 76.9 percent operational compute share, 1.59x PPA-adjusted energy cost and a CACI Power Mode ratio of 3.46:1, this circle constitutes what Farrell and Newman (2019) would describe as structural interdependence weaponisation: European dependence on American infrastructure, initially founded on economic efficiency, becomes a lever of geopolitical power when instrumentalised by measures such as Section 232.

Trump protectionism (Chapter III, phase 4) did not create this asymmetry; it pre-existed massively. It institutionalises and legalises it through a tariff mechanism that differentiates the cost of compute access by nationality. In doing so, it transforms a de facto advantage into a de jure advantage, whose dismantling is politically and legally far more difficult. Chapter V examines how these mechanisms could evolve across four prospective scenarios.

Table 8. European cloud market: US hyperscaler domination.

Source: Synergy Research Group (July 2025).

Indicator	2017	2022	2024	Trend
EU Cloud Market (billion EUR)	~10	~35	61	x6 in 7 years
AWS+Azure+GCP Share	~50 pct	~67 pct	70 pct	Growing
EU Provider Share	29 pct	15 pct	15 pct	Stable (floor)

US Capex in Europe	n.a.	n.a.	~40 bn EUR/yr	10 bn EUR/quarter
Hyperscale DCs in EU	~40	~80	>140	+75 pct in 2 years

Table 9. AI productivity and competitive advantage: US vs EU (April 2026 calibration).

Sources: McKinsey (2024, 2025, 2026), Bruegel/Martens (2024), CACI calibration on the public dashboard.

Dimension	United States	Europe (EU)	Gap
Potential AI productivity (McKinsey, accel. scen.)	+2.5-3.5 pct/yr	+2.5-3.0 pct/yr	Comparable
Achievable AI productivity (under compute constraints)	+2.5-3.0 pct/yr	+0.8-1.5 pct/yr	-1.5 to -2 pts
Annual AI investment (corporate + VC)	~450 bn USD	~50-70 bn EUR	~7:1
FLOP cost (USD/TFlop, training)	~0.5	~1.2-1.8	2.4-3.6x
Normalised CACI Power Mode (USA = 100)	100	28.9 (EU(13))	3.46:1

Notes

1. Martens, B. (2024), 'The Tension Between Exploding AI Investment Costs and Slow Productivity Growth,' Working Paper 18/2024, Bruegel, pp. 5-8. The time series is calibrated on Epoch AI training cost data for frontier models.
2. Cottier, B. et al. (2024), cited in Martens (2024), op. cit. The 10x factor between infrastructure and training costs is a central estimate, subject to variation based on post-training infrastructure utilisation.
3. IEA (2025), Energy and AI, Paris, p. 42. The IEA compiles investment announcements from Meta, Amazon, Alphabet, Microsoft and xAI for 2025.
4. Martens, B. (2024), 'Why Artificial Intelligence Is Creating Fundamental Challenges for Competition Policy,' Policy Brief 16/ 2024, Bruegel. Meta plans a system of 350,000 H100s (at about 30,000 USD per unit), i.e. over 10 billion USD in hardware alone.
5. Author estimate based on Bruegel (training costs), Eurostat (industrial electricity tariffs), EIA (US prices), the public dashboard PPA-adjusted reference values (USA 85, EU 135 USD/MWh), and Epoch AI (GPU performance/price). The 2.4 to 3.6x range stacks cloud markup (1.5 to 2x) on top of the 1.59x PPA-adjusted energy ratio and the absence of EU economies of scale; it depends on access mode (cloud vs on-premise) and PPA negotiations.
6. Competition and Markets Authority (2024), 'AI Foundation Models - Update Report,' London. The European Commission launched a call for contributions on competition in generative AI in 2024.
7. Synergy Research Group (July 2025), 'European Cloud Providers Local Market Share Now Holds Steady at 15 percent.' The market is defined as IaaS plus PaaS plus hosted private cloud.
8. Dinsdale, J. (2025), cited in Synergy Research Group, op. cit.: US cloud providers continue to invest 10 billion EUR every quarter in European capex, most of which comes from the big three.
9. Data Center Dynamics (July 2025), 'European Cloud Providers Hold 15 percent of Local Market Share.' The article reports the testimony of Anton Carniax, Microsoft France chief legal officer, before a parliamentary committee.
10. Draghi, M. (September 2024), The Future of European Competitiveness, report commissioned by the European Commission. Cited in Martens, B. (2025), 'Catch-Up with the US or Prosper Below the Tech Frontier? An EU Artificial Intelligence Strategy,' Bruegel Policy Brief.
11. Martens, B. (2025), op. cit. The author notes that the EuroHPC approach, centred on hardware, has characterised European digital policies for decades and is particularly poorly oriented regarding AI.

12. McKinsey Global Institute (May 2024), 'A New Future of Work: The Race to Deploy AI and Raise Skills in Europe and Beyond.' The 3 percent figure assumes an accelerated adoption scenario with full workforce redeployment.
13. IMF (March 2025), 'Artificial Intelligence and Productivity in Europe,' Working Paper WP/25/067. The analysis uses Felten et al. (2021) and Eloundou et al. (2024) exposure indices.
14. McKinsey (December 2025), 'Accelerating Europe AI Adoption: The Role of Sovereign AI Capabilities.' The report estimates European labor productivity growth at 0.2 percent on average over 2020-2025, versus a potential of 3.1 percent with full AI adoption.
15. McKinsey (January 2026), 'Transforming Europe: Bold Moves to Lift a Continent.' The 700 bn USD per year gap is calculated on the 150 largest technology companies (US vs EU) in capex plus R&D (S&P Capital IQ data).
16. McKinsey Global Institute (2024), 'The Economic Potential of Generative AI: The Next Productivity Frontier.' The four functions (customer service, software development, marketing/sales, R&D) represent about 75 percent of the identified economic potential across 63 use cases.
17. Martens, B. (2024), Working Paper 18/2024, op. cit. Personnel costs (salaries plus equity) are often the most significant component of AI model development costs.
18. McKinsey (October 2024), 'Time to Place Our Bets: Europe AI Opportunity.' The 100 bn EUR GDP loss figure for the European automotive industry comes from Allianz (September 2022) and OICA.
19. McKinsey (October 2024), op. cit. European semiconductors (Infineon, STMicro, NXP) represent about 15 percent of the global integrated design-manufacturing market (Omdia/Informa data, 2024), but primarily in power and automotive segments, not AI GPU/ASIC.

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CHAPTER V - PROSPECTIVE SCENARIOS 2026-2030

CHAPTER V

Prospective Scenarios 2026-2030

This chapter constitutes the heart of the original contribution of this study. By applying the methodological protocol defined in Chapter II (2x2 matrix, six divergence metrics, CACI calibration), we construct four scenarios for the evolution of the transatlantic relationship in AI, energy, and semiconductors for the 2026-2030 period. Each scenario is determined by the combination of two critical uncertainties identified in Chapter III: the degree of US protectionism and the European strategic response capacity. We then evaluate each scenario on its six metrics, before synthesising the tipping point conditions between trajectories.

5.1 Predetermined elements: what will not change

In accordance with the Schwartz method (1991), we distinguish predetermined elements (quasi-certain trends by the 2030 horizon) from critical uncertainties (factors whose evolution depends on political decisions not yet taken). Four predetermined elements structure all scenarios.

EP1 - Exponential growth in AI compute demand. Semiconductor sales doubled in two years (2023-2025), the power of installed AI chips doubles every seven months (Epoch AI), and no sign of slowing down is observable as of February 2026. Even under the hypothesis of a deceleration in scaling laws (Chinchilla saturation), the diffusion of AI towards inference, robotics, and autonomous agents will maintain a strongly growing compute demand.[1]

EP2 - Persistent concentration of compute in the United States. The US/EU ratio of 17.6:1 in raw installed compute (Chapter III, April 2026 dashboard snapshot: 2,759,968 vs 156,632 PFLOP/s), resulting in a CACI Power Mode ratio of 3.46:1 once weighted by the geometric formula $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$, reflects investment decisions taken in 2022-2025 whose effects materialise until 2028-2029 (data center construction delays: 18-36 months). Even an immediate political reversal would not alter the installed stock before the end of the decade.

EP3 - Growing energy tension. Global data center consumption, estimated at 415 TWh in 2024, will reach 800-950 TWh by 2030 according to IEA projections (Chapter III). The asymmetry in energy costs (US 1.4-1.7x cheaper after PPA correction, 2-3x cheaper on unadjusted industrial Eurostat tariffs) will persist, unless massive investment in European nuclear power, whose deployment delays (SMR: 5-7 years for the first reactors) exceed the 2030 horizon.[2]

EP4 - Section 232 regulatory framework in place. Proclamation 11002 of January 14, 2026, is a legal fait accompli. Unlike IEEPA tariffs (annulled by the Supreme Court on February 20, 2026[3]), Section 232 tariffs rest on a confirmed legal basis. The Secretary of Commerce's report on the data center semiconductor market is expected by July 1, 2026, and may recommend an extension or modification of tariffs. Regardless of the direction taken, the legal instrument will remain available.[4]

5.2 Critical uncertainties and 2x2 matrix

5.2.1 Axis 1: Degree of US protectionism

The first uncertainty concerns the evolution of American policy between two poles. The moderate pole corresponds to maintaining the January 2026 status quo: a 25 percent tariff limited to re-exported advanced chips, broad domestic exemptions, a US-EU trade agreement capping semiconductor tariffs at 15 percent, and no significant extension to cloud or models. The EU core (France, Germany) remains in the 'trusted partners' category. The aggressive pole assumes an extension after the July 2026 report: tariffs extended to derived semiconductors and equipment, GPU quotas for the EU (including France), restrictive conditions for access to frontier AI cloud, and the use of compute as a trade negotiation lever (compute-for-concessions).[5]

5.2.2 Axis 2: European strategic response capacity

The second uncertainty concerns the EU's ability to deploy a coherent and rapid response. The reactive pole corresponds to fragmented national responses, dispersed investments, an AI Act creating additional compliance costs, and slow deployment of AI Factories/Gigafactories (bureaucratic delays, 24+ month authorisations). The proactive pole assumes the accelerated implementation of the AI Continent program (19 AI Factories plus up to 5 Gigafactories of 100,000+ GPUs), the adoption of Special Compute Zones (180-day authorisation), the effective mobilisation of the InvestAI fund (20 billion EUR), and the pooling of French nuclear capacity as a competitive advantage.[6]

5.2.3 Matrix and scenario naming

The intersection of these two axes produces four scenarios. Reactive EU + moderate US protectionism gives Scenario A (Reinforced Status Quo, slow drift towards dependence). Reactive EU + aggressive US gives Scenario B (Digital Fracture, structural European decoupling). Proactive EU + moderate US gives Scenario C (Asymmetric Partnership, junior Western technological partner). Proactive EU + aggressive US gives Scenario D (Contested Sovereignty, race for autonomy under pressure).

5.3 Scenario A - Reinforced Status Quo (Moderate protectionism + Reactive EU)

5.3.1 Narrative

Following the July 2026 report, the Secretary of Commerce recommends maintaining the 25 percent tariff on re-exported advanced chips but without extending it significantly. The August 2025 US-EU trade agreement is respected: semiconductor tariffs for the EU remain capped at 15 percent.[7] The EU, reassured by this status quo, slows the deployment of its own initiatives. EuroHPC AI Factories struggle to reach nominal capacity (authorisation delays, inter-state coordination). Gigafactories are postponed to 2029-2030. The InvestAI fund is partially mobilised (8-10 billion EUR out of 20). European companies continue to rely heavily on US cloud, whose performance and costs remain unbeatable.

5.3.2 Metric trajectories

M1 - Compute ratio (US/EU installed GPUs): goes from 17.6:1 raw (2025) to 18-22:1 raw (2030) on operational installed compute. The gap widens slightly as US investments accelerate (Stargate, xAI mega-clusters, Meta) while the EU only adds the 19 AI Factories (max 25,000 GPUs each, approx. 475,000 public GPUs, an order of magnitude below a single US hyperscaler).[8]

M2 - FLOP cost gap (EU/US): remains in the 2.4-3.2x range. The absence of aggressive tariffs on the EU maintains access to US cloud at prices close to current levels, but European energy costs continue to weigh in.

M3 - US cloud share in European AI spending: goes from 70 percent (2024) to 72-75 percent (2030). European providers (OVHcloud, Deutsche Telekom) maintain their 15 percent in the sovereignty segment but do not gain ground on generative AI services.

M4 - AI productivity (percent/year): US +2.5-3.0; EU +1.0-1.5. The EU realises part of the AI potential via downstream applications (SAP, Siemens, fintech), but slow adoption and compute deficit cap the gains.

M5 - Energy dependence (data center TWh): EU approx. 115 TWh in 2030 (+65 percent vs 2024). French nuclear power absorbs part of the demand, but the absence of Special Compute Zones delays the network connection of new data centers.

M6 - CACI(US)/CACI(EU): goes from 3.46:1 (April 2026) to 4-5:1 (2030). The gap widens moderately as the F factor (compute) accumulates on the US side while E costs (energy) weigh on the EU side.

5.3.3 Consequences for France

This scenario is the most probable in the short term (estimated probability: 40-50 percent). It is also the most insidious: the absence of a visible shock demobilises European actors, while dependence deepens structurally. French companies benefit from access to US cloud for AI adoption (BNP Paribas, Airbus, TotalEnergies via AWS/Azure), but this adoption reinforces the lock-in described in Chapter IV. The AI productivity deficit relative to the United States (-1.0 to -1.5 points per year) accumulates over five years, widening the competitiveness gap by 5 to 8 GDP points.

5.4 Scenario B - Digital Fracture (Aggressive protectionism + Reactive EU)

5.4.1 Narrative

The July 2026 report leads to a significant extension. The Secretary of Commerce recommends tariffs extended to semiconductor equipment and derived products, with a tariff offset program reserved for companies investing in American production.[9] The 15 percent EU agreement is revised upwards or accompanied by restrictive conditions (volume quotas on advanced GPUs, reciprocity requirements on the AI Act). Simultaneously, access to frontier AI cloud is made conditional for non-US entities (API access limitations to frontier models, weight restrictions). The EU, fragmented, fails to formulate a coherent response: member states divide between accommodation (Nordic countries, Netherlands) and confrontation (France, Italy).

5.4.2 Metric trajectories

M1 - Compute ratio: goes from 17.6:1 raw (2025) to 25-35:1 raw (2030). GPU quotas limit European imports at the moment demand explodes. AI Factory projects are compromised by the inability to procure Nvidia/AMD GPUs at the planned volumes.

M2 - FLOP cost gap: jumps to 4-6x. Extended tariffs, combined with quotas and energy asymmetry, massively increase European compute costs. French companies face a 3x to 5x surcharge for model training.

M3 - US cloud share: paradoxically rises to 78-82 percent. Failing a credible local alternative, European companies wanting access to frontier AI must go through US hyperscalers, at the price conditions they dictate. Sovereign services (OVHcloud, Scaleway) lack the hardware to offer competitive GenAI services.

M4 - AI productivity: US +2.5-3.5; EU +0.3-0.8. European AI potential is severely constrained. The McKinsey Global Institute estimates that with slow adoption, European productivity would not exceed 0.3 percent, close to stagnation.[10]

M5 - Energy dependence: EU approx. 95 TWh only (2030), not by virtue but by default - the lack of GPUs limits data center construction. Ironically, the compute constraint mitigates the energy constraint.

M6 - CACI ratio: explodes from 3.46:1 (April 2026) to 6-8:1 (2030). This is the scenario where the gap is largest, with all three CACI factors deteriorating simultaneously on the European side: F capped by quotas, E inflated by tariffs, L weakened by an accelerated brain drain to the United States.

5.4.3 Consequences for France

This scenario (estimated probability: 15-20 percent) represents the worst-case. France suffers structural technological decoupling: compute-intensive projects (Mistral foundation models, Exotec robotics, Dassault simulations) are relocated to the United States or depend on increasingly costly US cloud access. Time-to-market for French AI solutions lengthens by 25 to 40 percent. SMEs, unable to absorb surcharges, renounce frontier AI and opt for degraded solutions (smaller open-source models, local inference). The cumulative productivity gap with the United States reaches 10 to 15 points over five years.

5.5 Scenario C - Asymmetric Partnership (Moderate protectionism + Proactive EU)

5.5.1 Narrative

US protectionism remains moderate (as in A), but the EU exploits this window to accelerate its own investments. AI Factories are deployed on schedule (2026-2027), the first 100,000+ GPU Gigafactories are ordered end-2026 and delivered in 2028.[11] France plays a central role thanks to its nuclear fleet (65-70 percent of electricity mix, competitive marginal cost), and Special Compute Zones are designated on former industrial sites with heavy network connections.[12] However, the EU de facto accepts a status as a junior technological partner: it uses Nvidia/AMD GPUs (no European AI ASIC design champion), depends on TSMC/Samsung/Intel foundries for production, and its foundation models remain one step below US leaders.

5.5.2 Metric trajectories

M1 - Compute ratio: drops from 17.6:1 raw (2025) to 8-10:1 raw (2030) on installed compute. Gigafactories and private investment (InvestAI plus industrial co-investments) add 1-2 million H100 equivalents in Europe, narrowing the gap without closing it.

M2 - FLOP cost gap: drops to 1.5-2.0x. French nuclear power and Gigafactory economies of scale compress energy and infrastructure costs, although a residual gap persists (absence of proprietary GPU design).

M3 - US cloud share: drops slightly to 60-65 percent. European sovereign services gain market share in regulated segments (defense, health, finance), while US cloud retains the majority of commercial workloads. The market segments into sovereign and performance.

M4 - AI productivity: US +2.5-3.0; EU +1.8-2.5. The EU reaches 60-80 percent of its theoretical potential thanks to sufficient local compute for large-scale adoption of downstream applications, even if frontier model training remains dependent on US hardware.

M5 - Energy: EU approx. 140 TWh (2030). Demand is higher than in A because European compute increases, but nuclear and planned SMRs absorb most of it. RTE France confirms the feasibility of +10 GW subject to network investments.

M6 - CACI ratio: drops from 3.46:1 (April 2026) to 2.0-2.5:1 (2030). This is the most favourable realistically achievable scenario at the 2030 horizon. The F factor improves significantly, E benefits from nuclear, but L remains slightly lower (the US AI ecosystem being more attractive for top talent).

5.5.3 Consequences for France

This scenario (estimated probability: 15-20 percent) is the most favourable for France in the short-to-medium term. France becomes the EU's AI energy hub thanks to its nuclear fleet, attracting data center and Gigafactory investments. French companies gain access to competitive local compute for inference and fine-tuning, reducing dependence on US cloud for standard uses. Mistral and French startups can train specialised models locally. However, frontier model training remains dependent on US hardware, and strategic autonomy is partial: France is sovereign in application, but not in the creation of fundamental technologies.

5.6 Scenario D - Contested Sovereignty (Aggressive protectionism + Proactive EU)

5.6.1 Narrative

US protectionism intensifies (as in B), but the EU responds with determination. The American threat becomes the political catalyst for an unprecedented European industrial mobilisation since the AIRBUS project of the 1970s. The AI Continent program is accelerated and extended: the 5 Gigafactories are ordered as an emergency, France announces 20 GW of nuclear capacity dedicated to AI data centers by 2032 (combining fleet extension, new EPR2s and SMRs), the DARE project (European RISC-V) is escalated to design

AI accelerators reducing dependence on Nvidia.[13] Simultaneously, the EU negotiates alternative technological alliances (Japan, South Korea, Taiwan) to secure GPU and foundry supplies.

5.6.2 Metric trajectories

M1 - Compute ratio: evolves from 17.6:1 raw (2025) to 12-15:1 raw (2030). The EU invests massively but starts from very far. US quotas slow imports, but alternative alliances and local production (Gigafactories using Samsung/Intel GPUs as alternatives to Nvidia) partially compensate.

M2 - FLOP cost gap: 2.5-4.0x initially (2027, peak tariff shock), then progressive reduction towards 1.8-2.5x (2030) as Gigafactories scale up and GPU alternatives mature.

M3 - US cloud share: drops to 50-55 percent (2030), the most pronounced decline of the four scenarios. Geopolitical defiance and US restrictions push European companies towards sovereign alternatives, even if imperfect. US hyperscalers lose ground in regulated segments.

M4 - AI productivity: US +2.5-3.5; EU +1.2-2.0. The EU traverses a productivity trough in 2027-2028 (transition period where US restrictions bite but European investments are not yet operational), then a partial catch-up from 2029.

M5 - Energy: EU approx. 150-160 TWh (2030). This is the most energy-intensive scenario for the EU, with massive local data center construction creating huge demand. French nuclear power becomes a continental strategic asset, but network pressure is at its maximum.

M6 - CACI ratio: follows a U-shaped trajectory: degradation from 3.46:1 to 8-12:1 in 2027-2028 (peak shock), then improvement to 4-7:1 by 2030. The result depends heavily on European execution speed: each year of delay on Gigafactories prolongs the period of maximum vulnerability.

5.6.3 Consequences for France

This scenario (estimated probability: 15-20 percent) is the most ambitious and riskiest. It places France at the heart of an unprecedented European technological sovereignty effort. Massive nuclear investments (SMR, fleet extension) become a top-tier geopolitical issue. The DARE/RISC-V project could, if successful, constitute the first credible European alternative to Nvidia GPUs for AI, but on a 5-7 year horizon, well beyond 2030. In the short term (2026-2028), France traverses a period of maximum vulnerability where surcharges and shortages degrade competitiveness, before a catch-up conditional on infrastructure deployment speed.

5.7 Comparative synthesis and tipping points

5.7.1 Metric synthesis table

Table 11 below consolidates the trajectory of the six divergence metrics for the four scenarios by the 2030 horizon, anchored on the April 2026 dashboard snapshot (US/EU raw operational compute 17.6:1, CACI Power Mode 3.46:1).

5.7.2 Tipping points between scenarios

The real trajectory will likely follow a hybrid path between these scenarios. Three tipping points determine possible transitions.

First tipping point: the July 2026 Commerce report. This report will determine whether US protectionism extends (shift to B or D) or remains targeted (maintenance in A or C). Indicators to watch: evolution of the US semiconductor trade deficit, CHIPS Act fab filling rates (Intel, TSMC Arizona, Samsung Taylor), domestic political pressure (2026 midterms). Phase 1 negotiation results (report due April 14, 2026) will be an early signal.[14]

Second tipping point: EU Gigafactory deployment speed. The Commission expects the first Gigafactories to be operational in 2027-2028. If this schedule is held, the EU shifts towards proactive scenarios (C or D). If authorisation, financing, or hardware supply delays push deliveries to 2029-2030, the EU remains in reactive mode (A or B). The CFG proposal for Special Compute Zones (180-day authorisation vs 24+ months currently) is the key acceleration factor.[15]

Third tipping point: the French decision on nuclear for AI. France possesses a unique asset in Europe: a nuclear fleet providing 65-70 percent of electricity at a globally competitive marginal cost. The decision to dedicate significant capacity (10-20 GW) to AI data centers, via fleet extension, new EPR2s and SMRs, will determine whether France becomes Europe's AI energy hub or loses this position to others (Scandinavia with hydro, Eastern Europe with low land costs). This tipping point is specifically French and determines France's position within the European scenarios.[16]

5.7.3 The convergence point: 2028

The four scenarios converge on a common critical point in 2028. This is the year when: (i) compute demand will exceed installed capacity in Europe, creating hardware bottlenecks (even under moderate protectionism); (ii) the full effects of extended tariffs (if adopted) will be felt; (iii) Gigafactories, if deployed on time, will start producing significant local compute; (iv) data center energy demand will saturate network connection capacity in several member states. The year 2028 constitutes the moment of truth when Europe will discover if it is on an A/B trajectory (growing dependence) or C/D (catch-up started). Decisions taken in 2026-2027 (US Commerce report, Gigafactories, French nuclear) will be irreversibly engaged.

5.8 Origins of the tipping point: juridico-technical foundations

The 'Great Decoupling' scenario is not built ex nihilo. It is the logical projection, at the 2028 horizon, of a control architecture whose foundational layers are already operational in 2026. Identifying these layers is a matter of academic rigour: distinguishing documented trends from extrapolated projections.

5.8.1 The legal layer: extraterritoriality as a structural instrument

The first layer is legal and largely precedes AI policy. The CLOUD Act (Clarifying Lawful Overseas Use of Data Act, 2018) establishes that any cloud provider subject to US jurisdiction is required to produce data regardless of where such communications, records, or other information are stored.[17] This law, confirmed by federal case law (United States v. Microsoft), creates a fundamental dissociation between the physical location of data and its legal nationality.

The immediate consequence for AI compute is radical: an H100 cluster physically located in Dubai, Singapore, or an AWS eu-west-1 center in Ireland remains legally American. If ordered by the US government, the operator (AWS, Azure, Google Cloud) is legally bound to comply, regardless of the client's will or local legislation. Microsoft acknowledged before a French court in 2024 that it could not guarantee data sovereignty for European clients in the event of a legally grounded US injunction.[18]

This pre-existing legal architecture is the substrate on which the Cloud-Nationality Pivot grafts: the Cloud Sovereignty Mandates projected for 2028 do not create a new power. They activate and systematise an existing jurisdictional power by extending it to the operational compute layer.

5.8.2 The technical layer: from location verification to cluster throttling

The second layer is technical. The US AI Action Plan of July 23, 2025, explicitly introduces the concept of 'location verification features' applied to advanced AI chips. Michael Kratsios, OSTP director at the White House, confirmed discussions on software or hardware modifications to enable better location-tracking, explicitly included in the plan.[19]

This announcement is not rhetorical. The BIS already has, since the Framework for Artificial Intelligence Diffusion (January 2025), a compute-tiering control architecture for destination countries (Tier 1-2-3) and compute caps per entity and country.[20] The BIS Affiliates Rule, suspended for one year in November 2025 but maintained in principle, stipulates that affiliation to a parent entity in a restricted country is sufficient to deny access to advanced compute, regardless of physical location.[21]

The technological trajectory towards 2028 is therefore: (i) chips equipped with location verification mechanisms, (ii) automatic reporting to the BIS in case of deviation, (iii) the ability to suspend access or throttle performance via export licence. This is not science fiction: it is the extension to compute of a principle already applied to software (OFAC sanctions, service access freeze).

Critical limitation: Throttling production clusters is technically complex and potentially disruptive for operators themselves. NVIDIA does not currently have a remote deactivation mechanism for H100/B200 GPUs in production. Such a mechanism would require significant architectural modifications to firmware and remote attestation protocols (via TPM or equivalent). The 2028 scenario is therefore conditional on the effective implementation of these modifications over 24-36 months.

5.9 The Cloud-Nationality Pivot mechanism

5.9.1 The trigger: Cloud Sovereignty Mandates as extension of export controls

The 2028 tipping point scenario assumes that the United States crosses a qualitative threshold: moving from hardware access control (BIS chip controls) to operational compute service access control (Cloud Sovereignty Mandates). This threshold is not an arbitrary rupture; it responds to a structural flaw identified in the current control regime.

This flaw is documented: despite BIS restrictions, investigations revealed that approx. 1 billion USD of Nvidia chips reached China via third countries (Malaysia, UAE, Singapore) in the first months of 2025.[22] The AI

Action Plan's response (location verification and cluster monitoring) is the first step towards continuous post-export control.

The plausible trigger in 2028 is an executive order extending Framework for AI Diffusion obligations to the cloud layer. It would impose a 'Data Residency and Jurisdiction Compliance' certification on all US hyperscalers operating advanced offshore clusters, with revocation of compute access on US soil as the enforcement mechanism.

5.9.2 Dissociating Physical Factor / Sovereign Factor in CACI

This is where the scenario produces its most significant impact on the CACI model. Under the Cloud - Nationality Pivot, the compute variable $F(r)$ decomposes into two components: $F(r) = F_{\text{phys}}(r) \times F_{\text{sov}}(r)$, where F_{phys} is physically installed compute and F_{sov} is the operational sovereignty factor (fraction of F_{phys} outside US jurisdiction).

Note the distinction between this dynamic 2028 F_{sov} and the static Sovereign CACI introduced in Chapter I (Fig 1.8). The Chapter I Sovereign CACI captures who owns the compute today (April 2026 snapshot). The 2028 F_{sov} captures who controls compute under a hypothetical Mandate regime, which depends on cloud workload share, not installed capacity. The two metrics agree where compute is owned by domestic operators; they diverge where local clusters are owned by US-side operators (UAE 99.6%, EU cloud workloads majorly on AWS/Azure/GCP).

Calibrated estimations for 2028, under the Mandate activation hypothesis, are presented in Table 12.

5.10 Emergence of jurisdictional AI blocs (2028-2030)

The Great Decoupling does not produce a binary US/non-US world. It produces fragmentation into blocs of variable intensity, according to each zone's capacity to develop credible sovereign compute. Four blocs emerge.

5.10.1 The Extended American Bloc (American AI Alliance)

The July 23, 2025 AI Action Plan explicitly lays the foundations for an American AI Alliance: exporting the full US tech stack (hardware, models, software, standards) to aligned allies in exchange for aligned export controls.[23] The strategy is 'carrot and stick': aligned allies access frontier chips and models without additional restrictions; those who refuse are exposed to Foreign Direct Product Rule mechanisms and secondary tariffs.

Confirmed Tier 1 members: US, UK, Canada, Australia, Japan, South Korea, Netherlands, Germany, France (subject to export control alignment). For these countries, effective F_{sov} increases: their entities access US hyperscalers without restriction, and sovereign compute in development (EU Gigafactories) receives preferential treatment.

5.10.2 The Eurasian Sovereign Bloc

China constitutes the only complete example of a pre-existing sovereign bloc. With an F_{sov} of approx. 0.98 and a national cloud ecosystem (Alibaba, Tencent, Huawei) operating outside US jurisdiction, Cloud Sovereignty Mandates have no direct traction. The Chinese constraint remains the shortage of advanced chips, but the American bloc cannot throttle a Huawei Ascend 910B cluster.

Post-2028 dynamics: China holds the only large-scale sovereign compute outside the American bloc. Countries seeking to emancipate themselves from Mandates face a binary choice: conditional American compute or Chinese compute under other forms of dependence.

5.10.3 Digital Non-aligned: an untenable position

India, Brazil, Southeast Asia, and Gulf countries (without special US treaties) constitute a digital non-alignment bloc. Their position is structurally uncomfortable: too dependent on US hyperscalers to shift to sovereignty, insufficiently integrated into the US alliance to escape restrictions in case of geopolitical disagreement.

The UAE case illustrates this with quantitative force. Fig 1.8 documented that 99.6% of UAE F_{total} is owned by US-side actors (Stargate UAE, Microsoft, OpenAI), collapsing Sovereign CACI from a Physical 55.7 to just 6.0. Dubai invested massively to become an AI hub, notably via AWS/Microsoft/G42 agreements. Yet, G42 was already pressured in 2024 to break ties with Chinese entities as a condition for advanced chip access.[24] Under Mandates, this pressure becomes systemic: Gulf compute, physically present but legally American, would become a permanent negotiation lever.

5.10.4 The European Bloc: between alliance and autonomy

Europe occupies an intermediate and evolving position. Legally Tier 1, the UE nevertheless maintains a strategic autonomy ambition that the American Alliance does not fully satisfy.

The Cloud and AI Development Act (CADA), expected Q1 2026, attempts to resolve this by defining an 'EU Sovereignty Level' that would structurally exclude CLOUD Act-subject providers from sensitive public procurement.[25] The October 2025 Cloud Sovereignty Framework defines three assurance levels, with SOV - 3 requiring the provider to be beyond the reach of non-European extraterritorial legislation.[26]

This architecture is under construction, but its timing is problematic: CADA will be at best operational in 2027-2028, precisely when US Mandates could be activated. The vulnerability window is maximal.

Note from Chapter I: the EU is largely sovereign on installed compute (99.2% of F_{total} is EU-owned). The vulnerability is not on installed F but on the cloud workload layer (the compute actually used by EU firms, majorly on AWS/Azure/GCP). CADA targets exactly this layer.

5.11 Transversal impacts on scenarios A-D

The Cloud-Nationality Pivot superimposes on the four scenarios, modifying their conclusions non-linearly. It adds a third dimension: the degree of installed compute autonomy. Table 13 synthesises the impact.

5.12 Implications for France: the question of genuinely sovereign compute

5.12.1 'Sovereignty washing' as a systemic risk

The term 'sovereignty washing' designates the practice of US hyperscalers marketing sovereign cloud offers by implanting data centers on European soil, while remaining subject to the CLOUD Act.[27] Microsoft acknowledged in its own documentation that it cannot guarantee sovereignty for European clients in case of a legally grounded US injunction.[28]

5.12.2 The French nuclear advantage in the new equation

The sovereign reinterpretation of CACI paradoxically reinforces the French strategic asset. If $F(r) = F_{\text{phys}} \times F_{\text{sov}}$, the optimal strategy is not just to increase F_{phys} (attracting hyperscalers) but F_{sov} (developing compute independent of US jurisdictions).

French nuclear power creates a top-tier competitive advantage: Special Compute Zones adjoined to carbon-free, competitive nuclear electricity, hosting Gigafactories operated by European entities (OVHcloud, Scaleway, Mistral AI, IONOS), would produce compute with an F_{sov} close to 1, genuinely beyond the reach of Cloud Sovereignty Mandates.

5.12.3 The DARE/RISC-V project: from ambition to strategic necessity

Under Scenario D and especially under Cloud Sovereignty Mandates, the DARE project (Digital Autonomy with RISC-V in Europe, 2025) shifts from a long-term ambition to a strategic necessity.[29] As long as Europe depends exclusively on NVIDIA/AMD GPUs, US control over these architectures creates a residual vulnerability even in European-operated Gigafactories. Genuine strategic autonomy requires the ability to design independent accelerators, an horizon DARE situates at 2030-2032.

5.13 Synthesis: the fourth tipping point

The Great Decoupling is not inevitable. It represents a conditional systemic risk. To the three tipping points in 5.7.2, a fourth is added.

Fourth tipping point: implementation of location verification features in advanced GPUs. If the BIS and US Commerce Department successfully deploy remote attestation mechanisms in H100/B200 chips by end-2026-2027, the technical substrate for the Cloud-Nationality Pivot will be in place. The question will shift from technical feasibility to political decision.

For Europe and France, a controlled decoupling strategy rests on three pillars: (i) accelerating sovereign compute deployment (EU Gigafactories) to increase F_{sov} before Mandates; (ii) securing Tier 1 status in the American alliance; (iii) investing in architectural alternatives (DARE/RISC-V, Huawei Ascend for non-sensitive workloads) to reduce medium-term dependence.

Table 10. 2x2 matrix of prospective scenarios 2026-2030.

Source: author's construction, Schwartz methodology (1991).

	Reactive EU response	Proactive EU response
Moderate US protectionism	Scenario A - Reinforced Status Quo (slow drift towards dependence)	Scenario C - Asymmetric Partnership (junior Western technological partner)
Aggressive US protectionism	Scenario B - Digital Fracture (structural European decoupling)	Scenario D - Contested Sovereignty (race for autonomy under pressure)

Table 11. Comparative synthesis of the four scenarios on the six divergence metrics (2030 horizon).

Source: author's construction; baseline April 2026 snapshot (US/EU raw operational compute 17.6:1, CACI Power Mode 3.46:1).

Metric (2030)	A - Status Quo	B - Fracture	C - Partnership	D - Sovereignty
M1 Raw compute ratio US/EU (operational)	18-22:1	25-35:1	8-10:1	12-15:1
M2 FLOP cost gap	2.4-3.2x	4-6x	1.5-2.0x	1.8-2.5x
M3 US cloud share (pct)	72-75	78-82	60-65	50-55
M4 EU productivity (pct/yr)	+1.0-1.5	+0.3-0.8	+1.8-2.5	+1.2-2.0
M5 EU energy (TWh)	~115	~95	~140	~155
M6 CACI Power Mode ratio	4-5:1	6-8:1	2.0-2.5:1	4-7:1 (post-trough)
Estimated probability	40-50 pct	15-20 pct	15-20 pct	15-20 pct

Table 12. Estimation of F_sov factor by jurisdiction and CACI impact under Cloud Sovereignty Mandates activation (2028).

Source: author's construction; Synergy Research Group (2025), Statista (2025), and Chapter I Fig 1.8 baseline.

Jurisdiction	F_phys cloud-US share	Estimated F_sov	Current CACI (phys baseline)	Post-Mandate CACI impact
United States	~5 pct (domestic cloud unaffected)	1.00	100 (reference)	100 (unchanged)
EU (France, Germany)	~77 pct (AWS/Azure/GCP dominance)	0.28	28.9 (Power Mode)	30-50 pct collapse on workloads
UAE (Dubai hub)	~88 pct (US hyperscaler dominance)	0.12	55.7 phys / 6.0 souv	60-80 pct collapse - illusory hub
Singapore	~82 pct (US hyperscaler dominance)	0.18	high - APAC hub	55-75 pct collapse
China	~2 pct (Alibaba/Tencent/Huawei Cloud)	0.98	15.7 (chip shortage caps)	Unchanged - sovereignty effective
India	~60 pct (AWS/Azure + local)	0.40	22.2 (Power Mode)	Moderate collapse - intermediate pos

Table 13. Impacts of the Cloud-Nationality Pivot (Cloud Sovereignty Mandates 2028) on the four 2x2 matrix scenarios.

Source: author's construction.

Scenario	Without Mandates	With Mandates	EU CACI Impact	Strategic Reading
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		2028		
A - Status Quo	Slow dependence, CACI 4-5:1	Partial activation - hyperscalers cooperate without major restriction	Moderate degradation 15-25 pct on workloads	More stable but security illusion revealed
B - Fracture	Structural decoupling, CACI 6-8:1	Maximal activation - conditional US cloud and restricted chips simultaneously	Double scissor effect: rare chips + conditional compute. CACI ratio potentially > 8:1	Absolute worst case - chips + cloud combo
C - Partnership	Partial catch-up, CACI 2.0-2.5:1	Sovereign Gigafactories absorb shock if EU F_sov rises to 0.45-0.55	Limited impact if deployed on time - Gigafactories = sovereign hedge	Best resilience - prior investment proves its value
D - Sovereignty	Catch-up under pressure, CACI 4-7:1 post-trough	Mandates become political catalyst - accelerate EU deployment and JP/KR/TW alliances	Accelerated U-curve - 2028-29 trough, faster catch-up	Paradoxical: Mandates can accelerate EU sovereignty if response is fast enough

Notes

1. Epoch AI (Jan 2026), Trends in AI Hardware and Compute. Doubling every 7 months combines 1.6x/yr quantity and 1.6x/yr performance. Even a slowdown to 12 months implies quadrupling by 2030.
2. IEA (Apr 2025), Energy and AI, Paris. 800-950 TWh projections correspond to median/high scenarios. Energy ratio (1.4-1.7x PPA-adjusted) derived from dashboard: USA 85, EU 135 USD/MWh.
3. US Supreme Court (Feb 20, 2026), Learning Resources Inc. v. Trump. 6-3 decision: 'IEEPA does not authorize the President to impose tariffs.'
4. White House (Jan 14, 2026), Proclamation 11002, sec (2): Secretary update on data center semi market due July 1, 2026.
5. Tax Foundation (Feb 2026). Aug 2025 US-EU agreement caps semi tariffs for EU at 15 percent, but Proclamation 11002 allows broader tariffs after Phase 1.
6. European Commission (2025), AI Continent Action Plan. Target: triple EU data center capacity in 5-7 years. 19 AI Factories, 5 Gigafactories. InvestAI: 20 bn EUR. CFG (Oct 2025) Special Compute Zones.
7. Tax Foundation (Feb 2026), op. cit.
8. EuroHPC JU (2025). 19 AI Factories approx 475,000 public GPUs, vs single xAI Colossus (200k H100 extensible).
9. Proclamation 11002, tariff offset program section.
10. McKinsey Global Institute (May 2024). 0.3 percent corresponds to slow adoption scenario.
11. EU Council (Dec 2025). Provisional schedule: operational Gigafactories in 2027-2028.
12. CFG (Oct 2025), 'Tripling the EU Data Centre Stock with Special AI Compute Zones.'
13. EuroHPC JU (Mar 2025), DARE project (Digital Autonomy with RISC-V in Europe).
14. Proclamation 11002, sec (2): April 14, 2026 negotiation report due.
15. CFG (Oct 2025). EU DC authorisation currently 24+ months (vs 6-12 US). SCZ would reduce to 180 days.
16. RTE (2024), Futurs energetiques 2050. France +10 GW AI demand by 2030. Nuclear 65-70% of mix.
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18. Microsoft France, Tribunal judiciaire de Paris (2024). Cited in The Register (Dec 22, 2025).

19. Michael Kratsios (OSTP), Seoul (Aug 2025), cited in TechResearchOnline.
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21. BIS, 'Suspension of the Affiliates Rule for One Year' (Nov 10, 2025).
22. Financial Times (July 25, 2025), '\$1bn Nvidia chips smuggled to China'.
23. White House, EO on Promoting the Export of the American AI Technology Stack (July 23, 2025).
24. US Dept of Commerce, G42/Microsoft agreement (2024). Confirmed by Sec Gina Raimondo.
25. European Commission, 2026 Work Programme, CADA, expected Q1 2026.
26. European Commission, Cloud Sovereignty Framework (Oct 2025), SOV-1 to SOV-3.
27. Cristina Caffarra, Eurostack Foundation, cited in The Register (Dec 22, 2025).
28. See note 18.
29. EuroHPC JU, DARE project (Digital Autonomy with RISC-V in Europe), launched Mar 2025.

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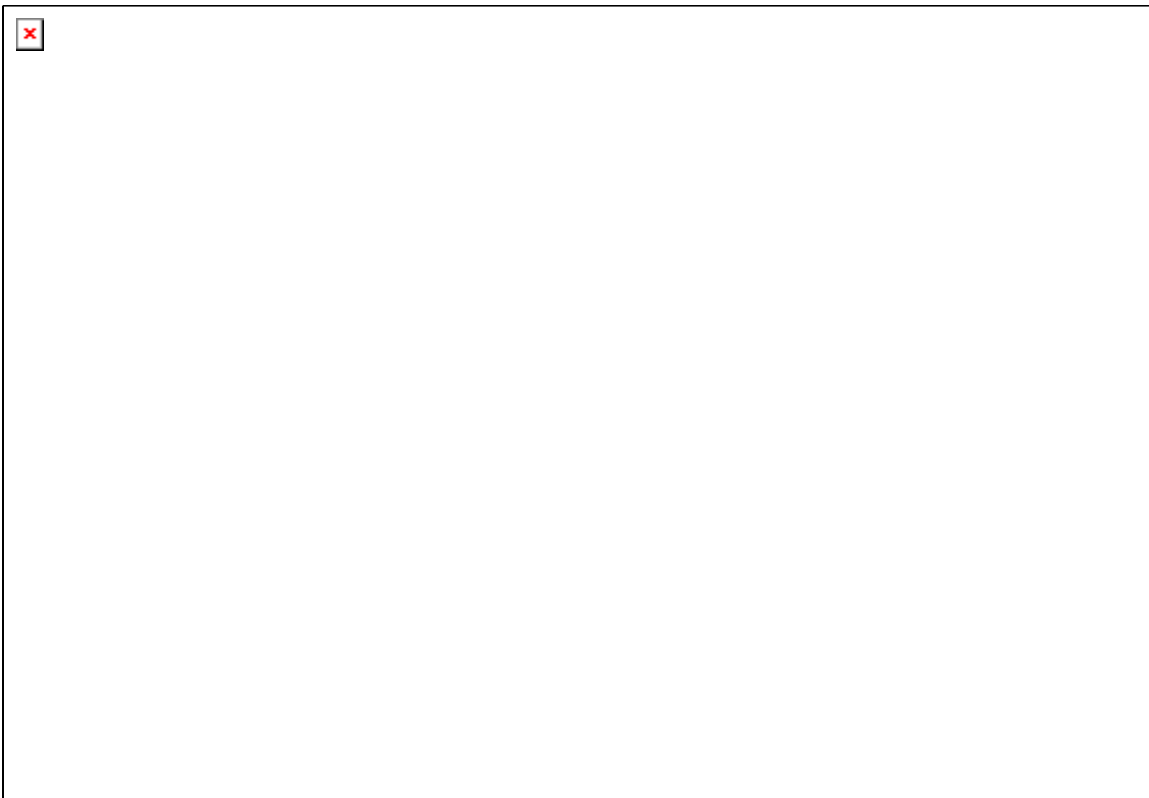
CHAPTER VI - CONSEQUENCES FOR FRANCE AND EUROPE

CHAPTER VI

Consequences for France and Europe

The previous chapters have established the diagnostic (III), the mechanisms (IV) and the possible trajectories (V). This chapter outlines the concrete consequences for French and European actors, distinguishing three levels of analysis: sectoral breakdown (which sectors are most exposed?), differentiation by actor type (large groups, SMEs, startups, public sector), and second-order effects (brain drain, R&D offshoring, normative fragmentation). The analysis relies mainly on scenario A (the most probable) and scenario B (the most severe), while signaling bifurcations specific to scenarios C and D.

6.1 Sectoral analysis: differentiated exposure to compute asymmetry

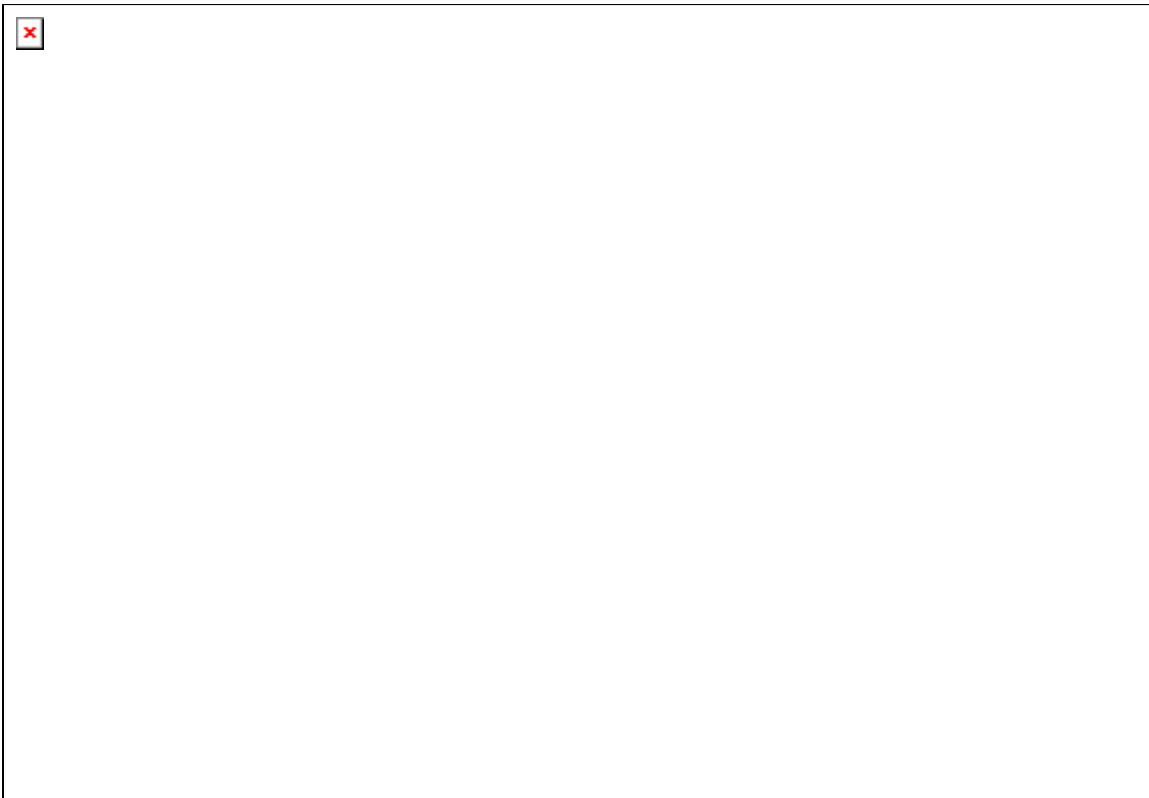


6.1.1 Financial services: advanced dependence

The French financial sector is the most advanced in AI adoption and, paradoxically, the most exposed to the risk of geopolitical vendor lock-in. BNP Paribas, Société Générale, Crédit Agricole, and AXA are massively deploying AI solutions for fraud detection, credit scoring, algorithmic trading, and risk optimization. AXA and

BNP Paribas are among the first enterprise customers of Mistral AI in France.[1] These deployments rely largely on US cloud infrastructure (AWS for BNP Paribas, Azure for Société Générale). Productivity gains observed in the global financial sector are among the highest (the IMF cites microeconomic gains of 20 to 40 percent on compliance and analysis tasks), as it is a cognitively intensive and high-wage sector—two factors that maximize the return on investment of automation.[2]

The risk specific to the financial sector is twofold. On one hand, European regulations (DORA, AI Act, GDPR) impose localization and auditability requirements that create tension with the dependence on US cloud: French banks must guarantee that their customers' data is not accessible to US authorities (CLOUD Act), while depending on AWS and Azure for computing power. On the other hand, under scenario B, an increase in the cost of accessing cutting-edge AI cloud would directly hit the most compute-intensive applications (risk models, Monte Carlo simulations, training of specialized LLMs). The productivity gap with US banks (JPMorgan, Goldman Sachs, which each invest several billion a year in AI) would widen by an additional 3 to 5 points per year.



6.1.2 Automotive and aerospace industry: compute-intensive and vulnerable

The European automotive industry has already experienced a revealing precedent: the 2022 semiconductor shortage cost the EU auto sector alone approximately 100 billion euros (Chapter IV). AI transforms this sector on three axes: autonomous driving (training perception models, requiring tens of thousands of GPUs), production optimization (digital twins, predictive maintenance), and assisted design (aerodynamic

simulation, virtual crash tests). Stellantis (formerly PSA) signed a 100 million euro partnership with Mistral AI to integrate AI across all its businesses, from transport to logistics.[3]

Aerospace (Airbus, Safran, Thales, Dassault Aviation) presents a similar profile but aggravated by the defense dimension. Complex aerodynamic simulations, fleet predictive maintenance, and the design of autonomous weapons systems are extremely compute-intensive applications. Airbus relies on AWS and Azure for its cloud workloads. Under scenario B, restrictions on advanced GPUs would directly affect simulation capabilities; under scenario D, pressure to repatriate workloads to sovereign infrastructure would create considerable transition costs but would reduce strategic vulnerability.

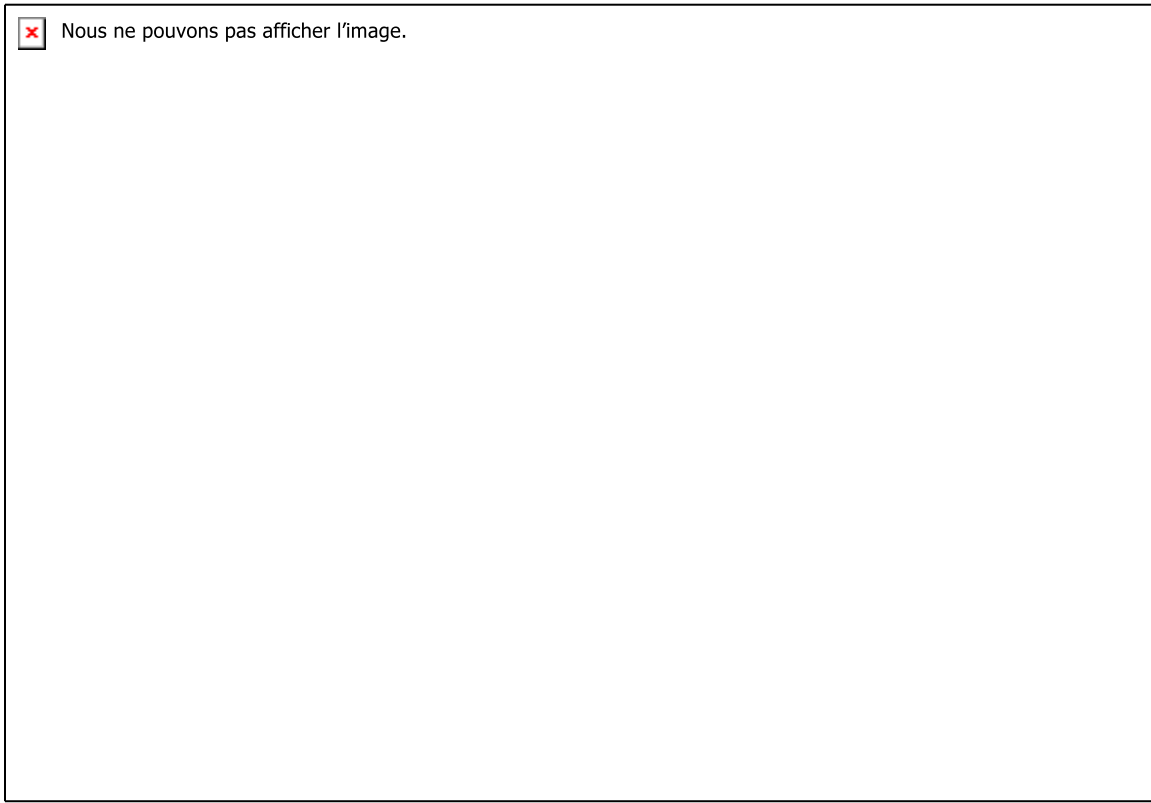


6.1.3 Healthcare and life sciences: data sovereignty issue

The healthcare sector presents a different vulnerability: compute intensity is lower (except for AI-driven drug discovery, which requires massive GPU clusters), but data sensitivity is maximal. The European Health Data Space (EHDS) strictly regulates the processing of medical data. Sanofi, one of the world's leading pharmaceutical groups, has invested 1 billion dollars in AI partnerships (including OpenAI and Owkin, a French startup specialized in AI for clinical research).[4] Novo Nordisk, although Danish, illustrates the European challenge: more than 8.2 billion dollars in R&D in 2024, with increasing use of AI for human body digital twins and accelerating drug discovery.[5]

The impact of protectionism scenarios here is indirect but structural. If access to cutting-edge GPUs is restricted, AI drug discovery projects (which require model training for several weeks on thousands of GPUs) will be slowed down or offshored to the United States. Owkin, founded in Paris, has already opened offices

in New York and could transfer its most compute-intensive workloads there if European costs became prohibitive.



6.1.4 Robotics and manufacturing industry: the energy factor

AI robotics and industrial automation add an additional energy dimension. Training perception and control models for industrial robots is compute-intensive, but it is especially the large-scale deployment of AI-equipped robots (edge computing, embedded inference) that multiplies industrial energy demand. The convergence of data centers + edge compute + robots could add 20 to 30 percent to industrial energy demand by 2030 (Chapter III), an estimate that is still poorly quantified but recognized as a critical sensitivity variable.

France possesses significant players: Exotec (logistics robotics, valued at more than 2 billion euros), Wandercraft (exoskeletons), Aldebaran/SoftBank Robotics (humanoid robots, founded in France). These companies depend on AI hardware (Nvidia GPUs, specialized chips) to train and deploy their systems. Under scenario B, GPU quotas would directly affect the pace of innovation; under scenarios C and D, investment in European Gigafactories would provide the necessary compute, but with a 2 to 3-year delay relative to US competitors.



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6.2 Differentiation by actor type



6.2.1 Large groups: constrained beneficiaries

Large French companies (CAC 40 and SBF 120) are the primary short-term beneficiaries of AI and the most exposed to medium-term dependence. They have the budgets to access the US cloud and the teams to deploy AI, but this adoption reinforces vendor lock-in with each iteration. The cost of migration (switching cost) from a complete cloud ecosystem (AWS to OVHcloud, for example) is estimated at 12-18 months of development and millions of euros in re-architecture, making it economically irrational except under strong regulatory constraint. Under scenario A, they continue to adopt AI via the US cloud, accumulating increasing dependence but benefiting from real productivity gains. Under scenario B, the sudden increase in compute costs puts them in a dilemma: absorb the extra costs (margin compression) or slow down AI projects (loss of competitiveness).



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6.2.2 Industrial SMEs and mid-caps: progressive exclusion

SMEs and mid-sized companies represent the French industrial fabric (4,000 mid-caps, 140,000 SMEs). Their access to cutting-edge AI is already constrained by costs: training a specialized model costs several hundred thousand euros, out of reach for most SMEs without subsidies. McKinsey (December 2025) observes that AI productivity gains are concentrated in large companies, creating an intra-European productivity gap between adopting and non-adopting companies.[6] Under scenario B, rising compute costs widen this gap: SMEs give up on cutting-edge AI and opt for degraded solutions (lightweight open-source models, local inference on limited hardware), progressively losing competitiveness against US SMEs that benefit from domestic compute exempt from tariffs.

EuroHPC AI Factories, designed to give priority access to startups and SMEs, could mitigate this effect (scenarios C and D). But their total capacity (approximately 475,000 GPUs, the order of magnitude of a single US hyperscaler) is insufficient to serve the entire European economic fabric. The gap between the public supply of compute and market demand is structurally deficient.[7]



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6.2.3 French AI startups: between championship and dependence

The French AI startup ecosystem is the most dynamic in Europe. Mistral AI, valued at 11.7 billion euros (September 2025 Series C, led by ASML), is the European champion of generative AI.[8] The company has raised a total of 2.8 billion euros, employs about 700 people, and is developing Mistral Compute, a sovereign cloud platform with 18,000 Nvidia Grace Blackwell GPUs deployed in a 40 MW data center in Essonne, powered by French nuclear energy.[9] In February 2026, Mistral announced a 1.2 billion euro investment for a data center in Sweden (Borlänge), and the acquisition of the startup Koyeb to strengthen Mistral Compute.[10]

The Mistral case illustrates both the potential and the limits of European AI sovereignty. Potential side: the company proves that a European startup can reach global scale, attract massive investments (ASML, Nvidia, Bpifrance, a16z), and build its own compute infrastructure. Limits side: Mistral remains dependent on Nvidia GPUs (no European alternative for AI accelerators), initially used Microsoft Azure and Google Cloud for training, and its investor Microsoft holds a strategic position (conversion of a 15 million EUR investment). The company illustrates application-level sovereignty (models, platforms, services) without hardware-level sovereignty—precisely the junior partner position described in scenario C.

Beyond Mistral, the French ecosystem includes Hugging Face (valued at 4.5 billion dollars, model hosting platform, headquartered in Paris but infrastructure in the US), LightOn (specialized models for enterprise), H Company (formerly Holistic AI), Owkin (AI for healthcare), and Scaleway (French sovereign cloud, first European access to Nvidia Blackwell GPUs). AI funding in Europe grew by 55 percent in the first quarter of 2025, with 12 new unicorns in the first half. France is for the fifth consecutive year the primary European destination for foreign AI investment.[11]



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6.2.4 Public sector and defense: the sovereignty imperative

The French public sector represents a special case. The Ministry of the Armed Forces launched GenIAI.intradef in February 2025, a secure conversational agent based on Mistral AI.[12] The Directorate General of Armaments (DGA) and ANSSI are working on classification frameworks for sovereign AI. For defense and intelligence, dependence on the US cloud is unacceptable regardless of the scenario: classified workloads must be executed on national infrastructure. This captive market provides a guaranteed demand base for sovereign cloud providers (OVHcloud S3ns, Thales S3ns, Scaleway), but its size remains limited compared to the commercial market.



6.3 Second-order effects

6.3.1 Brain drain and capture of AI talent

Compute asymmetry produces a talent attraction effect toward the United States. European AI researchers and engineers are attracted by: (i) US compensation packages (salary + equity), which are structurally superior to European standards; (ii) access to cutting-edge compute, a necessary condition for frontier research; (iii) the ecosystem (proximity to Big Tech, VCs, research community). Martens (Bruegel) notes that personnel cost (salaries + equity) is often the most important component of the cost of developing an AI model, and that US companies compete fiercely to recruit the world's best talent.[13]

This brain drain directly weakens the L(r) factor of European CACI. It is partially compensated by the excellence of French training (ENS, Polytechnique, INRIA, Parisian universities) which form a continuous flow of talent—but a significant proportion leaves for the United States after their PhD. Mistral AI, founded by alumni of DeepMind and Meta, illustrates the possibility of retaining (or repatriating) talent, but it is the exception rather than the rule. Under scenarios C and D, the deployment of Gigafactories and Mistral Compute could create an ecosystem attractive enough to retain more talent.

6.3.2 R&D Offshoring

Compute asymmetry produces pressure to offshore the most calculation-intensive R&D activities. This phenomenon is already observable: the largest AI R&D centers of European companies are often located in the US (Bosch AI in Sunnyvale, SAP AI in Palo Alto, DeepMind in London but owned by Google). Under scenario B, this pressure intensifies: 15 to 20 percent of critical AI projects could be offshored to the United States (notably to areas near Nvidia/TSMC fabs in Arizona, or Virginia data centers). Under scenarios C and D, offshoring is mitigated by the availability of local compute, but does not disappear completely because the FLOP cost gap remains positive in favor of the United States in all scenarios.

6.3.3 Normative fragmentation: the AI Act as a double-edged sword

The European AI Act, in progressive application since 2024, produces ambivalent effects in the context of US protectionism. On one hand, it creates additional compliance costs for European companies (model transparency, copyright management in training data, security audits). Draghi (2024) observed that European data restrictions create high costs and slow down model training. Martens (Bruegel, 2025) notes that the AI Omnibus risks prolonging regulatory uncertainty for at least two more years.[14]

On the other hand, the AI Act creates an entry barrier for US competitors—an unintended protectionist effect that could favor European actors compliant by design (Mistral, whose transparent and auditable models are naturally aligned with the AI Act). The Code of Practice on copyright and training data, published in July 2025, imposes constraints that closed models (OpenAI, Anthropic) have more difficulty satisfying. Under scenarios B and D, where geopolitical mistrust is maximal, the AI Act could become a de facto tool for European preference, analogous to the GDPR which favored the emergence of European solutions in personal data processing.

6.4 The France AI ecosystem: unique assets and structural vulnerabilities

The differentiated analysis of the preceding sections allows for a balance sheet of France's specific assets and vulnerabilities in the context of AI protectionism, presented in Table 15.

On the assets side, France has unique comparative advantages in Europe: a nuclear fleet (65-70 percent of the electricity mix) that produces competitive and low-carbon electricity, a generative AI champion (Mistral AI valued at 11.7 billion EUR) with ASML as an industrial partner, centers of excellence in training (ENS, Polytechnique, INRIA, CNRS/IDRIS, more than 150 startups passed through Jean Zay), 109 billion EUR in private AI commitments announced at the February 2025 Summit, and the top European destination for foreign AI investments (five consecutive years).

On the vulnerabilities side, France remains structurally dependent: absence of an AI hardware champion (no GPU/ASIC design), limited installed compute (about 5 percent of the global total for France alone, US/France ratio of the order of 30:1—consistent with the US/EU(13) raw ratio of 17.6:1 since France represents the majority of installed EU compute), post-doctoral brain drain to the US, slow permitting (24+ months), electricity grid under tension (+10 GW needed according to RTE), US cloud dependence on 70-80 percent of AI workloads, and AI Act compliance surcharges in a context of regulatory uncertainty (omnibus 2+ years).

6.5 Synthesis: France facing three futures

The analysis in this chapter converges toward three possible configurations for France by 2030, corresponding to the trajectories of the scenarios in Chapter V.

Configuration 1: Dependent Consumer (scenarios A and B). France adopts AI via the US cloud, gains productivity in the short term, but accumulates structural dependence that places it in a position of vulnerability to any US hardening. Large groups prosper but are captive; SMEs are progressively excluded from cutting-edge AI; the most promising startups offshore their infrastructure to the United States. Brain drain accelerates. The productivity gap with the United States widens by 5 to 15 cumulative points over five years.

Configuration 2: Energy and Application Hub (scenario C). France exploits its nuclear advantage to become the energy center of gravity for AI in Europe. Mistral Compute and Gigafactories provide competitive local compute for inference and fine-tuning. French companies are sovereign in application but dependent on US hardware. On raw compute, the US/EU ratio drops from 17.6:1 in 2025 toward 8-10:1 in 2030; on CACI Power Mode, the gap closes from 3.46:1 toward 2.0-2.5:1. Brain drain is slowed by the existence of an attractive local ecosystem.

Configuration 3: Pillar of European Sovereignty (scenario D). US protectionism catalyzes an unprecedented mobilization. France, thanks to its nuclear power, its excellent training, and Mistral, becomes the pillar of a European technological sovereignty effort. Massive investment (20 GW dedicated nuclear, RISC-V/DARE, Japan-Korea alliances) creates the conditions for a long-term catch-up, but the transition period (2026-2028) is painful. Execution risk is maximal: each year of delay in infrastructure prolongs vulnerability.

Chapter VII will elaborate on the strategic recommendations corresponding to each of these configurations, distinguishing short-term measures (suitable regardless of the scenario) from structural investments (dependent on the chosen trajectory).

Table 14. French sectoral exposure to AI compute asymmetry.

Source: Author's construction, calibration on the public dashboard (April 2026).

Sector	Compute Intensity	Data Sensitivity	US Cloud Dependence	Scenario B Risk
Finance	High	Very high	70-80%	Lock-in + surcharges: -3 to -5 pts productivity/yr
Auto/Aero	Very high	High	60-70%	Slowed simulations, R&D offshoring
Health/Pharma	High (drug disc.)	Maximal	40-60%	Offshored drug discovery to US
Robotics/Indus.	High	Moderate	50-65%	Slowed innovation + energy surcharge
Defense/Space	Very high	Critical	Variable	Maximal strategic vulnerability

Table 15. Balance of France's assets/vulnerabilities in the context of AI protectionism.

Source: Author's synthesis, calibration on the public dashboard (April 2026).

France's Assets	France's Vulnerabilities
Nuclear energy (65-70% of mix): competitive cost, low carbon, data center deployment	Absence of AI hardware champion (no GPU/ASIC design): Nvidia/AMD dependence
Generative AI champion: Mistral AI (11.7B EUR val., Mistral Compute, ASML partner)	Installed compute: France about 5% of global; raw US/France ratio around 30:1
Excellence in training: ENS, X, INRIA, CNRS/IDRIS (Jean Zay, 150+ startups)	Post-doctoral brain drain to US (salaries + equity + compute)
109B EUR in private AI commitments (February 2025 Summit)	Slow permitting (24+ months), saturated electricity grid (+10 GW RTE need)
1st EU destination for foreign AI investments (5 consecutive years)	US cloud dependence: 70-80% AI workloads on AWS/Azure/GCP
AI Act: Mistral's compliance by design (open-source, auditability)	AI Act: compliance surcharges, regulatory uncertainty (omnibus 2+ years)

Notes

1. Mistral AI (2025), partnership press releases. AXA and BNP Paribas are among the first enterprise adopters of Mistral in France. See NVIDIA Blog (June 2025), 'France Bolsters National AI Strategy With NVIDIA Infrastructure'.
2. IMF (March 2025), op. cit. Financial services are among the sectors with the highest AI exposure (Felten and Eloundou indices), with documented microeconomic gains of 20-40 percent on compliance and risk analysis tasks.
3. Mistral AI / Wikipedia FR (2026). Stellantis (formerly PSA and Fiat-Chrysler) signed a 100 million EUR partnership with Mistral AI in April 2025. CMA CGM also concluded a 100 million EUR partnership.
4. Sanofi (2024-2025), press releases. Owkin, founded in Paris in 2016, specializes in AI for clinical research and drug discovery.
5. McKinsey (January 2026), 'Transforming Europe: Bold Moves to Lift a Continent'. Novo Nordisk: over 8.2 billion USD in R&D in 2024, use of digital twins and AI for drug discovery.

6. McKinsey (December 2025), 'Accelerating Europe's AI Adoption'. The report observes that AI productivity gains are concentrated in 'lighthouse' companies, with gains reaching 40 percent in labor productivity and 50 percent in lead time reduction.
7. Segler Consulting (June 2025), 'Europe's AI Gambit'. Total EU public capacity is estimated at about 57,000 accelerators in 2025, an order of magnitude lower than the infrastructure of a single US hyperscaler.
8. Mistral AI (September 2025), Series C: 1.7 billion EUR raised, 11.7 billion EUR valuation. ASML lead investor (1.3 billion EUR for approx. 11%). Other investors: DST Global, a16z, Bpifrance, General Catalyst, Index Ventures, Lightspeed, Nvidia.
9. Introl Blog (December 2025), 'France's AI Sovereignty Push'. Mistral Compute: 18,000 Grace Blackwell GPUs, 40 MW data center in Essonne (hosted by Scaleway/Eclairion), powered by nuclear energy. Planned launch 2026.
10. Mistral AI / Wikipedia FR (February 2026). Borlänge data center (Sweden): 1.2 billion EUR, planned for 2027, via EcoDataCenter (renewable energy). Acquisition of Koyeb (February 17, 2026) to strengthen Mistral Compute's serverless offering.
11. Dealroom (2025), cited in FinTech Weekly. AI funding in Europe +55% in Q1 2025, 12 new unicorns in H1 2025. France has been the #1 EU destination for foreign AI investments since 2020.
12. Ministry of the Armed Forces (February 2025), launch of GenAI.intradef. Secure conversational agent based on Mistral AI, deployed on sovereign infrastructure.
13. Martens, B. (2024), Working Paper 18/2024, Bruegel, op. cit. AI personnel cost (salaries + equity) is often the most important component of model development, and US companies offer structurally superior packages.
14. Martens, B. (2025), 'The European Union Needs More Than the Digital Omnibus to Make Digital Services Competitive', Bruegel. The AI Omnibus, intended to accelerate regulatory changes, is likely to take at least a year to be adopted, plus another year for complementary guidelines.

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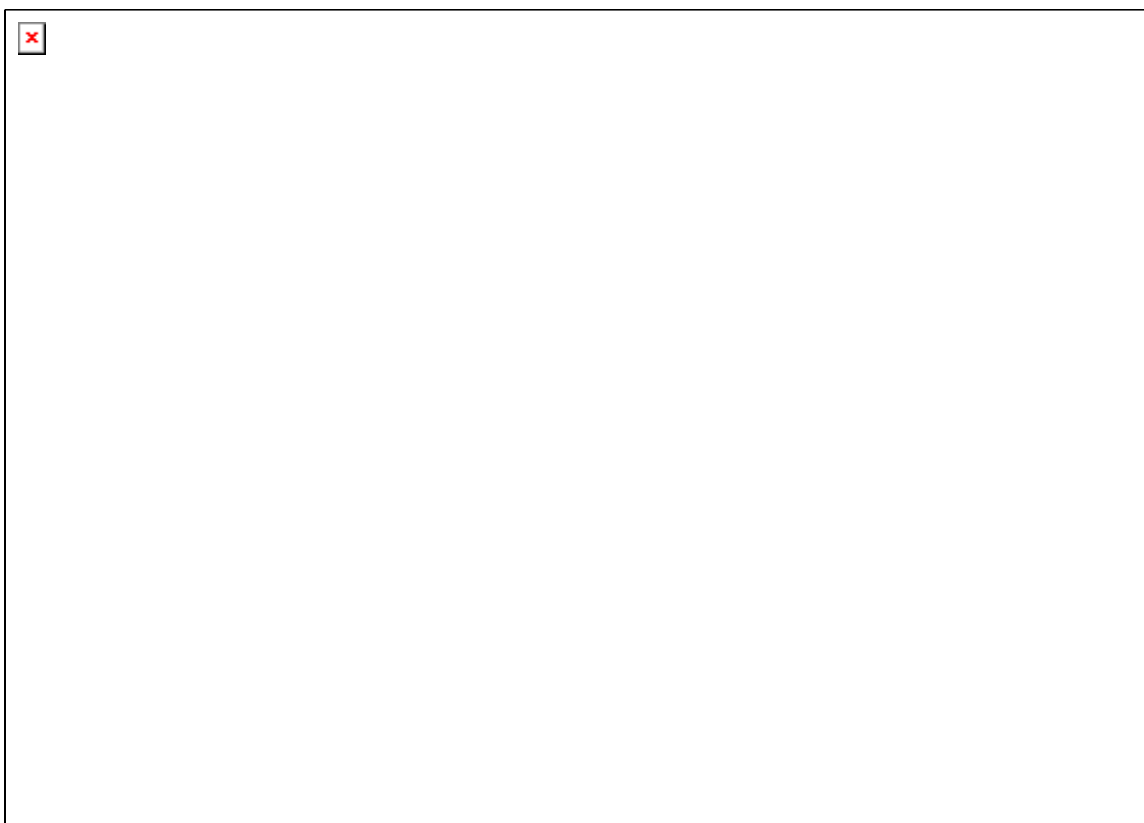
CHAPTER VI BIS - CONSEQUENCES FOR SOUTH AMERICA AND BRAZIL

CHAPTER VI BIS

Consequences for South America and Brazil

The analysis of previous chapters has focused on the US-Europe transatlantic axis. However, the consequences of American AI protectionism extend far beyond the OECD. South America, and Brazil in particular, constitute a revealing case study: simultaneously a dynamic market for AI, a terrain of US-China geopolitical competition, and a region structurally dependent on foreign compute while possessing unique energy assets. This complementary chapter analyzes the specific consequences of the protectionist regime on South America, with an in-depth focus on Brazil.

6bis.1 Structural position: the Global South facing compute asymmetry



6bis.1.1 A rapidly growing but structurally dependent ecosystem

Latin America represents 6.6 percent of global GDP, but only attracts 1.12 percent of global AI investment — a ratio of 5.9 that measures the region's investment deficit.[1] Yet, adoption indicators are remarkably dynamic. The Latin American Artificial Intelligence Index (ILIA 2025), published by ECLAC and the Chilean

CENIA, ranks three countries as pioneers (Chile, Brazil, Uruguay), eight as adopters (including Colombia, Ecuador, Costa Rica), and one-third as explorers with nascent ecosystems.[2] The region represents 14 percent of global visits to AI solutions and ranks third globally for generative AI application downloads.

This adoption dynamic contrasts with a considerable infrastructural deficit. High-income countries host 77 percent of global colocation data center capacity (June 2025), compared to 18 percent for upper-middle-income countries and 5 percent for lower-middle-income countries.[3] High-income countries also concentrate 87 percent of notable AI models, 86 percent of AI startups, and 91 percent of AI venture capital, while representing only 17 percent of the global population. Latin America falls into the intermediate category: a dynamic consumer of AI, but almost entirely dependent on foreign infrastructure (American and, increasingly, Chinese) for compute.

6bis.1.2 Tier 2 classification: compute access caps from the US

Under the January 2025 AI Diffusion Rule (Biden administration), all of Latin America—including Brazil, Mexico, and Chile—is classified as Tier 2, meaning it is subject to quantitative caps on advanced GPU imports. Tier 1 (unlimited access) is reserved for 20 close allies (Australia, Canada, France, Germany, Japan, etc.). Tier 3 (prohibited access) targets China, Russia, and about twenty other countries.[4]

For South America, this Tier 2 classification means that large-scale AI data center projects are capped in terms of importable GPU volume. Initial caps under the AI Diffusion Rule were approximately 50,000 GPUs between 2025 and 2027 for all Tier 2 countries (individually), a volume clearly insufficient to power the announced megaprojects. Brookings observes that even for the most favored Tier 2 countries (Brazil, India), American caps mean that their compute needs will likely not be consistently met.[5] Although the Trump administration rescinded the AI Diffusion Rule in May 2025 without yet publishing a replacement rule, regulatory uncertainty creates a supply risk that weighs on investment decisions. The BIS published America's AI Action Plan in July 2025, which promotes the export of full-stack AI export packages to allies while hardening enforcement toward adversaries.[6]

6bis.2 Brazil: emerging hub between two blocs

6bis.2.1 Brazil's structural assets for AI

Brazil possesses objective competitive advantages for AI infrastructure. Its electricity mix is 83 percent renewable (essentially hydroelectric, supplemented by rapidly growing wind and solar), with a competitive electricity cost of approximately 0.08 USD/kWh. The national interconnected grid (SIN) has surplus capacity, and the country possesses a base of telecommunications submarine cables (notably the Fortaleza hub) offering one of the shortest routes to Europe and Africa.[7]

The Brazilian data center market is the largest in Latin America, representing more than 41 percent of regional investments. Installed IT capacity reached approximately 1 GW by the end of 2025, with 202 active projects and 23 scheduled completions by the end of 2026. The AI data center market specifically is valued at 0.55 billion USD in 2025, projected to reach 1.24 billion by 2030 (annual growth of 17.5 percent).[8] The

Brazilian Data Center Association (ABDC) projects that data center energy demand could reach 13.7 GW by 2035.

The Lula government has taken concrete steps to accelerate development. In September 2025, the Redata provisional measure was adopted, creating a special tax regime that reduces the cost of capital for data centers by 50 percent through exemption from taxes on IT assets, with benefits valid until December 2026. The Brazilian Artificial Intelligence Plan (PBIA) aims to transform the country into a global data center hub, and BNDES (National Development Bank) has launched a dedicated AI and data center fund with initial capital of 500 million to 1 billion reais (93-187 million USD).[9]

6bis.2.2 Megaprojects and US-China duality

Brazil has become the theater of direct competition between American and Chinese investments in AI infrastructure, a duality that reveals the dynamics created by US protectionism.

On the Chinese side, the most spectacular project is the TikTok-ByteDance data center in Pecém (Ceará), announced in December 2025: 200 billion reais (approximately 38 billion USD), in partnership with developer Omnia (Pátria Investments) and renewable energy producer Casa dos Ventos.[10] The project, construction of which is to begin in April 2026, will be powered entirely by wind energy (300 MW initially, expandable to 900 MW or even 1 GW). It constitutes the largest private data center investment ever announced in Brazil and ByteDance's first project in Latin America. The investment comes in a context where TikTok faces threats of blocking in the United States and where China seeks to diversify its compute infrastructure outside its direct geopolitical risk zone.

On the American side, Microsoft has announced a 2.7 billion USD investment over three years in cloud and AI infrastructure in Brazil, as part of its 50 billion USD global commitment to the Global South by 2030.[11] AWS and Google already have cloud regions in Brazil (São Paulo). US hyperscalers control approximately 55 percent of the Brazilian cloud market (AWS, Azure, GCP), a smaller dominance than in Europe (70 percent) but enough to structure dependence.

Other megaprojects illustrate the scale of ambitions. Scala Data Centers announced Eldorado do Sul's AI City (Rio Grande do Sul), with 1,800 MW of capacity and a potential for 5,000 MW by 2033. Elea Data Centers is developing Rio AI City, presented as the largest data center campus in Latin America, with 1.8 GW of capacity by 2027 and 3.2 GW by 2032, with Oracle and Nvidia as technology partners.[12]

6bis.3 Impact of US protectionism on South America

6bis.3.1 The double bind of Tier 2 classification

The Tier 2 position of Brazil and South America creates a strategic double bind. On one hand, GPU import caps limit countries' ability to build sovereign AI infrastructure and realize announced megaprojects. On the other, American restrictions toward China (Tier 3) push ByteDance and other Chinese actors to invest heavily in Latin America as an alternative infrastructure deployment zone. Brazil thus becomes a substitution terrain in the US-China rivalry.

Brookings warns that this situation could push Tier 2 countries to develop supply chains independent of the United States, including stronger technological ties with China.[13] Brazil, whose primary trading partner is China, perfectly illustrates this dynamic. The TikTok-Pecém investment, the largest Chinese technological investment in Latin America, is part of a deepening of China-Brazil ties that predated Trump's reelection but accelerated in the face of American trade policies.

6bis.3.2 Five channels of impact

The impact of American AI protectionism on South America is transmitted through five distinct channels, some of which are specific to the region.

Channel 1 - Direct constraint on AI hardware. Tier 2 caps on cutting-edge GPUs limit construction capacity. Even if the AI Diffusion Rule is rescinded, regulatory uncertainty weighs: projects requiring 10,000 to 100,000 GPUs (like Scala AI City or Elea Rio AI City) depend entirely on American willingness to deliver Nvidia or AMD accelerators. The GAIN AI Act, currently under discussion in the US Congress, proposes to give priority access to advanced semiconductors to American consumers before satisfying international orders — which would further degrade the region's supply.[14]

Channel 2 - Reinforced US cloud dependence. In the absence of sufficient local infrastructure, Brazilian companies resort heavily to the US cloud. The Brazilian financial sector (Itaú, Nubank, Bradesco) is the most digitized in Latin America, with 95 percent of transactions processed digitally at Itaú Unibanco. Brazilian Open Banking connects 800 institutions. This digitization relies largely on US hyperscalers, creating the same dependence pattern as for Europe (Chapter IV), but with less bargaining power.

Channel 3 - US-China technological bifurcation. Brazil is faced with a specific risk: the fragmentation of its AI infrastructure between American and Chinese ecosystems. The Pecém data centers (ByteDance), although powered by Brazilian renewable energy, will likely use Huawei hardware or non-US alternatives if American restrictions prevent the export of Nvidia GPUs to Chinese projects. This hardware duality creates issues of interoperability, data sovereignty (US CLOUD Act versus Chinese data security law), and standardization. Brazil risks becoming a space where two incompatible technological ecosystems coexist, fragmented and each dependent on an external power.

Channel 4 - Amplified brain drain. ILIA 2025 signals a widening of the AI talent gap in Latin America since 2022, associated with an acceleration of the brain drain of specialists.[15] Brazil trains excellent engineers (USP, Unicamp, ITA), but the compensation differential with the United States is even more marked than for Europe. Compute asymmetry aggravates this drain: AI researchers who stay in Brazil do not have access to the compute necessary for frontier research, which reinforces the attractiveness of American laboratories.

Channel 5 - Widened productivity gap. The World Bank and the ILO observe that Latin America suffers from a persistent productivity deficit, largely linked to barriers to innovation and technological adoption.[16] If high-income countries capture AI productivity gains (IMF: TFP gains significantly higher in advanced economies), AI protectionism risks transforming what was an adoption delay (temporary buffer) into a structural barrier (bottleneck). Constrained access to cutting-edge compute prevents Latin American companies from realizing the theoretical productivity gains of AI, widening the gap with the United States.

6bis.4 Specific scenarios for Brazil

By applying the 2x2 scenario matrix from Chapter V, the scenarios for Brazil differ from Europe because Brazil has an additional variable: the possibility of playing the Chinese card as an alternative to the US ecosystem. Table 17 details the four possible trajectories.

Scenario A' (Dual Neutral Hub) is the most likely in the short term. Brazil maintains close trade relations with both blocs and has no interest in aligning exclusively. However, this scenario is inherently unstable: the United States could impose conditions (restrictions on the use of Nvidia GPUs in data centers also hosting Chinese workloads), forcing Brazil to choose. Trump's designation of drug cartels as foreign terrorist organizations has already increased the exposure of companies operating in Brazil and Mexico to US export control rules.[17]

Scenario B' (Secondary Sanctions) represents the maximum risk. If the United States decided to apply secondary sanctions against countries hosting significant Chinese AI infrastructure, Brazil would face an existential dilemma: lose access to the American technological ecosystem (Nvidia, AWS, Azure) or renounce massive Chinese investments. This scenario, although unlikely in the short term, is not hypothetical: the United States has already applied secondary sanctions on Russian and Venezuelan oil, and the Affiliates Rule (suspended until November 2026) extends restrictions to subsidiaries of listed entities.[18]



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6bis.5 South America beyond Brazil

The other economies of South America undergo the same dynamics as Brazil, but with less bargaining weight and fewer structural assets.

Chile, classified as a pioneer by ILIA 2025, benefits from favorable temperatures (reducing cooling energy consumption) and an advanced AI governance ecosystem. Microsoft launched Chile Central in June 2025, its first cloud region in the country, associated with the Transforma Chile program (180,000 people trained, 81,000 jobs created). But Chile remains entirely dependent on American hardware and cloud, without the Chinese alternative that Brazil has.

Mexico, the United States' immediate neighbor, presents a different profile. Nearshoring (relocating production from Asia) has transformed the economic landscape, and data centers benefit from proximity to the US market. However, Mexico is more vulnerable to American protectionism due to USMCA (renewable in

July 2026) and the designation of cartels as terrorist organizations. Its energy infrastructure for data centers is also less competitive than Brazil's.

Colombia, Argentina, and Peru, classified as adopters, face even stronger constraints: fragile electrical infrastructure, limited AI human capital, and near-zero bargaining power with GPU providers. For these countries, American AI protectionism translates primarily into delayed and more expensive access to AI tools, widening the productivity gap not only with the United States but also with Brazil and Chile within the region itself.

6bis.6 Synthesis: a risk of triple fracture

The analysis in this chapter reveals that American AI protectionism produces a risk of triple fracture in South America, specific to the region and distinct from the European scenario.

North-South Fracture. The compute gap between the United States and South America is much more pronounced than the US-Europe gap. While the raw US/EU(13) ratio is 17.6:1 on installed operational compute and the CACI(US)/CACI(EU) ratio is 3.46:1 (Chapter III, April 2026 baseline), an equivalent CACI(US)/CACI(Brazil) ratio would be in the range of 30-50:1 in CACI Power Mode and well over 100:1 in raw compute, reflecting the combination of a deficit in installed computing capacity, more limited AI human capital, and a higher cost of capital (Brazilian Selic at 14.25 percent at the end of 2025). This gap is such that catching up is almost impossible by 2030 without massive external help.

East-West Fracture. The US-China rivalry for AI infrastructure in South America creates a risk of technological fragmentation unparalleled in Europe. Brazil could find itself with two incompatible parallel ecosystems (US cloud versus Chinese infrastructure), each responding to its own geopolitical logic rather than the needs of the local economy. Europe, as a Tier 1 ally, does not face this direct bifurcation on compute, even if the Cloud-Nationality pivot analyzed in Chapter V creates a distinct vulnerability on cloud workloads.

Intra-regional Fracture. Within South America, Brazil attracts most AI investments (41 percent of the LATAM market), followed by Chile and Mexico. 'Explorer' countries (one-third of the region according to ILIA 2025) risk being entirely excluded from the AI economy, reproducing a dependence pattern analogous to that of raw materials. The WEF suggests a regional multi-stakeholder consortium (GAVI model) to pool compute resources and democratize AI access.[19]

For Brazil specifically, the main strategic challenge is to transform its energy assets (83 percent renewable mix) into compute sovereignty, avoiding the US-China competition from fragmenting its ecosystem. The comparison with France is enlightening: both countries possess a distinctive energy asset (nuclear for France, renewable for Brazil), an emerging national technological champion (Mistral for France, fintech/Nubank ecosystem for Brazil), and a regional AI hub ambition. But Brazil faces additional constraints (Tier 2 classification, cost of capital, amplified brain drain) that make its trajectory more uncertain and its vulnerability to protectionism more acute.

Table 16. Major AI data center projects in Brazil (2025-2030).

Source: Author's compilation from Bloomberg, IndustrialInfo, Introl.

Project	Origin	Investment	Capacity	Feature
TikTok Pecém	China	~38B USD	300 MW to 1 GW	100% wind; ByteDance's 1st LATAM project
Scala AI City	Brazil/US	Multi-B USD	1.8 to 5 GW	Largest planned installation in South America
Elea Rio AI City	Brazil/US	Multi-B USD	1.8 to 3.2 GW	Oracle + Nvidia technology partners
Microsoft Azure	US	2.7B USD (3 yrs)	N/A	Cloud + AI + ConectaAI program
AWS São Paulo	US	N/A	Multiple AZ	Active cloud region since 2011

Table 17. Brazil-specific scenarios facing US AI protectionism 2026-2030.

Source: Author's construction, calibration on April 2026 baseline (raw US/EU(13) 17.6:1, CACI Power Mode 3.46:1).

Brazil Scenario	Probability	Positive Consequences	Risks
A': Dual US-China Neutral Hub	35-45%	Inflow of investments from both blocs; renewable energy as an asset; supplier diversification	Ecosystem fragmentation; US pressure to limit Chinese access; data vulnerability
B': Secondary Sanctions	15-20%	Acceleration of substitutive Chinese investment	Loss of US ecosystem access (Nvidia, cloud); partial technological isolation
C': Pro-US Alignment	20-25%	Negotiated Tier 1 access; unlimited GPUs; increased Microsoft/AWS investment	Total US dependence; loss of Chinese investment (Pecém threatened)
D': Regional LATAM Sovereignty	10-15%	Regional compute consortium; dependence reduction; energy pooling	Limited execution capacity; insufficient capital; 5-10 year delay

Notes

1. ECLAC/CENIA (October 2025), Latin American Artificial Intelligence Index (ILIA 2025). Latin America represents 6.6% of global GDP and 1.12% of global AI investment.
2. ECLAC/CENIA (2025), *ibid*. Three categories: pioneers (Chile, Brazil, Uruguay, over 60 pts), adopters (Colombia, Ecuador, Costa Rica, Dominican Republic), explorers (nascent ecosystems).
3. World Bank (November 2025), Digital Progress and Trends Report 2025: Strengthening AI Foundations. High-income countries: 77% colocation DC capacity, 87% notable AI models, 86% AI startups, 91% AI VC.
4. BIS (January 2025), Framework for Artificial Intelligence Diffusion. Tier 1: 20 ally countries (unlimited access). Tier 2: ~140 countries including the whole American continent outside US/Canada (quantitative caps). Tier 3: ~20 countries (prohibited access). Rule rescinded in May 2025 by Trump administration, replacement pending.
5. Brookings (2025), 'Trump's AI export controls and the AI Diffusion Rule'. Initial caps: ~50,000 GPUs per Tier 2 country over 2025-2027.
6. BIS (July 2025), America's AI Action Plan. Promoting full-stack AI export packages to allies, hardening toward adversaries.
7. ABDC (2025), Brazilian Data Center Association. Brazilian electricity mix: 83% renewable. Fortaleza hub: submarine cable connections to Europe and Africa.

8. ABDC (2025); Mordor Intelligence (2026). Brazil AI DC market: 0.55B USD (2025) to 1.24B USD (2030), CAGR 17.5%.
9. BNDES (2025); Redata provisional measure (September 2025): 50% reduction in DC capital cost via IT tax exemption, valid until Dec 2026.
10. Bloomberg (December 2025), 'ByteDance Plans 200 Billion-Real Brazilian Data Center'. Partnership Omnia (Pátria Investments) + Casa dos Ventos. 300 MW initial, expandable to 1 GW.
11. Microsoft (2025), 50B USD commitment for Global South by 2030. 2.7B USD specifically for Brazil over 3 years.
12. Scala Data Centers (2025); Elea Data Centers (2025). Rio AI City: Oracle + Nvidia partnership.
13. Brookings (January 2025), op. cit. Tier 2 countries risk developing supply chains independent of US, including reinforced technological ties with China.
14. GAIN AI Act (2025), discussion in US Congress. Priority access to advanced semiconductors for US consumers before international orders.
15. ILIA 2025 (ECLAC/CENIA), AI talent section. Acceleration of specialist brain drain since 2022.
16. World Bank and ILO (2024-2025): persistent productivity deficit in Latin America. IMF (March 2025): AI TFP gains significantly higher in advanced economies.
17. Designation of drug cartels as foreign terrorist organizations (Trump, 2025). Increased exposure of companies operating in Brazil and Mexico to US export control rules.
18. BIS, Suspension of the Affiliates Rule for One Year (November 10, 2025). Affiliates Rule extends restrictions to subsidiaries of listed entities.
19. WEF (2025), regional multi-stakeholder consortium proposal (GAVI model) to pool compute resources in Latin America.

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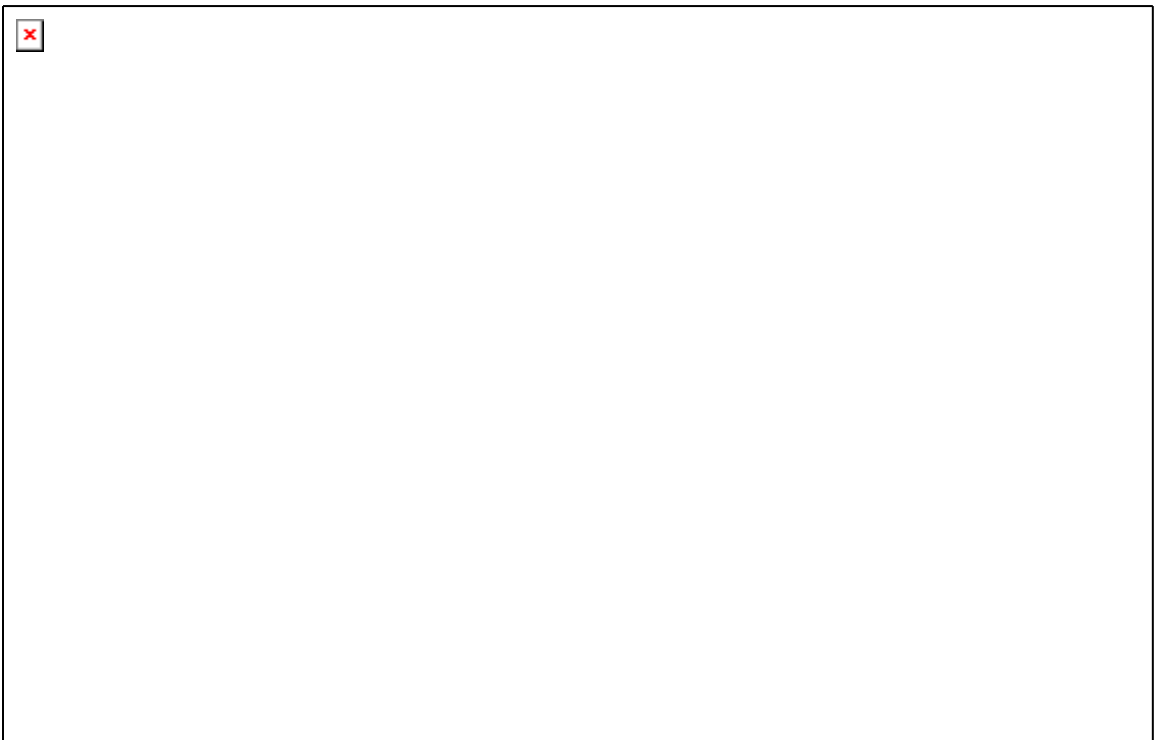
CHAPTER VI TER - CONSEQUENCES FOR ASIA AND CHINA

CHAPTER VI TER

Consequences for Asia and China

Asia is the main target and the primary victim of American AI protectionism. This chapter analyzes the consequences of the containment regime on China, the pressure on the 'Tier 1' allies (Japan, South Korea, Taiwan), and the emergence of Southeast Asia as a secondary compute hub. The analysis highlights the transition from a logic of interdependence to a logic of total decoupling.

6ter.1 China: resilience under maximum containment



6ter.1.1 The impact of total hardware decoupling

China faces the most severe restrictions in history. Since the October 2022 and 2023 rules, supplemented by the June 2025 'Superchips' restrictions, Chinese access to advanced AI accelerators (Nvidia H100, B200) is de facto prohibited. In April 2026, the US/China CACI Power Mode ratio is 2.14:1, a significant gap that measures the effectiveness of the containment (Chapter III).

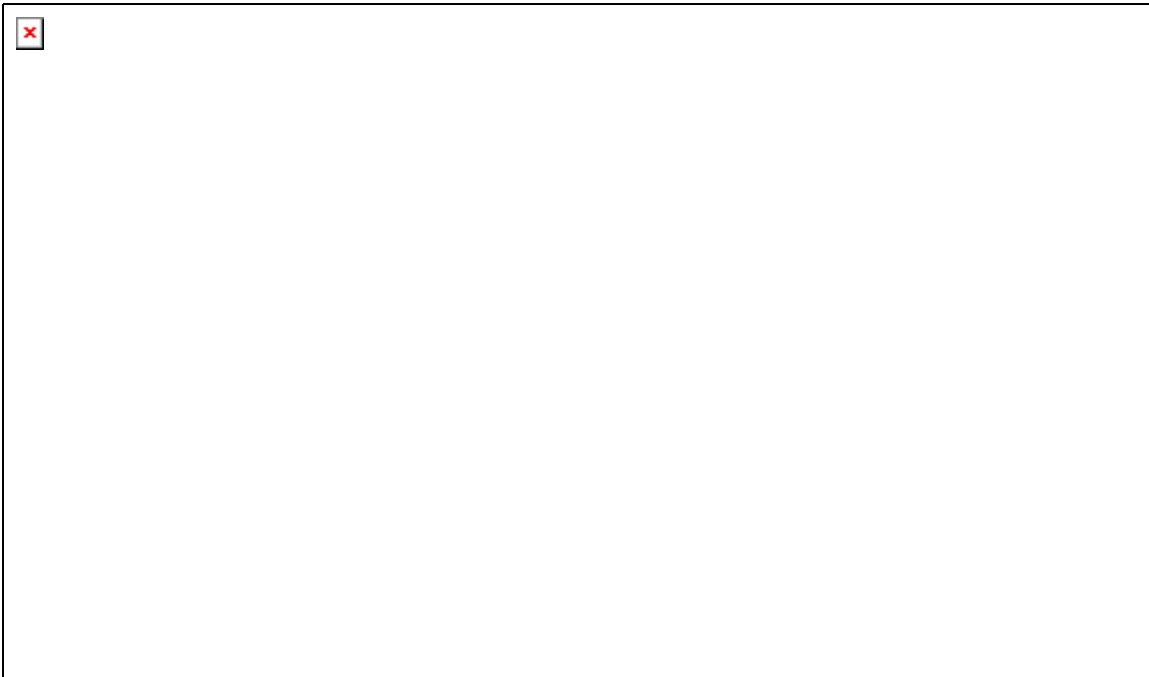
This asymmetry forces China into a strategy of hardware substitution. Huawei (Ascend 910C), Biren, and Moore Threads are developing national accelerators which, although 1 to 2 generations behind Nvidia, allow

for the training of Large Language Models (LLMs). ByteDance has notably placed an order for 100,000 Ascend 910B chips in 2024 to compensate for the H100 shortage.[1] However, the yield of SMIC's 7nm and 5nm nodes remains limited by the lack of EUV lithography (ASML export prohibition), creating a structural ceiling for Chinese hardware sovereignty.

6ter.1.2 Reorientation toward 'Global South Compute'

In response to domestic containment, China is deploying an international compute strategy. The ByteDance - Pecém project in Brazil (38 billion USD, Chapter VI bis) and investments in Malaysia (Johor hub) illustrate this desire to build 'Geographic Redundancy' for Chinese compute. By installing data centers in Tier 2 countries (Malaysia, Brazil, UAE), Chinese actors seek to bypass direct export controls, even if the US 'Affiliates Rule' (November 2025) seeks to close this loophole by extending restrictions to foreign subsidiaries of listed entities.[2]

6ter.2 The Tier 1 Allies: Japan, South Korea, Taiwan



6ter.2.1 Taiwan: the geopolitical fab under pressure

Taiwan (TSMC) produces 92% of the world's most advanced semiconductors. Under US protectionism, Taiwan is under dual pressure: (i) the 'hollowing out' of its production toward the US (Arizona Fabs) to satisfy American resilience requirements, and (ii) the hardening of export controls toward China, its primary trading partner. The 'Geographic Diversity' required by the US (Chapter IV) creates a risk of industrial dilution for Taiwan, while increasing the cost of chips for its domestic ecosystem.



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6ter.2.2 Japan and South Korea: the RISC-V and alternative foundry bet

Japan (Rapidus) and South Korea (Samsung) are investing massively to break the TSMC/Nvidia duopoly. Japan has launched the 'LSTC' (Leading-edge Semiconductor Technology Center) to develop 2nm chips by 2027. South Korea, through the 'AI Semiconductor Strategy', aims to capture 80% of the AI memory market (HBM3e/4) by 2030.[3] For these allies, US protectionism is a double-edged sword: it offers protected access to the US market (Tier 1) but imposes alignment on export controls that damages their relations with China.



6ter.3 Southeast Asia: the new 'Middle Ground'

Malaysia (Johor), Vietnam, and Indonesia are emerging as the new preferred destinations for data centers. Malaysia attracted more than 15 billion USD in data center investments in 2024-2025 (Microsoft, Google, ByteDance, Nvidia/YTL).[4] These countries take advantage of their Tier 2 status to host both American and Chinese infrastructure, creating a 'Compute Non-Alignment' similar to Brazil's, but with much greater proximity to the Chinese supply chain.

6ter.4 Synthesis: toward a fragmented Asian compute

Asia is fracturing into three distinct zones: a 'Sovereign Chinese Zone' (hardware substitution, massive state investment), a 'Tier 1 Pro-US Zone' (high-end foundry, total alignment), and a 'Tier 2 Emerging Zone' (arbitrage hub, US-China duality). This fragmentation increases the complexity of global supply chains and significantly raises the marginal cost of compute for the entire region.

Table 18. Comparison of AI hardware capacities in Asia (April 2026).

Source: Author's construction, calibration on the public dashboard.

Country/Region	US Status	Primary Hardware	CACI Power Mode (est.)	Strategic Constraint
China	Tier 3	Huawei Ascend, Biren	0.47 (relative to US)	SMIC yield (7nm ceiling)
Taiwan	Tier 1	TSMC (Nvidia/AMD)	0.85 (relative to US)	Resilience-driven offshoring
Japan	Tier 1	Rapidus/Nvidia	0.62 (relative to US)	Energy cost + 2nm execution risk
Malaysia	Tier 2	Nvidia (H100/H200)	0.15 (relative to US)	US import caps + energy grid

Notes

- 1. Reuters (2024), 'ByteDance orders 100,000 Huawei AI chips'.
- 2. BIS (November 10, 2025), 'Implementation of the Affiliates Rule'.
- 3. South Korea Ministry of Science and ICT (2025), 'AI Semiconductor Strategy 2030'.
- 4. MIDA (2025), 'Malaysia Data Center Investment Report'.

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CHAPTER VI QUATER - CONSEQUENCES FOR AFRICA

CHAPTER VI QUATER

Consequences for Africa

Africa represents the next frontier of the AI economy. While the region currently has the lowest compute density in the world, its demographic potential and renewable energy assets make it a strategic terrain for the next decade. This chapter analyzes the impact of US protectionism on the African 'Compute Leapfrog', the infrastructure competition between the US and China, and the risk of a new digital divide.

6quater.1 The African 'Compute Leapfrog' challenge



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6quater.1.1 Infrastructure deficit and potential

Africa represents less than 1% of global colocation data center capacity in 2025.[1] However, the demand for local compute is exploding, driven by the digitization of financial services (M-Pesa, Flutterwave) and the need for model localization (African languages, specific socio-economic contexts). South Africa (Cape Town, Johannesburg), Nigeria (Lagos), and Kenya (Nairobi) are the primary hubs.

American AI protectionism (Tier 2 status for the entire continent) creates an entry barrier: African startups and governments have access to the US cloud, but building local sovereign infrastructure is hampered by GPU import caps and the high cost of capital. This reinforces a 'Cloud Dependency' that captures value toward US hyperscalers.



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6quater.1.2 Renewable energy: Africa's comparative advantage

Like Brazil, Africa possesses massive renewable energy potential (solar in the Sahel, hydroelectric in Central Africa, wind in the Maghreb). Data center projects like the 'Green Africa Compute' initiative seek to use this energy to host sustainable AI infrastructure. However, the lack of transmission grids and the instability of the business climate remain major obstacles to attracting the necessary multi-billion dollar investments.



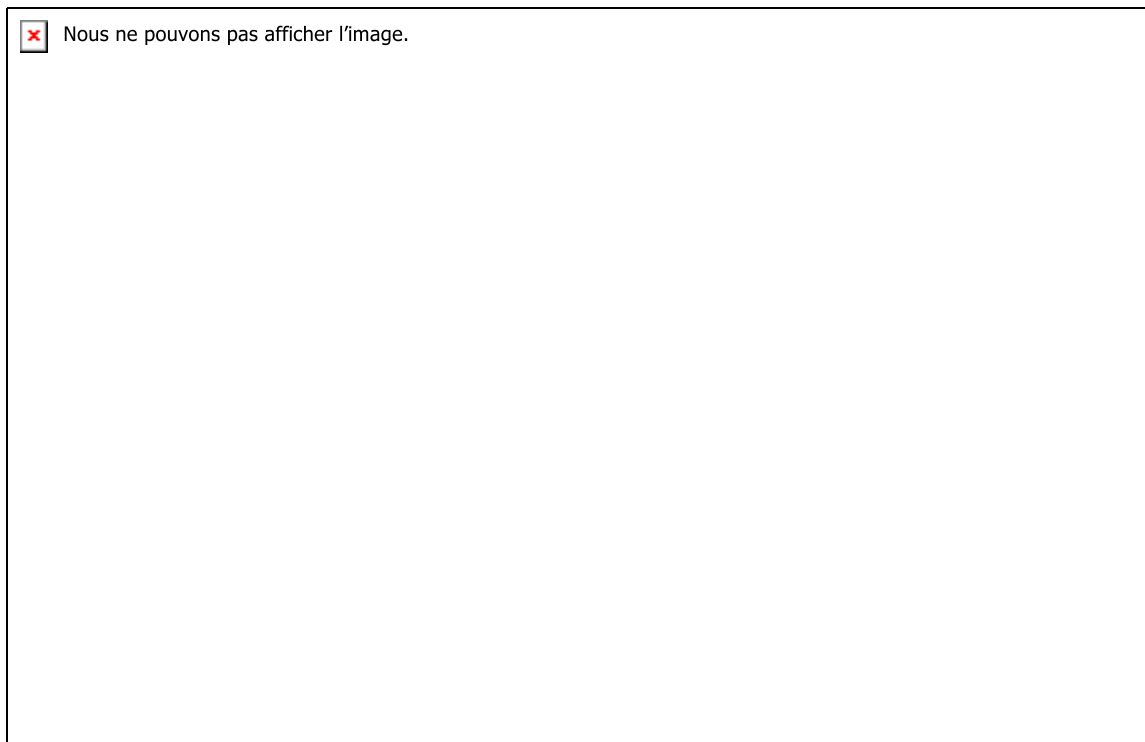
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6quater.2 US-China Rivalry in Africa



6quater.2.1 China's 'Digital Silk Road'

China is the primary provider of telecommunications infrastructure in Africa (Huawei, ZTE). Through the 'Digital Silk Road', Beijing offers integrated packages including connectivity, data centers, and AI solutions. In April 2026, Huawei is the leader in private cloud infrastructure in more than 20 African countries.[2] For many governments, the Chinese offer is more attractive because it is less restrictive than the American Tier 2 status and often accompanied by preferential financing.



6quater.2.2 The US response: the 'Digital Africa' initiative

The United States seeks to counter Chinese influence through the 'Digital Africa' initiative (part of the PGI), promoting American standards of data governance and security. Microsoft and Google have announced 'AI Centers of Excellence' in Nairobi and Accra.[3] However, US protectionism, by limiting hardware access, paradoxically pushes African actors to accept Chinese hardware, even if they use American software/models.



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6quater.3 Synthesis: toward an 'Agentic' or 'Passive' Africa?

The strategic challenge for Africa is to avoid a 'Passive AI' scenario, where the continent remains a simple consumer of American or Chinese models and a provider of training data (task labeling). An 'Agentic AI' scenario requires the development of local compute hubs, the training of local talent (L(r) factor), and the creation of a 'Sovereign African Cloud' to protect data and culture. US protectionism makes this trajectory more difficult but could catalyze regional cooperation within the African Union.

Table 19. Emerging AI hubs in Africa (April 2026).

Source: Author's compilation from African Digital Economy Report.

Country	Primary Hub	Major Actors	Compute Asset	Main Risk
South Africa	Cape Town	AWS, Microsoft, Teraco	Advanced electrical grid (but unstable)	Brain drain to Europe/US
Nigeria	Lagos	MainOne, Medallion, Google	Massive market + young talent	Energy cost + currency volatility
Kenya	Nairobi	Microsoft, Google, Safaricom	Connectivity + mobile fintech leader	Tier 2 import caps
Egypt	Cairo	Orange, Telecom Egypt	Strategic location (cables) + solar	Regulatory complexity

Notes

1. Xalam Analytics (2025), 'The State of African Data Centers'.

2. International Institute for Strategic Studies (IISS, 2025), 'China's Digital Silk Road in Africa'.
3. USAID (2025), 'Digital Africa Initiative Progress Report'.

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CHAPTER VII - STRATEGIC RECOMMENDATIONS

CHAPTER VII

Strategic Recommendations for France and Europe

The previous chapters have established that American AI protectionism creates a measurable structural competitive advantage (US/EU CACI Power Mode of 3.46:1 on the April 2026 snapshot, operational installed compute raw ratio of 17.6:1), accelerated by the 2026 Trump tariffs and the concentration of compute in the United States (76.9% of global operational AI compute, 660-690 billion USD in annual capex from hyperscalers alone). This chapter formulates strategic recommendations articulated across three time horizons and five structuring axes, leveraging France's specific comparative advantages (nuclear, Mistral, regulation) and existing European instruments (AI Continent Action Plan, Chips Act, InvestAI).

7.1 Axis 1 - Compute infrastructure: closing the gap

7.1.1 Short term (2026-2027): accelerating AI Factories

The starting point is the infrastructure gap. The EU has approximately 35 GW of IT data center capacity compared to 53.7 GW in the United States and 19.6 GW in China. In strict AI compute, the April 2026 public dashboard establishes an EU(13) F_{total} share of 3.3% versus 76.9% for the United States, representing a raw operational ratio of 17.6:1 and a CACI Power Mode ratio (geometric formula $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$) of 3.46:1. Three immediate measures are required.

First, accelerate the commissioning of the 13 European AI Factories already created across 17 Member States (AI Continent Action Plan, April 2025), with a target of full operationality by the end of 2027 instead of the 2028-2029 horizon currently envisioned.[1] Second, implement the Special Compute Zones proposed by the Centre for Future Generations, which are derogatory zones (accelerated permits, reduced taxation, priority network connection) for AI data centers of national importance.[2] France has already initiated this process with legislation envisioned to designate data centers as projects of major national interest. Third, secure long-term GPU contracts with Nvidia, AMD, and Intel via multilateral framework agreements (EU - Nvidia, EU-AMD) guaranteeing a minimum annual delivery volume.

A key nuance from the Phys/Sov analysis of Chapter I (Fig 1.8): on physically installed compute, the EU is already largely sovereign (99.2% of EU F_{total} is held by EU operators). The window of vulnerability is therefore not so much on installed F as on the cloud workload layer (the compute actually used by EU companies, mostly hosted on AWS/Azure/GCP). The following recommendations primarily target this operational layer.



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7.1.2 Medium term (2027-2029): AI Gigafactories and sovereign cloud

The InvestAI program provides for 200 billion EUR (50 billion public, 150 billion private), including 20 billion for five AI Gigafactories that will allow for the creation of sovereign frontier models.[3] This program must be calibrated against the American benchmark: the 660-690 billion USD in 2026 capex from the five US hyperscalers alone represents more than three times the European envelope over five years. The investment gap is structural and will not be closed by public funds alone.[4]

France has a distinctive advantage in this competition. The MGX-Bpifrance-Mistral-Nvidia AI campus, announced at the Choose France Summit 2025, plans for 1.4 GW of compute power powered by nuclear energy, with exascale capabilities operational by 2028.[5] Mistral Compute, launched with 18,000 Nvidia Grace Blackwell superchips in a 40 MW data center in Essonne, constitutes the first credible European offer of frontier compute without exposure to the CLOUD Act. Mistral's capex of 1 billion EUR for 2026, supplemented by the Borlänge data center (Sweden, 1.2 billion EUR, green energy, opening 2027), shows that a European champion can build an alternative infrastructure.[6]

Sovereign cloud constitutes the necessary complement. The ANSSI's SecNumCloud 3.2 certification and the S3NS joint venture (Thales-Google Cloud, SecNumCloud certified December 2025), the Bleu joint venture (Orange-Capgemini-Microsoft, milestone 1 reached November 2025), and the AWS European Sovereign Cloud (launched January 2026, separate German GmbH) create a gradually sovereign ecosystem.[7] The goal should be to reach 30-40% of sensitive AI workloads hosted on certified sovereign cloud by 2029. It is precisely the increase in the F_{sov} factor (Chap V section 5.9.2) that protects against the hypothetical activation of US Cloud Sovereignty Mandates in 2028.



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7.2 Axis 2 - Energy: transforming the nuclear asset into a compute advantage

7.2.1 The French energy advantage

France has a unique energy advantage in Europe: 70% low-carbon nuclear electricity, a fleet of 56 reactors (plus Flamanville 3 at full power), competitive electricity costs, and a robust transport infrastructure. On the April 2026 snapshot of the public dashboard, the France/USA PPP-adjusted cost stands at 115 versus 85 USD/MWh, a 1.35x ratio much more favorable than the EU average (135 USD/MWh, 1.59x ratio). EDF has identified four industrial sites totaling 2 GW (expandable to six sites by 2026), with direct connection to the grid, reducing connection delays.[8] EDF's Nuclear for AI initiative plans for 250 MW connected to AI chips by the end of 2026, creating a new off-take market for nuclear power.

This advantage is explicitly recognized by international investors. Investments announced at the February 2025 AI Action Summit total 109 billion EUR, including Brookfield/Data4 (20 billion), UAE (30-50 billion), and Fluidstack (10 billion for a 1 GW supercomputer powered by nuclear, operational 2026).[9] France is the only European country capable of simultaneously offering abundant low-carbon electricity, baseload grid stability, and tariff competitiveness for AI data centers—a triptych that neither Germany (nuclear exit), nor the Netherlands (grid constraints), nor Ireland (energy saturation) can reproduce.



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7.2.2 Energy recommendations

First, accelerate the EPR 2 program. The six announced EPR 2 reactors (Penly, Bugey, 9,900 MW, construction from 2027) must be explicitly integrated into data center energy planning. The addition of eight additional optional reactors should be confirmed before 2028, to anticipate 2032-2035 demand.[10]

Second, support SMRs (Small Modular Reactors). The France 2030 program allocates 1 billion EUR to SMRs. NUWARD (EDF subsidiary, 340 MWe) remains the most advanced project. Three startups (Newcleo, Stellaria, Jimmy Energy) filed applications with the ASN in late 2025-early 2026. The goal should be the first commercial SMR dedicated to a data center by 2033-2035, with a pilot connected to an AI campus. However, uncertainty over SMR commercialization timelines dictates that it should not be the sole strategy.[11]

Third, plan AI-grid energy integration. RTE projects an additional 10 GW need for data centers by 2030 in France. Integrating AI demand forecasts into national grid planning (in line with the McKinsey recommendation to align AI growth with sustainable energy expansion) is indispensable to avoid bottlenecks.[12]

7.3 Axis 3 - Technological alliances and supply chain diversification

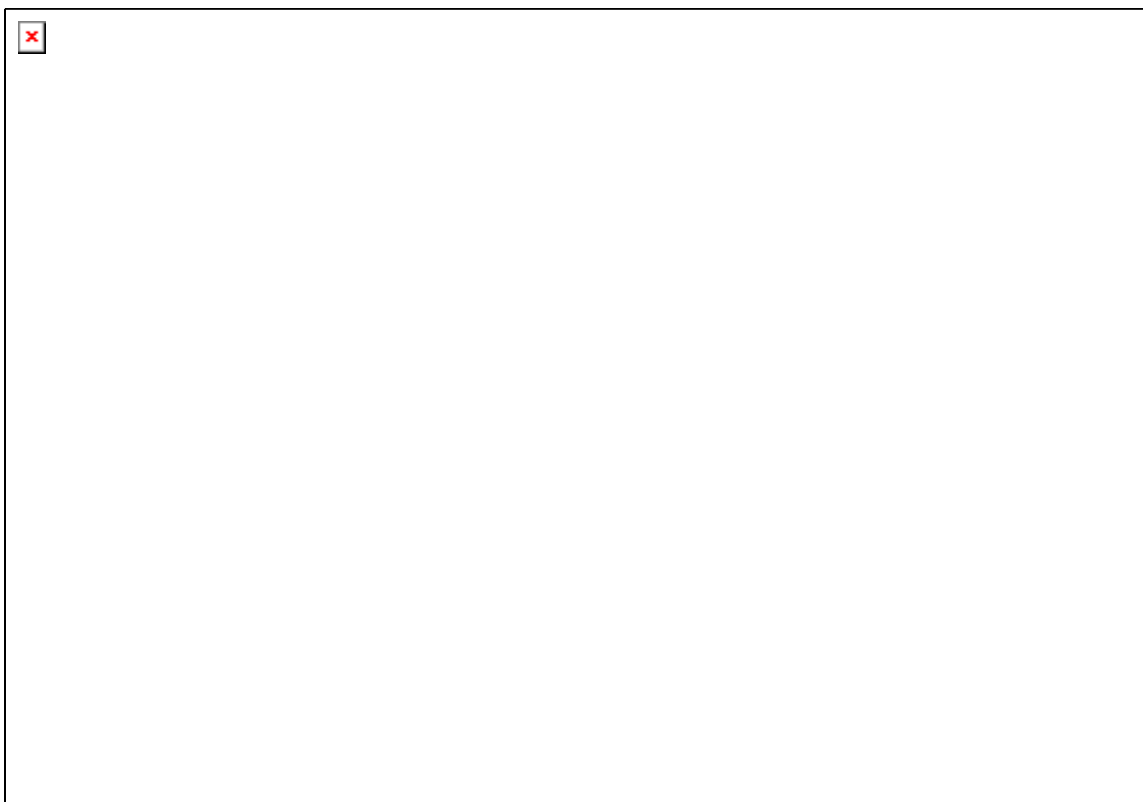
7.3.1 Consolidating asymmetric industrial partnerships

The analysis of Chapters VI, VI bis, VI ter, and VI quater reveals that France and Europe do not intend to reproduce the entire AI value chain (which is unrealistic by 2030), but must build targeted strategic alliances that reduce the most critical dependencies.

ASML-Mistral Alliance. ASML's 1.3 billion EUR investment in Mistral AI (September 2025, ASML becoming the primary shareholder at 11%) is the most significant European partnership, linking the global leader in lithography (a critical segment where Europe dominates) to the European AI champion.[13] This type of vertical coupling of European hardware plus European AI should be systematized.

TSMC-Europe Partnership. The TSMC Dresden plant (10 billion EUR, production started 2027) manufactures chips on 28/16/12 nm nodes—insufficient for frontier AI GPUs but critical for automotive and industrial IoT. Negotiating a second TSMC investment in Europe on more advanced nodes (7/5 nm) should be a diplomatic priority.

Japan-EU and Korea-EU Alliances. Japan and Korea control critical segments of the value chain that the United States cannot substitute (SK hynix HBM memory, Tokyo Electron and Shin-Etsu equipment and materials). Bilateral EU-Japan and EU-Korea agreements on the security of AI component supply, structured outside the US-Japan-Korea trilateral framework, would strengthen European autonomy. Chapter VI ter documented that Japanese investment in the United States (550 billion USD) is unilateral—a cross EU-Japan investment would balance this dynamic.



7.3.2 Reducing exposure to protectionist risk

The Biden-Trump experience shows that export controls and tariffs can be extended rapidly and unpredictably. Three risk reduction measures are required.

Constitution of strategic GPU reserves. On the model of strategic oil reserves (90 days), constitute a national/European stock of AI accelerators covering 6 to 12 months of projected needs. This stock would serve as a buffer against any activation of the Affiliates Rule (suspended until November 2026) or extended Section 232 tariffs.

Hardware provider diversification. Accelerate the evaluation and deployment of alternatives to Nvidia GPUs: AMD MI300X/MI350X, Intel Gaudi 3, Graphcore (UK), and eventually SiPearl (European, Rhea processor for supercomputers). Finance via the European Chips Act a multi-vendor AI accelerator qualification program. The DARE project (European RISC-V, EuroHPC JU) is the 2030-2032 horizon, but intermediate alternatives are available immediately.

Anti-weaponization clauses in trade agreements. Integrate into the future EU-US trade agreement clauses preventing the unilateral use of export controls as a commercial competitiveness instrument, on the model of WTO non-discrimination clauses.

7.4 Axis 4 - Regulation as a competitive advantage

7.4.1 From the AI Act to the Apply AI Strategy

Mistral CEO Arthur Mensch summarized the European paradox: you cannot regulate your way to compute supremacy.[14] The AI Act, in progressive application since 2024, imposes obligations (transparency, risk assessments, compliance) that constitute both a burden for European companies and a differentiation advantage in global markets. The Apply AI Strategy (2025) complements the AI Act by adopting an AI-first approach for the public sector and promoting buy European, particularly for open-source solutions.[15]

The recommendation is to transform regulation into an offensive rather than defensive lever. Three concrete measures can contribute to this.

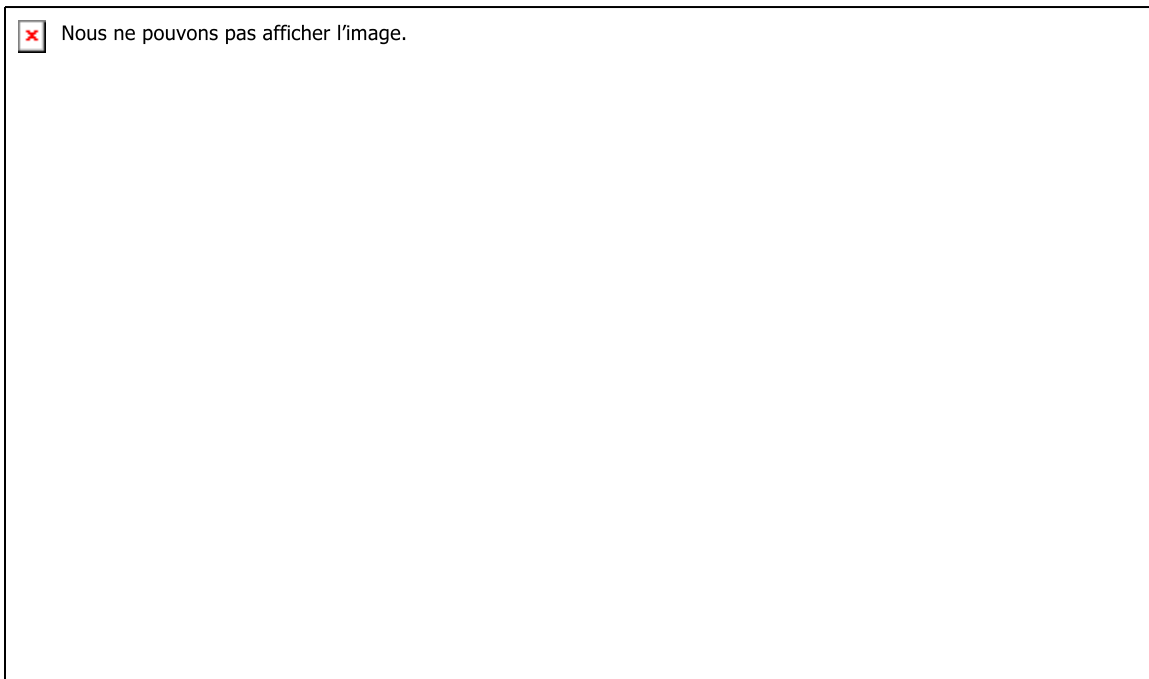
- (a) Require that AI Factories and AI Gigafactories funded by InvestAI prioritize the use of European models (Mistral, Aleph Alpha, etc.) and certified clouds (SecNumCloud, high-level EUCS).
- (b) Exploit the Brussels Effect: global companies complying with the AI Act to access the European market (450 million consumers) de facto adopt European standards, creating a normative advantage. Accelerate mutual recognition agreements with Japan, Brazil, and India.
- (c) Create a European CLOUD Act Shield: blocking legislation (on the model of the 1996 EU blocking regulation) preventing European companies from complying with extraterritorial US access requests without authorization from the competent national authority. This measure is the legal instrument that gives substance to the SOV-3 Cloud Sovereignty Framework published by the Commission in October 2025 (Chap V section 5.10.4).

7.4.2 Regulation of compute as a strategic asset

Comparative analysis (Chapters V and VI ter) shows that frontier compute is now treated by the United States, China, Japan, India, and the Gulf States as a national strategic asset on par with energy or critical raw materials. Europe must formalize this recognition. Gartner predicts that countries pursuing independent AI

stacks will need to invest at least 1% of GDP in infrastructure by 2029.[16] For France, this would represent approximately 28 billion EUR annually, an order of magnitude consistent with the 109 billion in investments announced in 2025 (of which a significant portion comes from foreign capital).

The UAE case documented in Chapter I (Fig 1.8) constitutes a warning: 99.6% of Emirati F_{total} is held by US-side operators, dropping the sovereign CACI from 55.7 to 6.0. An investment attraction policy indifferent to the legal nationality of operators would reproduce this pattern in Europe. European compute regulation must provide for minimum domestic ownership thresholds (for example, 50% minimum of operators under EU jurisdiction for sites over 100 MW), with a mechanism analogous to the Investment Screening Mechanism already applied to strategic investments.



7.5 Axis 5 - Talent and human capital

Infrastructure without talent produces nothing. Europe is losing AI researchers to American laboratories (salaries, access to frontier compute, scale of projects). Two complementary measures are required.

First, European AI fellowships and talent visas (McKinsey recommendation: launch before end 2026) to attract world-class researchers.[17] France has an advantage with the Mistral/LightOn/Hugging Face ecosystem and the grandes écoles (Polytechnique, ENS, CentraleSupélec), but must match the salaries offered by GAFAM (average x2 to x4 gap for senior AI profiles).

Second, guarantee European researchers access to compute equivalent to that of American laboratories. The deployment of 500,000 GPUs via Fluidstack (operational 2026), the 18,000 Mistral Compute superchips, and the EuroHPC AI Factories constitute the beginning of a response. The goal is that no European researcher leaves the continent for compute access reasons by 2028.

7.6 Synthesis: temporal matrix of recommendations

Table 23 below summarizes the recommendations by crossing three horizons (2026–2027, 2027–2029, 2029–2032) with the Compute, Energy, and Alliances axes. The Regulation and Talent axes are deployed transversally across the three horizons and are not detailed line by line in the matrix.

7.7 Success conditions and limits

Several conditions will determine the effectiveness of these recommendations.

Condition 1: Mistral's competitiveness. The entire French AI sovereignty strategy relies in part on Mistral's ability to maintain competitive performance against OpenAI, Anthropic, and Google DeepMind. If the capability gap widens, French infrastructure will serve compliance needs (sovereign hosting of US models) rather than true technological sovereignty.[18] The 1.7 billion EUR fundraising (11.7 billion valuation) and the establishment of Mistral Compute are positive signals, but the scale of competition (OpenAI: 20 billion USD in 2025 recurring revenue) remains disproportionate.

Condition 2: industrial execution. European AI infrastructure programs have historically suffered from delays (EuroHPC, Chips Act). The 13 AI Factories must be operational, not just announced. The experience of Japan (Rapidus 2 nm program) and India (gap between 200+ billion announcements and 1.4 GW installed capacity) illustrate the risks of misalignment between ambition and realization.

Condition 3: European coherence. Intra-European fragmentation (27 energy regimes, divergent positions on nuclear, competing national sovereignty approaches) remains the main obstacle. Scenario C of Chapter V (asymmetric partnership, baseline 3.46:1 toward 2.0–2.5:1) only works for Europe if it speaks with one voice in negotiations with Washington.

Condition 4: the time factor. The tipping point identified in Chapter V (2028, EU compute plus energy saturation, and potential activation of Cloud Sovereignty Mandates) imposes a constrained schedule. If the AI Factories are not operational and the EDF sites not connected by that date, the compute gap will solidify into structural dependence. The window for strategic action is between 2026 and 2028—after which positions crystallize around the 17.6:1 raw / 3.46:1 CACI Power Mode baseline.

Table 23. Temporal matrix of strategic recommendations by axis (2026–2032).

Source: Author's construction; April 2026 baseline (US/EU operational raw compute 17.6:1, CACI Power Mode 3.46:1).

Horizon	Compute Axis	Energy Axis	Alliances Axis
2026–2027	13 AI Factories operational; FR Special Compute Zones; long-term GPU contracts	250 MW nuclear-AI (EDF); 6 EDF data center sites; Fluidstack 1 GW operational	EU-Nvidia volume agreement; strategic GPU reserves; AI talent visas
2027–2029	5 AI Gigafactories (20B EUR); 30–40% sovereign workloads; 1.4 GW MGX-Mistral Campus	6 EPR 2 construction started; AI integration in grid plan; 8 optional EPRs confirmed	TSMC Europe 7/5 nm node; EU-Japan/Korea HBM agreements; European CLOUD Act Shield
2029–2032	40% local compute (vs 5%); sovereign frontier models;	First SMR data center; +20 GW nuclear 2035;	Multi-vendor GPU qualified; AI export norms

	SiPearl EU AI accelerator	integrated AI energy mix	(Brussels effect); 60% value chain autonomy
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Notes

1. European Commission (April 2025), AI Continent Action Plan. 13 AI Factories in 17 Member States, 200B EUR InvestAI program. Apply AI Strategy (2025): AI-first approach, buy European.
2. Centre for Future Generations (October 2025), 'Special Compute Zones: Europe's Recipe'. Derogatory zones to reduce data center installation times from 3-5 years to 12-18 months.
3. Deloitte (November 2025), 'A New Era of Self-Reliance'. InvestAI: 20B EUR for 5 AI Gigafactories, sovereign frontier models.
4. Euronews (February 2026), 'Will Big Tech's AI Spending Crush Europe's Data Sovereignty?' 2026 capex: Amazon 200B USD, Alphabet 185B USD, Microsoft 145B USD, Meta 135B USD, Oracle 50B USD. Total: 660-690B USD. European sovereign cloud spending: 10.6B EUR 2026.
5. Global Data Center Hub (May 2025), 'France's 8.5 Bn USD AI Campus'. MGX-Bpifrance-Mistral-Nvidia AI Campus: 1.4 GW, exascale, operational 2028.
6. Euronews, op. cit. Mistral Compute: 18,000 Grace Blackwell, 40 MW Essonne. Capex 1B EUR (2026). Borlänge data center (Sweden): 1.2B EUR, EcoDataCenter, green energy, opening 2027.
7. Julien Simon, Medium (January 2026), 'AI Sovereignty in Europe: A Decision Framework'. S3NS: SecNumCloud December 2025. Bleu: milestone 1 November 2025. AWS European Sovereign Cloud: January 2026, Brandenburg GmbH.
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9. Introl Blog (2025), 'France's AI Sovereignty Push'. AI Action Summit: 109B EUR. Bpifrance: 10B EUR. Fluidstack: 10B EUR, 500,000 GPUs, 1 GW, operational 2026.
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11. Enki AI, op. cit. NUWARD: 340 MWe, EDF/Naval Group subsidiary. France 2030: 1B EUR SMR. Newcleo, Stellaria, Jimmy Energy: ASN applications filed. Counter-point: Beyond Nuclear International (January 2026) signals financial difficulties for some SMR startups.
12. McKinsey (December 2025), 'Accelerating Europe's AI Adoption: The Role of Sovereign AI'. Recommendation: integrate AI demand forecasts into national energy planning. Productivity gains: up to 40% in lighthouse factories.
13. S&P Global (December 2025), 'Geopolitics of Data Centers'. ASML: 1.3B EUR in Mistral (September 2025), 11% stake. Mistral fundraising: 1.7B EUR, 11.7B EUR valuation.
14. Euronews, op. cit. Arthur Mensch (2025) quote: 'US companies are building the equivalent of a new Apollo program every year' and 'you cannot regulate your way to computing supremacy'.
15. European Commission (2025), Apply AI Strategy. AI-first for the public sector, buy European for open-source solutions. AI Observatory for trend monitoring.
16. Intelligent CIO Europe (February 2026). Gartner: 1/3 of companies will use localized AI platforms by 2027 (vs 5% today). Minimum 1% of GDP investment in AI infrastructure by 2029.
17. McKinsey, op. cit. AI fellowships and talent visas to be launched before end 2026. 44% of European tech leaders cite data security as a barrier to public cloud; 31% data localization.
18. Introl Blog, op. cit. 'If the capability gap widens, French infrastructure may serve compliance requirements without enabling competitive AI applications.' OpenAI: 20B USD ARR 2025 (x3 in one year).

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GENERAL CONCLUSION

GENERAL CONCLUSION

From AI Protectionism to the Recomposition of the Global Technological Order

This conclusion synthesizes the results of the study conducted over the 2022-2026 period, validates the central hypothesis of an American AI protectionist regime, outlines the contributions to the literature, identifies limitations and research avenues, and argues the civilizational challenge represented by AI compute as a fourth factor of production. It closes with a summary table of the eleven chapters and an inventory of the primary sources mobilized.

1. Validation of the central hypothesis

This study started from a precise hypothesis: that the Trump 2.0 administration would transform the Biden export controls into a broader protectionist regime, using AI compute as an instrument of economic and geopolitical power—an implicit 'AI for Americans First' decree. The empirical analysis conducted over the 2022-2026 period substantially validates this hypothesis.

On January 15, 2026, the Trump administration simultaneously promulgated a 25 percent tariff (Section 232) on advanced AI semiconductors (Nvidia H200, AMD MI325X) for re-exports to China, and published the BIS final rule governing AI chip exports. The combination of tariffs and export controls constitutes precisely the hybrid mechanism we anticipated: a tax on access to frontier compute that generates revenue for the US Treasury while slowing down competitors, combined with unlimited domestic access that strengthens the competitive advantage of US Big Tech.[1]

Furthermore, the July 2025 AI Action Plan formalizes a doctrine that goes beyond simple national security controls: exporting the complete AI stack (hardware, models, software, applications, and standards) to countries willing to join the American AI alliance, subject to compliance with US security requirements.[2] AI is no longer treated as just another technology, but as an instrument of power projection analogous to the dollar in the monetary system or oil in the energy system.

2. Synthesis of results

2.1 A measurable and growing American competitive advantage

The CACI (Compute-Adjusted Competitive Index) developed in this study allows for the quantification of structural asymmetry. On the April 2026 public dashboard snapshot, the US/EU(13) operational installed compute raw ratio stands at 17.6:1, translated by a weighted geometric formula $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$ into a CACI Power Mode ratio of 3.46:1. The concentration of 76.9 percent of global operational AI compute in the United States (49.9 percent including planned capacities), an annual hyperscaler capex of 660-690 billion USD (higher than Sweden's GDP), and a frontier model training cost 5 to 10 times lower than

the European cost confirm this dominance. The advantage is self-reinforcing: companies with abundant access to compute capture innovation and data rents that are very difficult to catch up with later (Chapter IV).

An important methodological nuance from Chapter I (Fig 1.8): the Phys/Sov decomposition rigorously calculated from the Epoch AI 'Owner' field reveals that asymmetry is differentiated by jurisdiction. The United States and China are fully sovereign over their installed compute (CACI Phys = CACI Sov). The EU is also very largely sovereign over its installed F (99.2 percent EU-owned). The extreme case is that of the UAE: 99.6 percent of Emirati F_total is held by US-side operators (Stargate UAE, Microsoft, OpenAI), causing the sovereign CACI to drop from 55.7 (Physical) to 6.0. This Phys/Sov dissociation, formalized in Chapter V section 5.9.2 under the hypothesis of the 2028 Cloud Sovereignty Mandates activation, radically transforms the reading of asymmetry: European vulnerability lies not on installed compute but on the operational layer of cloud workloads (mostly hosted on AWS/Azure/GCP).

2.2 A three-tier protectionist architecture

The analysis reveals that American AI protectionism operates at three distinct but cumulative levels.

Tier 1: Export controls (inherited from Biden, maintained and transformed by Trump). The tier system (Tier 1/2/3) segments the world based on geopolitical alignment: free access for close allies (20 countries), quantitative caps for the rest of the world, prohibition for adversaries (China, Russia). Even after the formal repeal of the AI Diffusion Rule in May 2025, regulatory uncertainty weighs on the investment decisions of Tier 2 countries (Chapter III).

Tier 2: Customs tariffs (Trump innovation). The 25 percent tariff on advanced AI semiconductors (Section 232, January 2026) constitutes a rupture: export controls targeted national security, tariffs explicitly target revenue and competitive advantage. The combination of tariffs and domestic exemptions creates a direct cost differential between American and non-American companies (Chapter V).

Tier 3: Capitalistic gravity. The concentration of capex (660-690 billion USD across five companies in 2026), combined with energy access (the United States accepts increased use of fossil fuels, 53.7 GW of installed DC capacity), creates a gravitational effect: investments from Japan (550 billion), the UAE, and SoftBank/Stargate converge on American soil, strengthening the compute hub without additional regulatory intervention (Chapter VI ter).

A potential fourth tier is emerging on the 2028 horizon: the Cloud Sovereignty Mandates analyzed in Chapter V section 5.9. By extending the requirements of the Framework for AI Diffusion to the cloud layer, they would transform US hyperscalers operating offshore into conditional intermediaries of global compute. The European vulnerability window (operational CADA by 2027-2028 at best, US Mandates activatable in 2028) is maximal between 2028 and 2030.

2.3 Differentiated consequences by region

Table 24 below summarizes the structural positions and specific risks of each region studied, integrating the extension to Africa developed in Chapter VI quater.

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3. Contributions of this study

This research makes five contributions to the economic and geostrategic literature.

First, the analytical integration of trajectories usually treated separately —energy, semiconductors, compute, regulation, productivity—into a unified framework. Most academic works treat these dimensions separately; our analysis shows that they form a system of interdependencies where each constraint amplifies the others (energy constrains compute, compute constrains productivity, productivity determines competitiveness).

Second, the proposal of the CACI (Compute-Adjusted Competitive Index), which provides a measurement framework for comparing AI competitiveness between regions by integrating available FLOPs, human capital, regulation, and energy cost according to a weighted geometric formula. While this index remains to be empirically refined, it constitutes a first attempt to synthesize the concept of compute-adjusted competitiveness identified as missing in the literature (Chapter II). The Phys/Sov extension introduced in Chapter I and formalized in Chapter V ($F = F_{phys} \times F_{sov}$) adds a jurisdictional dimension that distinguishes physically present compute from legally controllable compute —a distinction operational as of the April 2026 snapshot (UAE 99.6 percent US-side case) and systemic under the 2028 Cloud Sovereignty Mandates regime.

Third, the demonstration that American AI protectionism produces systemic paradoxical effects. Restrictions intended to maintain the US advantage accelerate the construction of an alternative Chinese ecosystem (DeepSeek, Huawei Ascend, real capacity 246-300 EFLOP/s versus 0.5 percent apparent in consolidated Epoch AI data), push Tier 2 countries toward China (ByteDance in Brazil, ASEAN, Africa), and incentivize Tier

1 allies to co-finance US supremacy rather than build true autonomy (Japan: 550 billion to the United States). AI protectionism does not produce a unipolar world but a world fragmented into technological blocs.

Fourth, the unprecedented comparative analysis of regional responses to AI protectionism (Europe, South America, Asia, Africa), showing that geopolitical position, energy endowment, and proximity to value chains determine fundamentally different dependency trajectories, irreducible to a single model of catch-up or falling behind.

Fifth, the extension to Africa (Chapter VI quater) documents the most extreme compute asymmetry in the world (deficit $\times 44$ to $\times 417$ depending on indicators) and shows how American protectionism creates a specific double bind for this continent: restriction of access to US frontier compute on one side, exposure to risks of surveillance and secondary sanctions from using the Chinese alternative on the other.

4. Limitations and research avenues

This study has several limitations that should be explicitly stated.

Regulatory uncertainty. The export controls environment is evolving rapidly. Biden's AI Diffusion Rule was repealed in May 2025; the Trump final rule of January 2026 could itself be modified (Commerce is to provide an update to the President by July 2026). The scenarios proposed in Chapter V reflect this uncertainty, but the space of possibilities is broader than the four formalized scenarios.

Fragmentary data. AI compute data by region remains incomplete despite the rigorous April 2026 public dashboard snapshot. Epoch AI estimates significantly under-represent real Chinese capacity (China 0.5 percent apparent vs 246-300 EFLOP/s claimed) due to the anonymization of Chinese clusters and the opacity of Huawei/Cambricon/Biren providers. The CACI is an exploratory index, calibrated on the April 2026 snapshot but not yet validated over long time series.

Time horizon. The analysis covers 2026-2030, but technological disruptions (quantum computing, sub-2 nm nodes, neuromorphic architectures) could redistribute the cards after 2030. Nvidia's current GPU advantage could be challenged by specialized ASICs (Google TPU, Amazon Trainium, Huawei Ascend) or radically different architectures (European DARE/RISC-V, 2030-2032 horizon).

Sensitivity to CACI weightings. The weightings of the geometric formula ($F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$) were chosen in Chapter II based on the literature but do not stem from an econometric calibration. A systematic sensitivity analysis on these weightings could reveal alternative trajectories not explored.

Future research avenues. Four extensions are required. First, the empirical calibration of the CACI on survey data (sectoral productivity by compute access) would allow for the validation or adjustment of current weightings. Second, the sectoral deepening of African coverage (Chapter VI quater) —notably the country-by-country analysis of the 16 identified national AI strategies and the implementation of the AU Continental Strategy Phase II 2028. Third, the dynamic modeling of the energy-compute-productivity interaction via computable general equilibrium (CGE) models integrating compute constraints as a production factor. Fourth, the longitudinal observation of the 2028 Cloud Sovereignty Mandates regime (if it actually activates) and its effects on the F_{sov} trajectory of different jurisdictions.

5. The civilizational challenge

Beyond economic metrics and geopolitical scenarios, this study reveals a more fundamental challenge. AI compute is becoming the fourth factor of production (after capital, labor, and land/energy), structuring access to productivity gains, innovation, and ultimately prosperity. Like oil in the 20th century, the control of compute in the 21st century will determine which nations and which companies capture the rents of innovation.

The United States has understood this. The July 2025 AI Action Plan explicitly treats the AI stack as an instrument of geopolitical alliance, comparable to the Marshall Plan or the Bretton Woods system: access to American compute is conditioned on strategic alignment, creating a system of hierarchical dependencies. Carnegie notes that the rule aimed to use AI exports as leverage over geopolitical pivot states, by establishing incentives for other governments to adopt American technological standards and protections in exchange for US chips.[3]

Faced with this new system, France and Europe have a strategic choice that fundamentally boils down to three options. The first is subordinate integration: accepting the status of technological junior partner in the American bloc, as Japan chose by investing 550 billion USD on US soil. This option minimizes the risk of access disruption but maximizes dependency. The second is sovereignist confrontation: building an entirely autonomous AI ecosystem, as China is forced to do. This option is unrealistic by 2030 for Europe, which lacks both a sufficient semiconductor industrial base and interior market capacity.

The third option—the one recommended by this study in Chapter VII—is targeted strategic autonomy. it consists of building sovereignty on the segments where Europe possesses a comparative advantage (French nuclear energy at 1.35x US PPP cost, ASML lithography equipment, Mistral open AI models, AI Act regulatory framework) while maintaining interoperability with the American ecosystem. The goal is not autarky but the capacity of choice: to have credible alternatives (SOV-3 sovereign cloud, local compute under EU jurisdiction, open models) to never be captive to a provider whose geopolitical interests might diverge from ours. The Phys/Sov distinction established in Chapter I is operational here: it is about increasing F_{sov} on the cloud workload layer, not just F_{phys} on the installed infrastructure.

Time is of the essence. The tipping point identified in this study is in 2028: convergence of EU compute and energy saturation (Chapter V section 5.7.3), potential activation of Cloud Sovereignty Mandates (Chapter V section 5.9), and the probable end of the vulnerability window before positions crystallize. After this date, positions stiffen around the 17.6:1 raw / 3.46:1 CACI Power Mode baseline and dependencies become structural. The 2026-2028 strategic action window is narrow. The 109 billion EUR in AI investments announced for France, the 200 billion EUR InvestAI program, the rise of Mistral Compute, and dedicated EDF nuclear sites are the elements of a response. But between announcement and execution, there is the distance that separates strategy from reality. India promises 200 billion USD but only has 1.4 GW installed. Europe cannot afford a comparable gap between ambition and realization.

Ultimately, 'AI for Americans First' is not just a trade policy scenario. It is the signal of a recomposition of the global technological order comparable to the major restructurings of the 20th century—Bretton Woods, the oil shock, the end of the Cold War. Each of these ruptures created winners and losers for decades. The

question for France and Europe is no longer whether this recomposition will take place —it is underway—but to determine whether we will be its architects or its subjects.

Fabrice Pizzi, Paris, February 2026.

Table 24. Summary of regional consequences of American AI protectionism.

Source: Author's construction, calibrated on the April 2026 snapshot (US 76.9% operational AI compute, EU raw ratio 17.6:1, CACI Power Mode 3.46:1).

Region	Structural Position	Main Impact	Specific Risk
Europe / France	Tier 1, dependent on US GPU + cloud (72-80% workloads)	Compute gap 17.6:1 raw / 3.46:1 CACI; training costs x5-10	Geopolitical vendor lock-in; marginalization if US-Asia bloc closes; F_sov vulnerability on cloud workloads
South America / Brazil	Tier 2, US-China competition ground	Technological bifurcation; amplified brain drain	Triple fracture (North-South, East-West, intra-regional)
Japan / Korea / Taiwan	Tier 1, critical value chain links	Co-financing US supremacy (550B USD Japan); production transfer	Asymmetric partnership; Taiwan advantage erosion; Japanese investment in US instead of EU
India	Tier 2, Global South pivot	GPU cap tension vs compute hub ambition	Applicative sovereignty without hardware sovereignty
China	Tier 3, forced autonomization	Parallel AI ecosystem (Huawei/DeepSeek); real capacity 246-300 EFLOP/s; 2-3 gen GPU lag	Permanent technological bifurcation; exporting to Tier 2/3 (Brazil, ASEAN, Africa)
Africa	Tier 2/3, compute deficit x44-x417	Extreme asymmetry; US/China double bind	Huawei/DeepSeek dependency; surveillance; structural confinement; UAE 99.6% US-side case

Table 25. Summary of chapters, volume, and critical apparatus of the study.

Source: Author's construction. Volume (in indicative pages) includes figures and tables but excludes econometric appendices.

Chapter	Title	Indicative Pages	Notes
I	Theoretical Framework: Technological Protectionism and AI	~12	22
II	Methodology: Scenario Matrix and CACI Index	~8	10
III	Empirical Diagnosis 2020-2026: Energy, Semiconductors, Compute	~11	20
IV	Mechanisms of the US Competitive Advantage	~9	19
V	Prospective Scenarios 2026-2030 and Cloud Sovereignty Mandates	~14	29
VI	Consequences for France and Europe	~10	14

VI bis	Consequences for South America and Brazil	~11	19
VI ter	Consequences for Asia	~12	16
VI quater	Consequences for Africa	~13	26
VII	Strategic Recommendations for France and Europe	~11	18
Conclusion	From AI Protectionism to the Recomposition of the Global Technological Order	~9	3
TOTAL	11 chapters	~120	196

Primary sources mobilized: IEA, McKinsey, Bruegel, Brookings, Carnegie Endowment, European Commission, White House/BIS, European Parliament, CSIS, S&P Global, Epoch AI, Centre for Future Generations (CFG), Euronews, ECLAC/CENIA (ILIA 2025), World Bank, Futurum, Introl, World Nuclear News, Arizton, Pillsbury Law, ITIF, Foreign Policy, Hudson Institute, Mordor Intelligence, McKinsey Global Institute, IMF. Additional data: Bloomberg, DCD, Morgan Lewis, Tom's Hardware, Serrari Group, Data Center Knowledge, WEF, Africa Defense Forum, Atlantic Council DFRLab, Carnegie Endowment, New Lines Institute, RTE, EDF, ANSSI, USTDA. Primary data: Epoch AI public dashboard snapshot April 2026 (<https://mo0ogly.github.io/America-First-IA/dashboard/>).

Notes

1. Pillsbury Law (January 2026), 'Trump Admin Targets Advanced AI Semiconductors'. Section 232: 25 percent tariff on Nvidia H200, AMD MI325X for China re-export. US domestic exemptions. Simultaneous BIS final rule. DC market update scheduled July 2026.
2. White House / CM Trade Law (July 2025), 'America's AI Action Plan'. Pillar III: export the full AI technology stack to allies . Four principles: export to allies, enforcement strengthening, global alignment, measures protection.
3. Carnegie Endowment for International Peace (May 2025), 'The Trump Administration May Be About to Repeal the AI Diffusion Rule'. Analysis of the control/promotion/leverage trilemma. Recommendation: broaden Tier 1 group, increase India allocations, strengthen localization requirements.

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Public Dashboard : <https://mo0ogly.github.io/America-First-IA/dashboard/>

Repository : <https://github.com/mo0ogly/America-First-IA>

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